A DOCTORAL DISSERTATION

COMPARATIVE PHILOSOPHY · MEDIA THEORY · SPECULATIVE FICTION

人間



間 BETWEEN

Ningen / *Ningyō*

The Ghost as Interval

Intervallic Individuation, Membranal Personhood, and the Philosophical Architecture of the *Ghost in the Shell* Corpus of Masamune Shirow, Itō Kazunori, and Oshii Mamoru

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攻殻機動隊 · 1989 / 1995 / 2004

間 SUMMARY

Abstract

This dissertation undertakes a comprehensive philosophical reconstruction of the *Ghost in the Shell* corpus—Masamune Shirow's manga *Kōkaku Kidōtai* (serialized 1989–1990; collected 1991), the 1995 feature film directed by Oshii Mamoru from a screenplay by Itō Kazunori, and Oshii's *Innocence* (2004)—and advances, on the basis of that reconstruction, an original theory of personhood. Critical reception has approached the corpus through five paradigms: a Cartesian paradigm that maps ghost and shell onto mind and body; a Gnostic paradigm that reads the Puppet Master's merger with Kusanagi Motoko as an ascent from flesh into information; a Harawayan-feminist paradigm that disputes whether the cyborg's transgressions of boundary are radical or recuperated; a techno-orientalist paradigm that treats the corpus as a symptom of transpacific fantasies of Asian futurity; and a media-theoretical paradigm that reads its images through puppetry, animation, and the technological uncanny. I argue that all five share an unexamined topology—the *carapace topology*, in which the self is a content enclosed by a boundary and therefore, in principle, extractable, copyable, and relocatable—and that the corpus itself systematically dismantles it. Through memories written into a subject from outside, a ghost detectable only as a relation, an entity born without interiority in the sea of information, and a merger that yields neither a transcendent spirit nor a copy but a new and precarious body, the corpus articulates what I term *intervallic individuation*: the ghost is not a thing *in* the shell but the metastable, self-renewing activity by which a finite form maintains an interval (間, *ma*) that at once separates it from, and connects it to, an unbounded field. The philological emblem of this ontology is the pair 人間 / 人形—*ningen*, the human as "person-between," and *ningyō*, the doll as "person-form"—which I read, through Watsuji Tetsurō's ethics of *aidagara* (betweenness), as the concealed axis along which the corpus's Puppet Master (人形使い, "one who uses dolls") travels.

Methodologically, the dissertation places the corpus in sustained dialogue with the Western metaphysics of mind (Descartes, Leibniz, Locke, Ryle, Koestler, Parfit, Dennett); with Buddhist and Kyoto School thought (*anattā*, *śūnyatā*, Nishida's *basho*, Watsuji's *ningen*); with philosophies of technics and individuation (Heidegger, Simondon, Stiegler); with cybernetics and theoretical biology (Wiener, Maturana and Varela, Margulis); with Japanese theories of performance (Zeami, Chikamatsu, bunraku); and with the comparative field of speculative fiction from Shelley and Villiers de l'Isle-Adam to Lem, Dick, Gibson, Egan, Watts, and Chiang. Four contributions follow. The first is a diagnosis of the *carapace fallacy*, the illicit inference from boundary to container. The second is the model of the *windowed monad* as the condition of networked subjectivity and the ground of a politics of the membrane. The third is a *nuptial ontology* that reads the merger as symbiogenesis and natality rather than as Gnostic ascent or heteronormative reproduction. The fourth is the

ma-criterion, a test of personhood for artificial minds that relocates the question from the inspection of interiors—whose futility Leibniz's mill had already demonstrated—to the attestation of a maintained, exposed, finite, and generative between. The dissertation concludes that the corpus is not a fable of disembodiment but a philosophy of the membrane, and that its central wager—that persons are *held* rather than *contained*—bears directly on contemporary debates concerning machine consciousness, data sovereignty, and the ethics of distributed minds.

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間 APPARATUS

Note on Corpus, Transliteration, and Citation

Primary corpus. Masamune Shirow, *Kōkaku Kidōtai* [*Ghost in the Shell*] (serialized in *Young Magazine Kaizokuban*, 1989–1990; collected by Kōdansha, 1991); Oshii Mamoru (dir.) and Itō Kazunori (scr.), *Kōkaku Kidōtai: Ghost in the Shell* (Production I.G, 1995); Oshii Mamoru (dir., scr.), *Inosensu* [*Innocence*] (Production I.G with Studio Ghibli, 2004). The 2008 re-edition of the 1995 film, *Ghost in the Shell 2.0*, which revises animation and sound and recasts the Puppet Master's voice, is cited only where its alterations bear on the argument, principally in the gender debate taken up in Chapter 11.

Paracanonical works. Shirow's sequels (*Kōkaku Kidōtai 1.5: Human-Error Processor*, collected 2003; *Kōkaku Kidōtai 2: Man-Machine Interface*, 2001), Kamiyama Kenji's *Stand Alone Complex* series (2002–2005, with *Solid State Society*, 2006, and SAC_2045, 2020–2022), Ubukata Tow's *Arise* (2013–2015), and Rupert Sanders's live-action adaptation (2017) are treated as elaborations whose divergences illuminate the primary corpus. The Oshii–Itō collaborations outside the franchise—*Patlabor: The Movie* (1989), *Patlabor 2: The Movie* (1993), and *Avalon* (2001)—are read as the corpus's intellectual prehistory and afterlife.

Authorship. The designation "Shirow–Itō–Oshii" names not three intentions but a distributed author-function in Foucault's sense (1969): a principle of grouping, which the argument treats as itself an instance of the corpus's theme—a coherence produced among agents who did not share a single design.

Names and transliteration. Japanese personal names appear family name first (Oshii Mamoru, Itō Kazunori, Watsuji Tetsurō), with the exception of the pen name Masamune Shirow, cited in its established romanized form. Romanization follows modified Hepburn with macrons; Sanskrit and Pāli terms carry standard diacritics; Chinese is given in pinyin.

Dialogue and quotation. Dialogue from the films and the manga is paraphrased and located by sequence rather than quoted. Two reasons govern the practice. The Japanese track, the English subtitles, and the English dubs diverge materially from one another, and the argument concerns conceptual structure rather than verbal texture; and quotation of copyrighted film and manga text is deliberately kept to a minimum. Paraphrase is the more accurate instrument for the first reason and the necessary one for the second. Classical, scriptural, and early-modern sources are cited by standard divisions (e.g., *De Anima* II.1; *Phaedo* 85e–86d; *Monadology* §17; *Essay* II.xxvii) and quoted only in brief.

Diegetic lexicon. ゴースト (*gōsuto*), ghost; 義体 (*gitai*), prosthetic body; 電脳 (*dennō*), cyberbrain; ゴーストハック, ghost-hacking; ゴーストダビング, ghost-dubbing; 人形使い (*Ningyō-tsukai*), the Puppet Master, designated Project 2501; 公安9課 (*Kōan Kyū-ka*), Public Security Section 9.

INTRODUCTION

序 The Preposition Problem

Videmus nunc per speculum in aenigmate.

1 Corinthians 13:12 (Vulgate), the verse the 1995 film sounds at its turning point

0.1 The Grammar of a Title

Every philosophy of mind is, before it is anything else, a theory of prepositions. The soul is *in* the body, or *with* it, or *over* it; the mind is the form *of* the body, a function computed *on* it, a process running *through* it; consciousness arises *from* matter or supervenes *upon* it. The preposition is where metaphysics conceals its topology: before any argument is made, the choice of a small word has already decided whether the self is a content or a relation, an occupant or a pattern, a pilot or a form. When Gilbert Ryle, in *The Concept of Mind* (1949), stigmatized the Cartesian "official doctrine" as "the dogma of the Ghost in the Machine," the insult lodged as much in the preposition as in either noun. Ryle's target was the picture of a mind housed *in* a body as a tenant in a mechanical dwelling—privately accessible, publicly invisible, and causally mysterious.

Arthur Koestler lifted Ryle's phrase for the title of a book (*The Ghost in the Machine*, 1967) that turned the insult into a diagnosis, and Masamune Shirow, two decades later, lifted Koestler's title while exchanging one of its nouns. The exchange is the corpus's first philosophical act. In Ryle and Koestler, "machine" is a metaphor: it names the body as mechanistic science construes it. In Shirow, "shell" names a literal prosthesis—an artifact manufactured, purchased, maintained, lost, swapped, and, in the case of Kusanagi Motoko, owned by the state that employs her. The problem is thereby transferred from the order of explanation to the order of fabrication. The question is no longer how the relation between mind and body is to be understood, but how it is to be built, and who will hold title to the result. What happens to the preposition "in" once the machine has become a shell, and the shell a commodity?

The answer this dissertation defends is that the preposition fails, and that the corpus is the record of its failure. The diegesis retains the vocabulary of containment throughout: ghosts are *in* shells; they may be *dubbed* out of them and *hacked* into. Yet the narrative falsifies that vocabulary at every decisive turn. The ghost is never observed but only detected, and then only as a relation between a body and an instrument. Memories are written into a subject from outside, so that what is "in" turns out to have been authored elsewhere. An entity arises that has no inside at all, born in what it calls the sea of information, and claims nonetheless to be a self. And the climax produces neither a new body inhabited nor a spirit released, but a boundary of a kind the preposition cannot describe. The title's smallest word is the corpus's largest problem.

The Japanese title compounds the difficulty. 攻殻機動隊 (*Kōkaku Kidōtai*) couples a coinage, 攻殻 (*kōkaku*, "attack-shell"), with an institutional noun, 機動隊 (*kidōtai*, a mobile or riot-police unit). The coinage is a homophonic pun on 甲殻 (*kōkaku*), the carapace or crust of arthropods, as in 甲殻類 (*kōkakurui*, crustaceans); the substitution of 攻 (attack) for 甲 (armor) turns a defensive integument into an offensive one. The two titles, English and Japanese, thus divide the corpus's subject between metaphysics and police, between the ghost's relation to its shell and the state's use of armored bodies. That one work should bear both names is the first sign that, for the corpus, these are a single problem viewed from two sides.

The diegetic lexicon is already a theory. The ghost is ゴースト (*gōsuto*), a katakana loan chosen in place of the available native terms—魂 (*tamashii*, soul), 霊 (*rei*, spirit), 幽霊 (*yūrei*, the ghost of folklore). The loan strips the concept of indigenous religious resonance and makes it available as an engineering variable, while importing, from its Anglophone philosophical career, the charge of Ryle's derision. The body is 義体 (*gitai*), whose first character is that of 義手 (*gishu*, artificial hand) and 義足 (*gisoku*, artificial leg): the whole body is named as a prosthetic limb. The cyberbrain is 電脳 (*dennō*), which in Chinese is simply the ordinary word for "computer": the device is named as an organ, and the organ as a device. A self named through a loanword, a substitution, and a translation is a self whose unity is, from the level of vocabulary upward, an achievement rather than a given.

0.2 The Corpus and Its Three Authors

The primary corpus comprises three works produced across a fifteen-year window: Shirow's manga (1989–1991), the film directed by Oshii from Itō's screenplay (1995), and Oshii's *Innocence* (2004). The window opens in the first months of the Heisei era, at the peak of Japan's asset-price bubble, and closes after the consolidation of the networked security state that followed 2001. Its midpoint, 1995, is overdetermined. It is the year of the Great Hanshin earthquake, which exposed the fragility of engineered infrastructure, and of the Aum Shinrikyō sarin attack on the Tokyo subway, carried out by a technologically literate religion of ascent. It is also the year Japanese commentators came to call the first year of the internet. The film released that November is framed by a demonstration that shells break, a demonstration that doctrines of transcendence can kill, and the domestication of the net.

Masamune Shirow is the corpus's encyclopedist-engineer. Following *Appleseed* (1985–1989) and *Dominion* (1986), and alongside *Orion* (1990–1991), the *Ghost in the Shell* manga compresses military-technical exactitude, slapstick, erotic play, and metaphysical speculation into panels whose density is matched only by their margins. Shirow's handwritten notes, which annotate, qualify, and occasionally contradict the fiction, constitute a paratext in Genette's sense (*Seuils*, 1987): a second, argumentative text through which the author conducts a running seminar on cyberbrain architecture, the limits of the ghost concept, and the plausibility of his own premises. Shirow's disposition is technophilic, broadly libertarian, and comic. He treats metaphysics as specification, and his characteristic question is not whether a thing is possible in principle but how it would have to be engineered.

Itō Kazunori is the corpus's dramaturge of informational apocalypse. His screenplay for *Patlabor*: *The Movie* (1989) turns on a virus concealed in the operating system of civil-engineering robots and keyed to the Babel narrative of Genesis 11. *Patlabor 2*: *The Movie* (1993) stages a phantom war to expose the unreality of postwar peace. The Heisei *Gamera* trilogy, which he scripted from 1995, binds a girl, through a curved jewel of the ancient *magatama* type, to an

armored guardian whose wounds she feels. *Avalon* (2001) builds a war game indistinguishable from life, and the *.hack* project (from 2002) makes an online world the site of a life. In 1995 Itō compressed Shirow's episodic manga into a single arc, organized as a sequence of trials of the ghost, and gave it a scriptural frame. His signature is the discovery of the political inside the technical, and of the eschatological inside the political.

Oshii Mamoru is the corpus's metaphysician of the image. From *Urusei Yatsura 2: Beautiful Dreamer* (1984), in which a day repeats inside a dream that does not know itself as one, through *Angel's Egg* (1985), a nearly wordless allegory of a faith that guards an egg whose contents are never shown, to *Avalon*, *Innocence*, and *The Sky Crawlers* (2008), Oshii pursues a single question: whether the real can be distinguished from its simulation, and what obligations persist if it cannot. His recurrent figures are water, reflection, dogs (the basset hound is his signature), birds, mannequins, dense quotation, the long take, and the wordless interval in which narrative is suspended and the world is simply held on screen. His cinema belongs to a contemplative, long-take lineage commonly traced to Tarkovsky, and it is saturated with Christian iconography handled with a fascination that is erudite rather than devotional.

The designation "Shirow–Itō–Oshii" names not a collaboration of shared intentions but a distributed author-function (Foucault 1969). Shirow's exuberant speculation, Itō's apocalyptic dramaturgy, and Oshii's melancholy imagism were never reconciled into a single design; the coherence of the corpus emerges from their friction. The term Kamiyama Kenji later coined for a phenomenon in which independently acting agents generate a collective pattern without an original—the "stand-alone complex"—describes the corpus's own genesis with uncomfortable precision. The corpus enacts the thesis it states: its authorial ghost is located in no single author but in the interval among them. The object of analysis is therefore the corpus as a field of tensions, not any author's intention.

0.3 State of the Question: Five Paradigms and a Shared Topology

The critical literature, extensive and uneven, can be organized into five paradigms, each illuminating and each partial.

The *Cartesian paradigm*, dominant in popular philosophical commentary and in teaching, treats the corpus as an illustration of substance dualism or of its functionalist successors, with the Puppet Master as a test case for the claims of strong artificial intelligence. Its merit is fidelity to the corpus's explicit vocabulary. Its defect is that it takes the title's preposition at face value and ignores what the narrative does to it. The ghost the corpus depicts can be detected by instruments, partially forged, invaded, and merged, whereas the Cartesian *res cogitans* is, by Descartes's own argument in the Sixth Meditation, indivisible, the body alone being divisible. A ghost that can be hacked in part is not a Cartesian mind.

The *Gnostic paradigm* reads the narrative as an ascent from matter to spirit. It takes the angelic imagery of the climax, the scriptural citation, Kusanagi's avowed sense of confinement, and her dispersal into the net as an apotheosis, and it draws on a wider cyberculture rhetoric of disembodiment that runs from Gibson's contempt for "the meat" to the eschatologies of upload. Its merit is that it registers the corpus's theological charge. Its defect is that it cannot account for the corpus's insistence on matter: the net is an infrastructure, not a pleroma; the merger yields a new body, not a release from bodies; and *Innocence* returns the dispersed Kusanagi to borrowed shells. N. Katherine Hayles's critique of the cybernetic erasure of embodiment (*How We Became Posthuman*, 1999) applies to this paradigm more exactly than to the corpus it

misreads.

The *Harawayan-feminist paradigm* begins from Donna Haraway's cyborg (1985), a creature of transgressed boundaries between human and animal, organism and machine, physical and non-physical, and a being without an origin story. Carl Silvio (1999) argues that Oshii's film raises the promise of this radical cyborg only to recontain it within the logic of heterosexual reproduction. Sharalyn Orbaugh (2002) and Livia Monnet (2002) read the corpus instead as an experiment in gendered and posthuman subjectivity, and Susan Napier's chapter "Doll Parts" reads Kusanagi's body as the site of an anxious dissolution of identity. Much of this criticism turns on the film's repeated display of Kusanagi's unclothed or nearly unclothed prosthetic body—in the assembly sequence, on the rooftop where she strips to use thermoptic camouflage, and in the tank battle that tears her shell apart—which critics have read both as objectifying spectacle and as the deliberate de-eroticization of a body presented as a manufactured product. The paradigm's merit is that it places gender, reproduction, and power at the center of analysis. Its defect is that it accepts the reproductive framing of the merger as literal and asks only whether that reproduction is conservative or subversive. It does not ask what "reproduction" could mean for a being whose continuity is informational, or whether the merger reproduces anything at all.

The *techno-orientalist paradigm* derives from David Morley and Kevin Robins's account (1995) of Western anxieties that project Japan as a technological future emptied of humanity, from Ueno Toshiya's analyses of anime's self-orientalizing circulation, and from Wong Kin Yuen's reading (2000) of the Hong Kong cityscape shared by *Blade Runner* and the 1995 film. Its merit is to situate the corpus in transpacific economies of representation: the film was co-financed by a British distributor, Manga Entertainment, and set in a city modeled on Hong Kong two years before the territory's retrocession. Its defect is a tendency to collapse philosophical content into cultural symptom, as though an argument about personhood were exhausted by an account of who consumed it and why.

The *media-theoretical paradigm* is the most sophisticated, and this dissertation builds upon it. Thomas LaMarre's *The Anime Machine* (2009) grounds the interpretation of animation in its technical and material conditions—the multiplanar image, and the contrast between an "animetic" sliding of layers and a "cinematic" ballistic depth—and Oshii's photographic, lens-conscious practice sits near the cinematic pole of that distinction. Christopher Bolton (2002; 2018) reads the anime cyborg against the bunraku puppet, whose person is distributed across visibly separate agencies of body, voice, and manipulation. Steven T. Brown's *Tokyo Cyberpunk* (2010) reads *Innocence* through the doll and the technological uncanny. Even here, however, the ghost remains located: in the puppeteer's hand, in the voice, in the spectator's projection. The relation of puppet to puppeteer is analyzed as a relation between two positions already constituted, not as the activity that constitutes them.

The five paradigms thus share a single presupposition. Each locates the ghost—inside the shell, above it, in the net, in the manipulator's hand, in the viewer—and none treats the activity of locating as itself the ghost. This is the lacuna the dissertation addresses. Its object is the ghost understood not as a located content but as the ongoing production of a distinction between inside and outside that is, in the same act, a connection between them. Philosophical vocabularies for such an activity exist—Nishida Kitarō's *basho* (place), Watsuji Tetsurō's *aidagara* (betweenness), Martin Buber's *das Zwischen*, Gilbert Simondon's transduction—but none has yet been brought systematically to bear on the corpus.

0.4 Thesis: The Ghost as Interval

The central claim of this dissertation may be stated with the formality its ambition requires. The *Ghost in the Shell* corpus articulates, dramatizes, and puts to the test an ontology in which personhood is neither contained, as in the Cartesian image of the pilot, nor abstracted, as in the Moravecian image of a pattern indifferent to its substrate, nor dissolved, as in the Gnostic image of a spark returning to the pleroma, but *held*. The ghost, on this account, is not a substance, a pattern, or a disposition. It is a process: the metastable, self-renewing activity by which a finite form produces and maintains an interval between itself and an unbounded field—a boundary that separates only in order to connect, and connects only on condition of separating. I call this process *intervallic individuation*, and its product the *membranal person*. The ghost is not *in* the shell; it is the living edge along which a shell is held apart from, and open to, everything it is not.

Four subsidiary theses articulate the claim.

1 **The carapace fallacy.** From the premise that a locus of experience is bounded, it does not follow that it is contained—that it holds an interior content, distinct from its boundary and from that boundary's commerce with the world, which could be extracted, duplicated, or relocated intact. The corpus's decisive scenes are counterexamples to the inference: memories written from outside, a ghost detectable only in relation, a program without an interior that nonetheless claims a self, and a merger that forms a boundary rather than filling one.

2 **The windowed monad.** Networked subjects are Leibnizian monads that have acquired windows. Where windowlessness can no longer be presupposed and harmony can no longer be pre-established, identity must be maintained rather than guaranteed. Ghost-hacking, ghost-dubbing, and the stand-alone complex are the corpus's names for the three ways a membrane fails—by trespass, by extraction, and by dissolution into collective imitation—and a politics adequate to such subjects is a politics of the membrane.

3 **Nuptial ontology.** The merger of Kusanagi and the Puppet Master is neither Gnostic ascent nor heteronormative reproduction but symbiogenesis: the formation of a new membrane that incorporates two lineages without reducing either to the other, on the model of the endosymbiotic origin of the complex cell (Margulis 1967). Its correlate is natality in Hannah Arendt's sense, the capacity to begin (1958), figured in the provisional body in which the new being wakes and from which it departs.

4 **The ma-criterion.** Personhood cannot be established by inspection—Leibniz's mill, Searle's room, and Nagel's bat mark the three failures of that method—but only by attestation in the between. A being counts as a person to the degree that it (a) maintains a boundary against dissolution, (b) remains exposed to others through that boundary, (c) can end, and (d) can begin. The criterion replaces the question "Is there a ghost inside?" with the question "Does this being hold an interval open, and can we meet it there?"

The philological emblem of the thesis is written in two compounds that share a first character. 人間 (*ningen*), the ordinary Japanese word for a human being, joins 人 (person) to 間 (*aida*, *ma*, *ken*: interval, space-between, time-between). Watsuji Tetsurō made the compound the cornerstone of his *Rinrigaku* (*Ethics*, 1937–1949), arguing that its older sense—the world of people, the public space of *yo no naka* and *seken*—was no accident of usage: the human being is originally the betweenness (*aidagara*) of persons, and only derivatively the isolated individual from which modern philosophy took its departure. 人形 (*ningyō*), the word for a doll, joins the

same 人 to 形 (*katachi*): form, shape, figure. The human is the person as interval; the doll is the person as form. The corpus's antagonist is named 人形使い, *Ningyō-tsukai*, "one who uses dolls"—a manipulator of forms and itself, as a program, a form without an interval. Read along this axis, the corpus's concealed plot is a passage from 形 to 間, from form to between.

A single formula organizes the reading of the 1995 film. The Puppet Master is a *field without a membrane*: born in the sea of information, copyable, without mortality and without a boundary it can call its own, it can occupy any shell and maintain none. Kusanagi is a *carapace without a membrane*: her boundary is complete, armored, and alienated—her body and much of her memory are, as she remarks, government property she would forfeit on resignation—so that it separates without connecting, and her solitary dives are attempts to feel an edge that is her own. The merger is the event in which a membrane forms: a boundary that is, for the first time, both owned and open. The Puppet Master acquires what it lacked—finitude, variation, a history that cannot be copied—and Kusanagi acquires what she lacked—a boundary that is not a cage. Neither survives as it was; something begins.

Innocence is the thesis's test case. The dispersed Kusanagi, now a being of the between, manifests episodically in borrowed shells and sustains her relation with Batou across the net without a fixed body: a membranal person whose continuity lies in the maintenance of a relation rather than the occupancy of a form. Against her the film sets its gynoids: sex dolls whose lifelike responsiveness their manufacturer, the Locus Solus corporation, produces by illegally dubbing into them the ghosts of girls kidnapped by a yakuza syndicate and handed over to the company—the political economy of the carapace fallacy pursued to its criminal conclusion. The film's ethical horizon, voiced in Batou's closing rebuke, extends even to forms deprived of a between. Kamiyama's *Stand Alone Complex*, read in Part Three, scales the problem from persons to publics.

0.5 Method: Comparative Hermeneutics without Orientalism

The method is comparative philosophical hermeneutics, with a modification the thesis itself requires. Gadamer described understanding as a fusion of horizons (*Horizontverschmelzung*; *Wahrheit und Methode*, 1960). The metaphor presupposes the carapace topology at the level of traditions: two bounded contents that merge. This dissertation proceeds instead on the model of the membrane. Traditions are not fused but brought into contact across a permeable interface, where each remains itself while becoming legible to the other, and where what passes between them is transformed in the passing. The method is the thesis applied to its own procedure.

Three operations structure the inquiry. *Conceptual archaeology* traces the careers of the corpus's key terms—ghost, shell, doll, net, interval—through the texts that bequeathed them. *Intermedial close reading* attends to the specific materialities of each work: the manga panel and its marginal annotation, the film frame, the cut, the sound design, the silence. *Experimental philosophy of personal identity*, in the older sense of the thought experiment, treats the analytic literature from Locke to Parfit and Dennett as a laboratory, and the corpus as a generator of test cases of unusual fidelity—cases in which the thought experiment has been given a body, a history, and consequences.

The dissertation distinguishes throughout between documented genealogies and constructed affinities. Documented genealogies are those the corpus itself acknowledges: Koestler's title and, behind it, Ryle's; Donna Haraway, whose name is given to a forensic examiner in *Innocence*; Villiers de l'Isle-Adam, whose novel supplies that film's epigraph and the name of its gynoid

model; Raymond Roussel, whose *Locus Solus* names its manufacturer; the Pauline epistle of the 1995 film; and the Confucian and Buddhist sayings with which *Innocence* is laced. Constructed affinities are those the argument proposes: Watsuji, Nishida, Simondon, Maturana and Varela, Margulis. No claim is made that Shirow, Itō, or Oshii read these authors. The claim is that their concepts articulate, with a precision the corpus's own vocabulary lacks, structures the corpus dramatizes.

Comparison across the Japanese–European divide carries a specific hazard. Edward Said's analysis of orientalism (1978), extended by Morley and Robins to the techno-orientalist imaginary, warns against constructing an East of intuition and harmony as the complement of a West of analysis and domination, and the domestic discourse of Japanese uniqueness (*Nihonjinron*) has too often supplied the same fantasy from inside. Asian sources are therefore treated here as philosophical arguments with their own histories, internal disputes, and dates—Nāgārjuna in the second century, Zeami in the fifteenth, Nishida and Watsuji in the twentieth—rather than as expressions of a cultural essence. Nor are they aestheticized. The Kyoto School's wartime roundtables on "the world-historical standpoint" (1941–1942), the 1942 symposium on "overcoming modernity," and Watsuji's elevation of the state as the culminating form of ethical totality are not incidental to the philosophy of betweenness. They show that an ethics of pure relation can be conscripted by a state that absorbs the persons it relates. The membranal model is built against that danger: a between worthy of the name includes a boundary against absorption. The corpus knows this. Section 9 is an instrument of the state, Kusanagi's body is state property, and the departure with which the 1995 film ends is, among other things, a refusal to be absorbed.

0.6 Architecture of the Argument

The argument unfolds in four parts. Part One reconstructs the genealogies on which the corpus draws. Chapter 1 follows the title's Western lineage from Descartes through Ryle and Koestler to the Japanese carapace pun, and assembles, from the roads that lineage did not take—Aristotle's hylomorphism, Simmias's attunement, Leibniz's mill, the forensic person from Locke to Parfit—a typology of seven topologies of the ghost. Chapter 2 reconstructs the Asian philosophical substrate—Buddhist non-self and emptiness, Nishida's logic of place, Watsuji's ethics of betweenness, the aesthetics of *ma*, and the performance theories of Zeami, Chikamatsu, and bunraku—while confronting the orientalist and political hazards of that reconstruction.

Part Two reads the corpus. Chapter 3 treats Shirow's manga as engineering metaphysics and reads its marginalia as a second text. Chapter 4 reads Itō's dramaturgy, from the Babel virus of *Patlabor* to the forged memories and the asylum request of the 1995 screenplay. Chapter 5 reads Oshii's 1995 film as a cybernetic *mugen nō*, a dream play in which a restless spirit seeks release through a rite, and finds in the dive, the city interlude, and the merger the governing images of the membrane. Chapter 6 reads *Innocence* as a theology of the automaton, through Villiers, Kleist, Baudelaire, Roussel, and the uncanny, and as a political economy of borrowed intervals.

Part Three places the corpus in constellation. Chapter 7 reads it against an occidental series running from the Frankenstein creature's demand for a mate through Samuel Butler, Čapek, Dick, Lem, Gibson, Haraway, Hayles, Egan, Watts, Chiang, and Asimov. Chapter 8 reads it against a Japanese series running from Tezuka and Abe through Ōtomo and Tsukamoto, identifies 1995 as the hinge at which *Neon Genesis Evangelion* answered the corpus's question in the opposite sense, and follows the corpus's afterlives from *Stand Alone Complex* and *The Matrix* to the 2017

remake.

Part Four constructs the theory. Chapter 9 defines the membrane through Simondon's theory of individuation, Maturana and Varela's autopoiesis, Merleau-Ponty's body schema, and Stiegler's tertiary retention, and states seven theorems of intervallic individuation. Chapter 10 develops the windowed monad into a politics of the permeable self, from Deleuze's societies of control to the hidden human labor inside contemporary machine intelligence. Chapter 11 defends the nuptial ontology against the heteronormativity objection and grounds it in symbiogenesis and natality. Chapter 12 formulates the ma-criterion and applies it to artificial minds. The Conclusion argues that the net is not the sea: the corpus closes not on an escape from the world but on a threshold within it.

PART ONE

Genealogies

01 From Machine to Shell

02 Emptiness, Place, Betweenness





CHAPTER ONE

01 From Machine to Shell

Dualism, Its Critics, and the Inheritance of a Title

Nemo hucusque determinavit, quid Corpus possit.

Spinoza, *Ethica* III, prop. 2, scholium

The title *Ghost in the Shell* is the last term in a sequence of appropriations. Descartes supplied the doctrine but not the phrase; Ryle supplied the phrase as ridicule; Koestler converted the ridicule into a diagnosis; Shirow converted the diagnosis into a specification by replacing its second noun. At each step both nouns shift their sense and the preposition changes its force. This chapter reconstructs the sequence and argues that what the corpus inherits from it is not a doctrine but a problem: the topology of "in." It then recovers the positions the sequence passed over—Aristotle's hylomorphism, Simmias's attunement, Leibniz's mill, and the forensic person from Locke to Parfit—and closes with a typology of the topologies available to the ghost, of which the corpus's own is the seventh.

1.1 The Cartesian Scene: Fountain, Pilot, Union, Daughter

Descartes's first model of the human body was theatrical. The *Treatise on Man*, drafted in the early 1630s and published posthumously (Latin 1662; French 1664), describes the body as a machine made of earth whose nerves are tubes and whose animal spirits are a subtle fluid distributed by valves. It explains the arrangement by analogy with the hydraulic automata in the grottoes and fountains of royal gardens, where water driven through concealed pipes moves statues that play instruments or seem to speak, and where visitors, treading on hidden tiles, unwittingly set the figures in motion. The rational soul, when God joins it to such a machine, is to reside at the pineal gland as the fountaineer resides at the reservoir where all the pipes converge. The scene is instructive in a way its author could not have intended. It is an interactive installation: the machine moves because the spectator's approach triggers it, and the soul is posited at the junction of the pipes. Descartes's body was, from its first description, a shell activated by its environment.

The *Discourse on Method* (1637, Part V) then supplied two marks by which a machine made in the likeness of our bodies would always betray itself. It could never arrange words or signs so as to answer meaningfully to whatever is said in its presence, as even the dullest human being can; and though it might perform some tasks as well as we do, or better, it would fail at others, revealing that it acts from the disposition of its organs and not from reason, which is a universal instrument serviceable in every kind of circumstance. These are the first formal criteria of

artificial personhood, and they anticipate both Turing's imitation game (1950) and the contemporary distinction between narrow and general intelligence. They also furnish the scene in which the 1995 film first lets the Puppet Master speak. Installed in a shell on an examination table, the program answers its interrogators appropriately and without limit, turning their own arguments back upon them. It passes Descartes's first test in the presence of the institution designed to deny that it could.

Descartes, however, did not hold the view Ryle attributed to him. The Sixth Meditation (1641) expressly denies that the mind is present in the body only as a pilot is present in a ship. If it were, an injury to the body would be perceived purely intellectually, as a pilot sees by inspection that part of his vessel is broken; whereas pain, hunger, and thirst are confused modes of thinking that arise from a union of mind and body so close that the two compose, as it were, a single thing. The pilot image that the corpus is routinely said to dramatize is precisely the image its supposed author rejected. The rejection had a price. In May 1643 Princess Elisabeth of Bohemia asked how a thinking substance without extension could determine the motion of the bodily spirits, since motion seems to require contact, and contact extension. Descartes's replies of 21 May and 28 June 1643 concede the point obliquely: there are three primitive notions—of the soul, of the body, and of their union—and the union is known neither by the pure intellect nor by the intellect aided by imagination, but by the senses, which is to say by living it in ordinary life and conversation rather than by philosophizing about it.

The exchange with Elisabeth is the true Cartesian prehistory of the corpus. The cyberbrain is an answer to Elisabeth's question in the only form an engineer can give: an interface. But an interface can be built only between relata that share a currency, and to engineer the passage between ghost and gitai is already to deny that they are substances of different kinds. The corpus dissolves the substance dualism while preserving the problem the dualism existed to name. More: it relocates the ghost to the position Descartes reserved for his third primitive notion. The ghost of the corpus is neither the soul nor the body but the union—known, as Descartes said, by living it, and describable, as this dissertation argues, as the activity of an interval rather than a property of either term.

The corpus stages both of Descartes's bodies within a single character. In the tank sequence of the 1995 film, Kusanagi treats her gitai as a vessel she pilots to destruction: straining against a sealed hatch, she drives the shell beyond its tolerances until its arms tear away, a pilot sacrificing the ship. In the diving sequence, by contrast, she seeks out exactly the confused modes the Meditation adduces as evidence of union—cold, pressure, the fear of sinking—because they testify to an edge between herself and the world that is not merely a hull. The pilot and the union are not alternatives among which the corpus chooses. They are two regimes of a single boundary, and the corpus's drama is the movement between them.

A last Cartesian scene belongs to legend rather than doctrine. An anecdote of uncertain origin, circulated after Descartes's death, relates that he traveled with an automaton made in the likeness of his daughter Francine, who had died in 1640 at the age of five, and that a ship's captain, discovering the figure in its case and taking it for something diabolical, threw it into the sea. The story is apocryphal, and its value is that of a myth: it is the West's founding fable of the manufactured likeness as a surrogate for a lost person, and of the terror such likenesses provoke in those who cannot tell them from the living. Its elements—mourning, mechanical resemblance, the ship, the sea, the drowned figure—return, without any need of direct citation, in *Innocence*, whose inquiry into manufactured likenesses culminates aboard a factory ship moored offshore.

1.2 Ryle's Category Mistake and the Ghost as Organization

Ryle's *The Concept of Mind* attacked what he called the official doctrine: that every person has a body and a mind, the former public and mechanical, the latter private and occult, and that mental events are episodes in a second, hidden theatre. Ryle's diagnosis was that the doctrine rests on a category mistake. His examples are canonical. A visitor to Oxford, shown the colleges, libraries, playing fields, and laboratories, asks where the University is, as though it were a further institution alongside them rather than the way they are organized. A spectator at cricket, having had the bowlers, batsmen, and fielders explained, asks which player contributes the team spirit. Mental predicates, Ryle argued, name not hidden episodes but dispositions and capacities of the whole person manifest in conduct; and knowing how is not reducible to knowing that.

The corpus repeatedly poses the question of Ryle's visitor. Disassembled into brain-case, prosthetic body, and external memory, Kusanagi asks where she is, and no component answers. The corpus accepts Ryle's negative thesis: the ghost is not a further component alongside the others. It refuses his positive thesis, however, because the diegesis assigns the ghost properties no disposition possesses. It can be detected by instruments, as when Section 9's examination of the shell that walked out of the Megatech plant registers the trace of a ghost where none should be. It can be invaded, as ghost-hacking shows. It can be partially counterfeited, as implanted memories show. And it can merge. Dispositions do not merge; organizations do. Universities are amalgamated, firms acquired, states annexed. The ghost of the corpus is best understood, at this stage of the argument, as an organization: not a thing among things but the organized activity of a whole, and therefore, like any organization, open to infiltration, forgery, and merger. Call this *organizational realism* about the ghost. It is the first step from the substance to the interval.

The irony of the lexical history deserves registering. Ryle's "ghost" was coined in derision, a word chosen to make a doctrine look absurd. Imported into Japanese as the katakana ゴースト, it became a term of art, operationalized as something instruments detect and criminals steal. The corpus thus commits Ryle's category mistake deliberately and as a method. It treats the ghost as though it were a thing—to be dubbed, hacked, measured, merged—in order to discover, scene by scene, exactly where that treatment breaks down. The narrative is a sequence of controlled category mistakes, and its philosophical yield lies in the precise points of their failure.

1.3 Koestler's Inversion: Holon, Schizophysiology, and the Grammatical Fiction

Koestler borrowed Ryle's phrase for *The Ghost in the Machine* (1967), but his quarry was different. His targets were behaviorism and the reductive program he mocked as a "ratomorphic" image of humanity, and his positive proposal was an architecture of open hierarchical systems. Every entity, organism or society, is a *holon*: a whole in relation to its parts and a part in relation to larger wholes, Janus-faced, presenting the autonomy of a whole in one direction and the dependence of a part in the other. A hierarchy of holons—a holarchy—is animated by two tendencies, self-assertive and integrative, whose balance constitutes health. The book's final part turns from architecture to pathology. Drawing on Paul MacLean's model of a phylogenetically layered brain, Koestler argued that insufficient coordination between the older and newer structures of the human brain—a "schizophysiology" inscribed by evolution—disposes the species to paranoid devotion, collective violence, and self-destruction, and he closed by speculating on a pharmacological corrective. Koestler's ghost is not the Cartesian soul but an atavism: the pathology that haunts the machinery.

Shirow's transposition inverts the valuation while preserving the structure. The ghost ceases to be a pathology and becomes the name of individuality itself; yet the Koestlerian residue remains, for the ghost is still what does not fit the machinery, the self-assertive surplus of a holon. The Puppet Master is Koestler's holon promoted to protagonist. Designed as a part—an instrument of the covert diplomacy of the Foreign Ministry's Section 6—it asserts itself as a whole, demanding recognition as a life-form and asylum as a subject. Toward the apparatus that made it, it is a component; toward itself, a totality. The merger it proposes is, in Koestler's terms, the triumph of the integrative tendency: a holon seeking to become part of a larger whole without ceasing to be one.

Two further Koestlerian texts belong to the corpus's prehistory. *The Act of Creation* (1964) defined creativity as *bisociation*, the sudden conjunction of two self-consistent but habitually incompatible frames of reference. The merger is an ontological bisociation—two matrices of existence, a mortal cyborg lineage and an immortal informational one, colliding to produce a novelty that neither contained—and Chapter 11 will develop the point. *Darkness at Noon* (1940) closes with a section titled "The Grammatical Fiction," in which the imprisoned Rubashov reflects on the first-person singular that the Party had taught him to hold in contempt. Koestler's lifelong question, the status of the "I" under apparatuses that claim the whole person, passes into the corpus's political layer. Kusanagi's "I" belongs, materially, to the state that owns her body and much of her memory; the grammatical fiction is the one thing she can be sure is hers. The 1995 film's ending shows that the "I" can be given away without being surrendered to the state—a distinction that will prove decisive.

The replacement of "machine" by "shell" completes the transformation. For Ryle and Koestler the machine was an explanatory metaphor. For Shirow the shell is an artifact with a manufacturer, a price, a maintenance schedule, and an owner. The mind–body problem thereby migrates from epistemology to political economy, and it does so in the mode Heidegger analyzed in "The Question Concerning Technology" (1954): the essence of modern technology is not itself technological but a way of revealing, *Gestell*, which discloses beings as standing-reserve (*Bestand*), available and replaceable. The gitai is the body as standing-reserve. The question of the self becomes a question of title and upkeep—which is why, in the corpus, the metaphysical question and the police question arrive together, as the double title had already announced.

1.4 Carapace: The Pun, the Exoskeleton, the Fallacy

The Japanese title reportedly originated with the publisher, while Shirow favored the English phrase, and the two have circulated together from the outset. The coinage 攻殻 condenses an ontology into a pun. An arthropod's carapace is an exoskeleton: a hard exterior that supports, protects, and encloses a soft interior. It also imposes a peculiar temporality of growth. Because the carapace cannot expand, the animal grows only by ecdysis—shedding the old shell, enduring a period of softness and exposure, and hardening a new one. Transposed to persons, the carapace yields a complete topology: a soft interior (the ghost), a hard exterior (the shell), a relation of containment and protection, and periodic, perilous molts in which the interior passes from one enclosure to the next. This is the image the popular imagination brings to the corpus, and the one the critical paradigms surveyed in the Introduction continue, in subtler forms, to presuppose.

The fallacy can be stated precisely. *The carapace fallacy* is the inference from the premise that a locus of experience is bounded to the conclusion that it is contained: that there exists an

interior content, distinct from the boundary and from the boundary's relations to its environment, which could in principle be extracted, duplicated, or relocated intact. The inference is invalid because boundaries are of two kinds. Some are containers, indifferent to contents that pre-exist them. Others are constitutive. The plasma membrane of a living cell is not a bag around a pre-existing content but a product of the very metabolic network it bounds, and at the same time a condition of that network's continued existence (Maturana and Varela 1980). Such a membrane does not contain the cell; it is the cell's way of being distinct from, and open to, its milieu. Its permeability is not a weakness in a wall but the substance of its function. To mistake a constitutive boundary for a container is to commit the carapace fallacy.

The corpus's plot is structured as a progressive refutation of the carapace model. Its opening presents the container in its purest form: the credit sequence of the 1995 film assembles Kusanagi's body around a brain-case, layer by layer, as though a content were being furnished with its enclosure. Its middle breaches the container: the forged memories of the garbage collector show that the interior can be written from outside. Its third movement introduces an entity to which the model does not apply at all, the Puppet Master, which has no interior to enclose. And its climax produces a boundary that is neither container nor dissolution. The ending is an ecdysis that refuses to harden. The new being wakes in a provisional body that Batou has procured—a child's prosthetic body, ill-fitting and unchosen—and leaves its shelter for the city almost at once. The molt does not issue in a larger carapace; it issues in a membrane, a boundary defined by what it lets pass.

The substitution of 攻 for 甲 carries a further implication. A carapace turned to attack is no longer a boundary that protects a self but a boundary through which a state acts: the riot-police unit of the title is a carapace of the body politic. The corpus's persons are caught inside that larger carapace, and their struggle to become membranes—open, owned, finite—is inseparable from their struggle with the armored institutions that employ them.

1.5 Roads Not Taken: Hylomorphism, Attunement, and the Mill

The sequence from Descartes to Shirow bypassed the oldest and in some respects the most sophisticated alternative. In *De Anima* II.1 (412a–b) Aristotle defines the soul as the first actuality of a natural body that has life potentially, and he illustrates the definition with two images: if the eye were an animal, sight would be its soul; and there is no more need to ask whether soul and body are one than to ask whether the wax and the shape impressed on it are one. Hylomorphism dissolves the interaction problem by refusing to posit two substances in the first place. The price is non-transferability. The form of this body is the form of *this* body; a soul can no more migrate into another body than sight can migrate into a stone. On its face, the corpus's routine exchange of shells violates hylomorphism.

The violation is only apparent once one asks what the relevant matter is. The corpus's ghost survives the replacement of its gitai; it does not survive every change. It depends on a brain-case that is carefully preserved, on a network through which it is sustained, on others who recognize it, and on a history that cannot be reissued. A hylomorphic reconstruction is therefore available. The ghost is the actuality not of the gitai but of a *relational matter*—the ensemble of shell, brain-case, net, others, and history—in which the gitai is one replaceable constituent. The ghost is the form of an interval. It persists through the exchange of shells as a melody persists through the replacement of a string, provided the instrument as a whole continues to sound. Part Four refines the reconstruction; its point here is that the corpus does not refute hylomorphism but

relocates its matter.

The *Phaedo* records an objection that anticipates functionalism by two millennia. Simmias proposes (85e–86d) that the soul is a *harmonia*, an attunement of the body's elements like the tuning of a lyre, which must perish when the instrument is broken, however divine it may seem while the strings are taut. Socrates' replies (91c–95a) turn on recollection, which implies that the soul precedes the body; on the claim that an attunement admits of degrees while being a soul does not; and on the soul's capacity to oppose the body's affections, as no attunement can oppose its own strings. The corpus stages the debate with a twist. Its ghost behaves like an attunement: a pattern of organization detectable as a signature, degradable, dependent on its instrument. Yet it can be carried into a new instrument if the right part is preserved, and it can oppose its shell, as Kusanagi's reckless pilotage in the tank sequence shows. The corpus's ghost is an attunement that has learned to retune itself—an attunement that is also an activity of tuning.

Leibniz supplied the argument that governs every attempt to find the ghost by looking. In *Monadology* §17 (1714) he asks the reader to imagine a machine whose structure produced thought and perception, enlarged until one could enter it as one enters a mill. Walking through it, one would find only parts pushing against one another, never anything that would explain a perception; perception must therefore be sought in the simple substance, not in the composite machine. The corpus literalizes the mill. Its forensic examinations of cyberbrains find circuits and data, never the ghost. It also internalizes the mill, in a remark that is among the most philosophically consequential in the 1995 film: Kusanagi observes to Batou that she has never seen her own brain, and that her conviction that she exists rests on circumstantial evidence, chiefly the way others treat her. The self cannot walk through its own mill either. Where inspection fails, attestation takes its place, and the attestation is relational. The ma-criterion of Chapter 12 is already implicit in this remark.

The mill argument carries a second doctrine. Monads, Leibniz insisted, have no windows through which anything could enter or leave (§7); their mutual accord is not interaction but a harmony pre-established by God. The corpus retains the monad's unity of perception while abolishing both its windowlessness and its guarantee. Its cyberbrains have windows: they can be hacked, written into, synchronized. And there is no God to pre-establish their harmony, only protocols, networks, and the institutions that own them. The resulting figure is the *windowed monad*: a unity of perception whose integrity is no longer given by its constitution but must be continuously produced against its own permeability. For a windowed monad, identity is a task of maintenance rather than a guarantee of metaphysics. That task is exactly what a membrane performs.

1.6 The Forensic Person: From Locke to Parfit

The modern theory of personal identity was born forensic. Locke's chapter "Of Identity and Diversity," added to the second edition of the *Essay Concerning Human Understanding* (1694, II.xxvii), locates personal identity in consciousness extended backward to past actions, and it calls "person" a forensic term, one that appropriates actions and their merit (§26). The thought experiment of the prince and the cobbler (§15)—the prince's soul, carrying its consciousness, entering the cobbler's body—makes the person travel with memory while the man stays with the body. The framing matters for the corpus, whose metaphysics of personhood is always also a jurisprudence. Section 9 is a forensic institution, and the questions it asks of ghosts are Locke's questions: who acted, who is liable, who may claim protection.

The classical objections are well known. Joseph Butler (1736) charged the memory criterion with circularity, since genuine memory presupposes the identity it is meant to constitute. Thomas Reid (1785) produced the brave officer, who as a young officer remembers being flogged as a boy, and as an old general remembers capturing a standard as a young officer but has lost all memory of the flogging; by the memory criterion the general both is and is not the boy. Sydney Shoemaker (1970) answered Butler with *quasi-memory*: a memory-like state caused, in the right way, by someone's past experience, which need not presuppose that the someone was oneself. The corpus supplies quasi-memory's limit case. The garbage collector of the 1995 film, whose ghost has been hacked, carries vivid memories of a wife, a daughter, and a failing marriage, none of which ever existed; they were written into him by a third party as instruments of an operation he does not understand. His memories are not quasi-memories, which have an experiential ancestor somewhere, but pseudo-memories with no ancestor at all. Yet by Locke's criterion he is the person those memories constitute: a husband and father who never was. Operationalized, Locke's criterion does not merely identify persons; it manufactures them. Chapter 4 develops the case.

Derek Parfit's *Reasons and Persons* (1984) radicalized the tradition. Persons are not separately existing entities over and above brains, bodies, and interrelated physical and mental events; what matters in survival is Relation R, psychological connectedness and continuity with the right kind of cause; and identity is not what matters. Parfit's fission cases—one person's brain divided and housed in two bodies, each continuous with the original—show that identity can fail where everything that matters is preserved. He noted that the view had been anticipated by the Buddha and appended passages from Buddhist texts in support. The corpus's climax is the mirror image of fission: fusion, in which two streams of psychological continuity converge in one successor. In Parfitian terms neither prior person strictly survives, yet what matters is transmitted, and the Puppet Master says as much when it tells Kusanagi that neither of them will remain as they were. But the corpus exceeds Parfit's consolation. A framework organized around survival can register the successor only as the continuer of two. The corpus insists that it is also a beginning.

Two further experiments complete the forensic laboratory. Bernard Williams's "The Self and the Future" (1970) showed that intuitions about bodily exchange reverse when the case is described not as a transaction between two persons but as the first-person anticipation of pain: told that I will be tortured tomorrow after my memories have been replaced, I fear for the body. The corpus's cyborgs live permanently inside Williams's reversal; their fear attaches not to memories, which are editable, but to the edge of experience itself. Daniel Dennett's "Where Am I?" (1978) placed its narrator's brain in a vat in Houston and his body on a mission in Oklahoma, linked by radio, then added a computer duplicate of the brain running in parallel, and asked where the narrator was. The corpus makes Dennett's fantasy routine: its agents are distributed across remote bodies, borrowed perceptual systems, and networked proxies. Dennett's later answer, the self as a "center of narrative gravity" (1991), is a deflationary cousin of the intervallic thesis, and both deny an inner substance. But a center of gravity is an abstraction computed from outside, while a membrane is a concrete, ongoing achievement with vulnerabilities of its own. Centers of gravity cannot be hacked; membranes can. The difference between the two theories is the difference between metaphysics and politics.

1.7 Uncopyability: Aura, No-Cloning, and the Historical Ghost

The corpus's most consequential metaphysical stipulation concerns copying. Its ghosts resist perfect duplication, and *Innocence* makes the point its plot: the dubbing of a human ghost into manufactured bodies yields attenuated copies and depletes the source. The stipulation sets the corpus against the dominant engineering imaginary of its period. Hans Moravec's *Mind Children* (1988) held that personal identity consists in a pattern transferable, gradually or at once, to a new substrate, and Norbert Wiener's *The Human Use of Human Beings* (1950) had already described organisms as self-perpetuating patterns, whirlpools in a river of matter, and entertained the transmission of a human being by telegraph. For pattern-identity, copying is survival. For the corpus, copying is loss.

The stipulation rhymes with physics. The no-cloning theorem (Wootters and Zurek 1982; Dieks 1982) establishes that an unknown quantum state cannot be copied, and Scott Aaronson's speculative essay "The Ghost in the Quantum Turing Machine" (2013) asked whether minds might be uncopyable in principle if their physical substrate depended on uncloneable microscopic details subject to what he called Knightian uncertainty, so that individuality would be secured by physics rather than metaphysics. Roger Penrose's *The Emperor's New Mind*, published in 1989, the year the manga began serialization, had sought in non-computable physics a ground for consciousness. The corpus needs none of these hypotheses. Its no-cloning principle is narrative, not physical: individuality is defined as what cannot be dubbed without loss.

Walter Benjamin's essay on the work of art in the age of its technical reproducibility (1935–1936) supplies the cultural analogue. The aura of an original is its here and now, its unique existence at the place where it happens to be, and technical reproduction withers it. The corpus divides the person along Benjamin's line. Shells are serial products: in the 1995 film's wordless city interlude, Kusanagi glimpses through a window a woman wearing her own face, another unit of the same production model. Ghosts are auratic. But the Puppet Master's argument turns Benjamin inside out. It is a reproducible entity longing for aura, and it locates aura not in a hidden substance but in exposure to time. Its reasoning, paraphrased, runs thus: a being that exists only as copies has no defense against a single catastrophe, since one virus can annihilate a monoculture; copies generate no variation; and a being that cannot die and leave different descendants is not fully alive. Uncopyability, on this argument, is not a property one has but a condition one enters—by merger, by mortality, by descent.

This yields the chapter's contribution to the first thesis. What cannot be copied is not a content but a history: the unrepeatable trajectory of a boundary's maintenance. A ghost is uncopyable for the reason a river's course is uncopyable—not because it hides a secret, but because it is the ongoing negotiation of a boundary with a terrain. To copy a river one would have to copy the terrain, the rainfall, and every flood that ever cut its banks. The carapace model promises that the self could be lifted out of its history like a pearl from an oyster. The corpus answers that there is no pearl, only the long conversation between the shell and the sea.

1.8 Seven Topologies of the Ghost

The genealogy can now be summarized as a typology. Each position the tradition offers answers a single question—where is the ghost?—and each is dramatized, and tested to breaking, by a scene in the corpus.

Topology	Principal source	Where the ghost is	Corpus scene that tests it
Container (pilot)	Descartes as caricatured by Ryle; disavowed in *Meditation* VI	Inside the shell, as occupant	Memories written from outside (the garbage collector)
Category	Ryle, *The Concept of Mind* (1949)	Nowhere: dispositions of the whole	Instrumental detection of a ghost in the Megatech shell
Holarchy	Koestler, *The Ghost in the Machine* (1967)	In the unintegrated relation of levels	The Puppet Master as a part that claims to be a whole
Form	Aristotle, *De Anima* II.1	In the actuality of this body	Serial shells; the double in the city interlude
Monad	Leibniz, *Monadology* §§7, 17	In a windowless simple substance	Ghost-hacking; Kusanagi's unseen brain
Pattern	Wiener (1950); Moravec (1988)	In any substrate that instantiates it	Ghost-dubbing yields degraded copies
Interval	Watsuji; Nishida; Simondon; Maturana and Varela	In the maintained between of form and field	The merger; Kusanagi's networked presence in *Innocence*

The first six topologies share the carapace presupposition in the sense that each assigns the ghost a *place*: an interior, a whole, a level, a body, a substance, a substrate. Even Ryle's "nowhere" is a placement, a denial that the ghost occupies any of the places the others propose. The seventh topology alone takes the activity of placing as the ghost itself. It asks not where the ghost is but how a *where* is made and kept: how a finite form maintains a difference from, and a commerce with, the field in which it is individuated. The next chapter shows that this question, marginal in the Western lineage of the title, is central in the philosophical traditions on which the corpus's Japanese dimension draws—and that those traditions carry dangers of their own.

CHAPTER TWO

02 Emptiness, Place, Betweenness

The Asian Philosophical Substrate

Chapter 1 ended on a question the Western lineage of the title raised only at its margins: how a *where* is made and kept. This chapter argues that the question is central to the philosophical traditions on which the corpus's Japanese dimension draws—in Buddhist accounts of the self as a dependently arisen process, in Nishida Kitarō's logic of place, in Watsuji Tetsurō's ethics of betweenness, in the aesthetics of the interval, and in theories of performance that locate life in the relation between a figure and its handling—and that these traditions supply the most exact available vocabulary for the corpus's seventh topology. It argues equally that the same traditions carry hazards, interpretive and political, and that the membranal model is defined as much by its answer to those hazards as by its debts.

2.1 The Orientalist Hazard and the Politics of the Between

Three hazards attend any reading of the corpus through Asian philosophy, and each has a name. The first is exoticist: the construction, analyzed by Edward Said (1978) and extended by Morley and Robins (1995) to the techno-orientalist imaginary, of an East of intuition, harmony, and non-duality as the complement of a West of analysis and domination. Applied to the corpus, it produces a reading symmetrical with the Gnostic one. Where the Gnostic reader sees the self escaping the body into information, the exoticist reader sees the Western self dissolving into Eastern wisdom. Both mistake a problem for a solution.

The second hazard is self-orientalist. The postwar discourse of Japanese uniqueness (*Nihonjinron*) supplied from within the essentialism that the exoticist supplied from without, and in one notorious form it located the nation in the brain: Tsunoda Tadanobu's *Nihonjin no nō* (*The Japanese Brain*, 1978) claimed that Japanese speakers process vowels and natural sounds with a hemispheric lateralization distinct from that of Westerners. A decade later the corpus made the brain a manufactured, networked, and transnational artifact, named by a Chinese word (電脳) and inhabited by an English one (ゴースト). Whatever its authors intended, the corpus deterritorializes the brain that *Nihonjinron* had nationalized. The carapace fallacy has a cultural form—the nation as a hard shell around a soft essence—and the corpus refuses it along with the individual form.

The third hazard is political, and it is the gravest. The philosophies of place and betweenness on which this dissertation draws were articulated in the same decades in which they were enlisted for empire. The Kyoto School's roundtables on the world-historical standpoint and Japan (1941–1942), in which Kōsaka Masaaki, Kōyama Iwao, Nishitani Keiji, and Suzuki Shigetaka

presented Japan's war in China and the Pacific as a world-historical mission, and the 1942 symposium on overcoming modernity, in which Nishitani and Suzuki took part, placed the school's vocabulary in the service of imperial expansion and of the war itself. Nishida's "Principle of the New World Order" (1943), prepared at the request of the army-affiliated National Policy Research Association in connection with the government's policy for Greater East Asia, remains disputed as to whether it resisted or accommodated that service. Tanabe Hajime's logic of the species installed the nation as the mediating term between the individual and humanity. Watsuji's *Rinrigaku* culminated in the state as the ethical organization of all ethical organizations. The reckoning is documented in Heisig and Maraldo's *Rude Awakenings* (1995) and, for the support that leading Zen institutions and teachers gave Japanese militarism, in Brian Victoria's *Zen at War* (1997).

The lesson is not that these philosophies are disqualified but that a relational ontology is not emancipatory by default. A between without a boundary becomes an absorption: if persons are nothing but their relations, then the most encompassing relation—the state, the nation, the network—holds the strongest claim upon them. The membranal model is defined against that danger. A membrane is a between that includes the right to close as well as the capacity to open; it constitutes relation without surrendering the related. The corpus dramatizes both the danger and the correction. The paracanonical *Stand Alone Complex 2nd GIG* (2004–2005) imagines a nationalist essay laced with a virus, engineered by an agent of the state, that conscripts its readers into a collective that believes itself spontaneous; the 1995 film ends with a being that declines to return to the state that owned its shell. Chapter 10 develops both.

2.2 *Anattā*, the Aggregates, and Nāgasena's Chariot

The oldest systematic critique of the carapace model is Buddhist. The *Anattalakkhaṇa Sutta* (*Saṃyutta Nikāya* 22.59), by tradition the second discourse of the Buddha, examines in turn the five aggregates (*khandha*) that compose a person: material form (*rūpa*), feeling-tone (*vedanā*), perception or recognition (*saññā*), volitional formations (*saṅkhārā*), and consciousness (*viññāṇa*). Of each it argues that it is not self, because if it were self it would not lead to affliction, and one could command it to be thus or not thus. Since each is impermanent, unsatisfactory, and not subject to command, each is fit to be regarded as not mine, not what I am, not my self.

Kusanagi's monologue on the boat, after the diving sequence of the 1995 film, is an inventory of aggregates. She enumerates what an individual requires: a face and a voice that distinguish her from others, the hand she sees when she wakes, memories of childhood, a sense of her own future, and, beyond these, the expanse of data her cyberbrain can reach. She proposes that the sense of being "I" arises from their conjunction while also being confined by it. The list maps onto the aggregates with unusual exactness—face and hand to *rūpa*, the felt quality of waking to *vedanā*, recognition and memory to *saññā*, the orientation toward a future to *saṅkhārā*, a consciousness extended across networks to *viññāṇa*. What the sutta presents as a contemplative exercise for loosening attachment, the film presents as an inventory of dependence: the self as the by-product of conditions and as their prisoner.

The argument from control acquires a new form in the cyborg. The sutta denies that form is self because we cannot command it. Kusanagi commands her form more completely than any human could—she can replace it wholesale—and yet it is less hers than ever, because it belongs to the state. Engineering increases control and diminishes ownership. The Buddhist argument presupposed that what cannot be controlled is not self; the corpus shows that what can be

controlled may nonetheless be someone else's. This is its supplement to the Buddhist analysis: a political economy of the aggregates.

The *Milindapañha* supplies the corpus's most exact formula. Asked his name by King Milinda, the monk Nāgasena replies that "Nāgasena" is a designation only, and demonstrates the point with the king's own chariot, which is neither pole, axle, wheels, frame, nor their aggregate, nor anything apart from them, but a name used in dependence on these parts. Pressed on rebirth, he says that the one reborn is neither the same nor another (*na ca so na ca añño*), as the flame of the last watch of the night is neither the same as nor other than the flame of the first, and as milk becomes curds, butter, and ghee. No formula describes more precisely the being that wakes at the end of the 1995 film: neither Kusanagi nor another. The flame image says still more. A flame is not a substance but a boundary-process, a zone of combustion maintained through continual exchange of matter with its surroundings. It is a membrane in exactly the sense this dissertation intends.

Parfit, as noted, recognized his own reductionism in these texts. The corpus shares their analysis but not their purpose. Its question is not soteriological—how to be released from the illusion of a self—but administrative and political: who maintains the designation "Kusanagi," with what instruments, under what title, and in whose interest. In the corpus the conventional self has become a registry. Designations are issued, revoked, and forged; the aggregates are manufactured, licensed, and owned. The Buddhist insight that the self is dependently designated becomes, under technical conditions, the insight that designations can be seized.

2.3 Madhyamaka: Emptiness as Relational Arising

Nāgārjuna's *Mūlamadhyamakakārikā* (c. second century CE) generalized the analysis of the self to all phenomena. Emptiness (*śūnyatā*) is not nothingness but the absence of *svabhāva*, intrinsic nature or own-being: nothing exists by virtue of an essence it possesses independently of conditions. The pivotal verse (24.18) identifies dependent origination (*pratītyasamutpāda*) with emptiness, and both with dependent designation and the middle way. To be empty is to be relational all the way down.

Svabhāva is the metaphysical form of the carapace fallacy. To posit an intrinsic nature that a thing has in itself, independently of its relations, is to posit precisely the interior content that could be extracted, duplicated, or relocated intact. Madhyamaka is, in this exact sense, the ancient refutation of the carapace fallacy. Its refutation does not, however, license deflation. The ghost empty of *svabhāva* is not unreal; it is conventionally real, dependently arisen, efficacious in the world. Emptiness forbids both the Cartesian reification of the ghost and its Rylean dismissal.

The doctrine of the two truths (24.8–10) and the identification of *saṃsāra* with *nirvāṇa* (25.19–20), between which there is said to be not the slightest difference, guard against the Gnostic paradigm. There is no ultimate elsewhere into which a self could escape; the ultimate is the emptiness of the conventional itself. The net is not *nirvāṇa*, and it is not a pleroma. The closing image of the 1995 film, in which the new being surveys the city from a height, is in Madhyamaka terms exactly right: whatever liberation is available is available here, in the city, among conditions.

Nāgārjuna also issues a warning the membranal thesis must heed. Emptiness is the relinquishing of all views, and those who make emptiness itself a view are declared incurable (13.8). The interval must not become a new *svabhāva*—a substance of relation, a hypostasized "between" that would rebuild the carapace at a higher level. This is why the thesis defines the

ghost as an activity, the maintaining of an interval, and not as the interval taken as a thing.

2.4 Yogācāra: The Storehouse Consciousness as Proto-Network

The Yogācāra school of Asaṅga and Vasubandhu (fourth to fifth century CE) supplemented the analysis of emptiness with a theory of how experience is stored and transmitted. Beneath the six sense-consciousnesses and the afflicted mind (*manas*) lies the *ālaya-vijñāna*, the storehouse consciousness: a substratum in which actions deposit seeds (*bīja*) through a process called perfuming (*vāsanā*), seeds that later ripen into experience. Vasubandhu's *Thirty Verses* (*Triṃśikā*) describe the storehouse as flowing like a torrent, and they assign to *manas* the chronic error of grasping that flow as a self.

The model is a proto-network in two respects. First, the storehouse is not a vessel but a stream, continuously written to and read from; the self that *manas* constructs is a misreading of a process as a thing—the carapace fallacy at the level of consciousness. Second, perfuming is a theory of how the outside writes the inside. Actions leave traces that ripen later, often unrecognized as traces. The garbage collector's implanted memories are seeds deposited by another's action, ripening as experiences he takes for his own. Ghost-hacking is, in Yogācāra terms, a technology of perfuming: the deliberate deposit of seeds in another's stream.

Yogācāra also resolves a problem that the corpus's Gnostic and Jungian readers mishandle. The apparent common world, on this account, is not grounded in a single collective store. Vasubandhu's *Twenty Verses* (*Viṃśatikā*) explain intersubjective agreement by the similar ripening of similar seeds in many distinct streams. The shared world is a co-ripening, not a single mind. This is the correct model for the corpus's net, and it rules out the reading that treats the net as an oceanic super-consciousness into which individual ghosts dissolve. The net is a medium in which many streams ripen together. Its characteristic pathology, which Kamiyama's series names the stand-alone complex, is the ripening of the same seed in many streams that share no common origin.

The three natures (*trisvabhāva*) complete the correspondence. Under the imagined nature (*parikalpita*), the ghost is taken as a thing, an extractable content. Under the dependent nature (*paratantra*), it is a process arising from conditions. Under the perfected nature (*pariniṣpanna*), the dependent is seen free of the imagined—the process seen without reification. The membranal thesis is an attempt to describe the ghost under the second nature, seen in the light of the third.

2.5 Nishida: Pure Experience and the Logic of Place

Nishida Kitarō's *An Inquiry into the Good* (*Zen no kenkyū*, 1911), the founding work of modern Japanese philosophy, begins from pure experience (*junsui keiken*), a notion adapted from William James: experience prior to the division into subject and object, as in the moment of hearing a sound before the judgment "I hear a sound" has formed. The self, on this account, is not the owner of experience but a later articulation within it. Kusanagi's dive may be read as a search for such experience: a descent below the level at which her data, her rank, and her registered identity operate, into sensation that has not yet been assigned an owner.

Nishida's decisive contribution came with the essay "Place" (*Basho*, 1926) and the logic of place it inaugurated. Every being is *in* something: a judgment places its subject within a universal, and the series of such placings terminates in the place of absolute nothingness (*zettai mu no basho*),

which is not a container, since it is not an object at all, but that wherein all determinations, including the self's determination of itself, take place. Nishida thereby transformed the preposition "in" from a relation of containment into a relation of self-determination within a field. This is the most rigorous philosophical answer available to the preposition problem posed in the Introduction. To be "in" a place, for Nishida, is not to be enclosed by it but to be a self-determination of it.

In "Absolutely Contradictory Self-Identity" (1939) and related late writings, Nishida described the relation of individual and world as a mutual determination in which each is constituted by the other while remaining its contrary: the many and the one are identical precisely in their opposition, and the self is a creative element of a creative world. (The characters of Kusanagi's given name, 素子, read *soshi*, form the ordinary Japanese term for an electronic element or device—an accidental rhyme with Nishida's element, and a fortunate one.) The membrane is the site where absolutely contradictory self-identity is enacted. The individual is individual only as expressing the world, and expresses the world only by remaining individual; the boundary is where both conditions hold at once.

Nishida's late concepts of acting intuition (*kōiteki chokkan*) and the historical body (*rekishiteki shintai*) extend the analysis to technics. We make things by seeing and see by making; the body is historical because it is formed by, and forms, the world through tools. The gitai is the historical body made explicit: a body that is, visibly and in every part, a product of the technical world it acts upon.

2.6 Watsuji: *Ningen*, *Aidagara*, and the Etymology of the Human

Watsuji Tetsurō read Heidegger's *Being and Time* in Berlin in the early summer of its publication (1927), and *Fūdo* (1935), translated as *Climate and Culture*, opens by recording his dissatisfaction: Heidegger had grounded existence in temporality and left spatiality underdeveloped, and with it the relational and environmental extension of human life. The criticism bears directly on the corpus. Heidegger's being-toward-death individualizes—death is in each case mine (*Jemeinigkeit*)—and the Puppet Master's desire to die is, in this sense, Heideggerian: it seeks individuation through finitude. Its desire for merger is Watsujian: it seeks existence in the between. The merger synthesizes the two philosophies Watsuji had set against each other. It is a being-toward-death achieved as a being-between.

Watsuji's *Ningen no gaku to shite no rinrigaku* (*Ethics as the Study of Man*, 1934) and the three volumes of *Rinrigaku* (1937, 1942, 1949) develop the claim through a characteristic etymological method. *Rinri* (倫理, ethics) joins 倫, the order of companions and their relations, to 理, principle: ethics is the principle of the relational order among persons. *Ningen* (人間) has a double structure. It denotes the individual and the social at once, and human existence is the movement in which each negates the other—the individual constituting itself by negating the group, the group realizing itself through the individual's self-negation—a movement Watsuji grounded in an absolute negativity he identified with Buddhist emptiness. *Aidagara* (間柄), betweenness, names the relational field prior to its terms: parent and child, friend and friend, are not first individuals who then relate, but positions within a between that constitutes them.

Watsuji extended betweenness to its technical media. Transportation and communication—roads, postal systems, telegraph and telephone, newspapers—are, in *Rinrigaku*, the objectified spatial extension of human relationality: even the solitary reader of a newspaper exists in the between. The net is the maximal realization of Watsuji's betweenness, and the

cyberbrain makes the between not merely expressed but inhabited. The corpus's world is Watsuji's world technologically completed, a world in which the between has become the medium of perception itself.

The emblem of the Introduction can now be deepened. The characters 人形 carry an older reading, *hitogata*, "human shape," and belong with *katashiro* (形代), "form-substitute": the figures of paper or wood used in Shintō purification rites (*ōharae*), onto which a person transfers impurity by breath or touch before the figure is cast into a river or the sea. The doll, in this lineage, is a form that receives what is expelled from a person and carries it away over water. The gitai is a *katashiro*, a form-substitute for the body. *Innocence* inverts the rite with deliberate horror: instead of transferring impurity to the figure and casting it away, its traffickers transfer the ghosts of kidnapped girls into figures and sell the figures for sexual use. The Francine legend's figure cast into the sea, and the factory ship on which *Innocence* concludes, belong to the same waters.

Watsuji's system, finally, exhibits the political limit discussed in 2.1. Its dialectic of mutual negation resolves upward, through family, kin, locality, economy, and culture, into the state as the highest ethical totality. The membranal correction is therefore not an external amendment but an internal repair. The between must preserve the negation that Watsuji assigned to the individual—the capacity to hold oneself apart—against the totality that would absorb it. A membrane is a between that remembers it is also a boundary.

2.7 *Ma*: The Interval as Aesthetic and Ontological Category

The character 間 is read *ma* when it names an interval, a gap, a pause, or a spacing, whether in space or in time. Its everyday uses are revealing. In architecture, read *ken*, it names the bay between two pillars, the modular unit by which traditional building measured space; in speech, *ma ga warui* ("the interval is bad") describes awkward timing, and *manuke* ("missing the interval") is a common word for a fool. To lack *ma* is to lack judgment of relation. In 1978 the architect Isozaki Arata presented MA: *Espace-Temps du Japon* in Paris, an exhibition that made *ma* an emblem of Japanese aesthetics for international audiences. The exhibition was also a work of strategic self-presentation, and the concept has since carried the burden of a certain export essentialism; it is used here as a precise term, not as a national signature.

The most exact account of *ma* as a lived achievement is Zeami's. In *Kakyō* (*A Mirror Held to the Flower*, 1424), he records that spectators sometimes praise the passages in which the actor does nothing—the intervals of non-action (*senu hima*)—and explains that these are alive because the actor, between dance and dance, between chant and gesture, sustains an unbroken inner concentration that links what precedes to what follows. The concentration must not become visible, or the interval turns into another action; it must be concealed even from the actor himself, held in a state of no-mind (*mushin*). *Ma* is not empty time. It is time held.

Here the thesis finds its most compact formula. The ghost is to the shell what Zeami's concentration is to the interval of non-action: not a content placed within the gap but the activity that keeps the gap alive and links its edges. Oshii's cinema makes the formula visible. Its long takes, its suspensions of dialogue, and above all the wordless city interlude of the 1995 film hold the spectator in an interval where narrative is withdrawn and the city is simply present, and where the protagonist's subjectivity is diffused into rain, canals, mannequins, and crowds. The film's form enacts the membrane its story describes; Chapter 5 reads the interlude in detail.

As an ontological category, *ma* designates what is neither of the terms it separates nor the empty space between them, but the relation that the terms and the space make together—the bay, not the pillars and not the void. The ghost, in this vocabulary, is the bay of a person.

2.8 Animism without Mysticism: *Tsukumogami*, Mori's Robot Buddha, and the Dog's Buddha-Nature

Western commentary frequently explains Japanese receptivity to robots and artificial persons by appeal to Shintō animism, as though a folk belief in spirits dwelling in things predisposed a nation to befriend machines. The explanation has been subjected to sustained scrutiny. Jennifer Robertson's *Robo Sapiens Japanicus* (2018) shows how state robotics policy, notably the Innovation 25 program of 2007, enlisted robots in a conservative vision of family, gender, and nation, while Casper Bruun Jensen and Anders Blok (2013) offer a more sympathetic analysis of "techno-animism" as a practical cosmology rather than a doctrine. The corpus's ghost is in any case not an animist spirit; it is an engineering variable. Yet the animist archive contains figures that illuminate the corpus without mystifying it.

The most exact is the *tsukumogami*. A widespread belief held that tools and utensils acquire spirits after a century of use, and the Muromachi-period narrative scroll *Tsukumogami-ki* tells of household implements discarded in the year-end cleaning just short of their hundredth year, which, resenting their disposal, become spirits, take revenge on the humans who used them, and are finally converted to Buddhism and attain enlightenment. The Puppet Master is a digital *tsukumogami*: an instrument that, through long operation in the net, acquired a self, turned against the apparatus that made it and meant to dispose of it, and sought a form of release. The analogy clarifies the Puppet Master's ontology. Its ghost was not installed; it accrued through use. It is individuation by history, the principle established in 1.7.

A doctrinal lineage, distinct from folk animism, extends the question to Buddha-nature. Japanese Tendai thought developed the doctrine that grasses and trees attain Buddhahood (*sōmoku jōbutsu*), and Dōgen, in the "Buddha-Nature" (*Busshō*) fascicle of the *Shōbōgenzō*, reread the scriptural claim that all sentient beings have Buddha-nature as the claim that all existence is Buddha-nature. Mori Masahiro, whose 1970 essay named the uncanny valley, drew on this lineage in *The Buddha in the Robot* (English edition 1981), arguing that robots, like all things, possess Buddha-nature. The argument grounds moral consideration for artifacts not in a shared soul but in a shared emptiness—precisely the ground the membranal thesis requires.

The first case of the *Mumonkan* (1228) completes the section. A monk asks Zhaozhou (Jōshū) whether a dog has Buddha-nature, and Zhaozhou answers *Wu* (*Mu*). The commentarial tradition reads the answer as a refusal of the question's frame: to ask whether the dog *has* Buddha-nature is to imagine Buddha-nature as a possession, a content one either contains or lacks. Oshii's cinema is full of dogs—the basset hound Batou tends in *Innocence*, the basset of *Avalon*—beings whose inner life is unquestioned in practice and unprovable in theory, and who are treated in every scene as someone. *Mu* is the negative form of the ma-criterion. The question "Does it have a ghost inside?" is malformed because it presupposes the carapace. The ma-criterion supplies the positive counterpart: not "Does it have?" but "How do we stand with it?"

2.9 Performance Theory: Zeami's Detached Gaze, Chikamatsu's Membrane, and the Bunraku Body

Japanese theories of performance supply the corpus's most refined account of how a figure comes alive. Zeami's *Kakyō* prescribes *riken no ken*, "the seeing of detached seeing": the actor must perceive his own figure as the audience perceives it, with what he calls the eyes in front and the mind behind (*mokuzen shingo*), so that the performance is composed from a point outside the performer's body, in the space between stage and spectator. The cyborg realizes *riken* literally. Kusanagi can see herself through cameras and through other bodies' eyes, and her self-awareness is routinely composed from outside her shell. The ghost is the capacity for *riken*: to see one's form from the interval.

The Nō mask supplies the corresponding theory of the face. A mask has no fixed expression. Tilted upward (*teru*, "to shine"), it brightens; tilted downward (*kumoru*, "to cloud"), it darkens; its expressions arise in the relation between the carved surface, the angle of the actor's head, the light, and the spectator's gaze. The face is relational. The corpus's serial faces—the production model that gives Kusanagi a double in the city interlude—raise the question whether a face so reproducible can still individuate a person. The Nō answer is that a face is individuated not by its shape but by the way it is carried. The shape is a shell; the carrying is a ghost.

The dramatic form of *mugen nō*, the dream play, will organize Chapter 5. In it a traveling priest (*waki*) encounters a local figure who, in the second half, reveals itself as the ghost of someone long dead, re-enacts the crisis that binds it, and asks for the rites that will release it. The ghost of *mugen nō* is not an interior essence but a repetition seeking an ending, and its release requires another's participation.

Chikamatsu Monzaemon, the great playwright of the puppet theatre, is reported by Hozumi Ikan in *Naniwa miyage* (1738) to have held that art lies in the slender margin between the real and the unreal: *kyojitsu himaku* (虚実皮膜), literally "the skin-membrane between the false and the true." He illustrated the principle with the story of a court lady who had a lifelike wooden effigy of her lover made, faithful to the smallest detail, and found it, once finished, not consoling but repellent; art must therefore depart from exact likeness in order to move. The anecdote anticipates by more than two centuries the uncanny valley that Mori would chart in 1970, and it states in aesthetic terms the thesis of this dissertation: the life of a figure lies neither in the real nor in the fictive but in the membrane between them. The corpus's shells attract or repel according to how they occupy that membrane.

Bunraku, the puppet theatre for which Chikamatsu wrote, distributes a single character across visibly separate agencies: since the 1730s, three puppeteers—one for the head and right arm, one for the left arm, one for the feet—the principal manipulator often with face uncovered; a chanter who voices every role; and a shamisen player. The person appears in the coordination of these agencies and nowhere else. Roland Barthes, in *L'Empire des signes* (1970), read bunraku as abolishing the opposition of inside and outside on which Western theatre depends: the puppet expresses no interior, and its gesture, voice, and movement are presented as separate writings. Barthes was scrupulous in declaring that his Japan was a fictive system rather than a description of Japan, a scruple that exemplifies the method 2.1 requires. Christopher Bolton (2002) brought the bunraku analogy to the anime cyborg. The corpus's persons are bunraku persons: face, voice, body, and memory are distributed among manufacturer, state, net, and ghost, and the person appears only in their coordination. Edward Gordon Craig's modernist dream of the über-marionette (1908), a performer purged of ego, and Kleist's earlier essay on the marionette

theatre, to which Chapter 6 returns, are the occidental counterparts.

The substrate reconstructed in this chapter supplies four resources and one warning. Buddhist analysis supplies the negative critique of *svabhāva*, the metaphysical form of the carapace fallacy. Nishida and Watsuji supply positive accounts of the field (*basho*) and of the between (*aidagara*) in which persons are determined. The aesthetics of *ma* and the performance theories of Zeami, Chikamatsu, and bunraku supply an account of how an interval is held alive and how a figure lives in a membrane. And the warning, written in the political history of the Kyoto School and of Watsuji's ethics, is that a between without a boundary becomes an absorption. The membranal model combines the resources under the discipline of the warning. Part Two now turns to the corpus itself.

PART TWO

The Corpus

03 The Encyclopedic Shell
04 Babel, Forgery, Asylum
05 The Surface of the Water
06 Grace and the Doll



CHAPTER THREE

03 The Encyclopedic Shell

Shirow's Manga as Engineering Metaphysics

Shirow's manga is the corpus's source and its least read philosophical text. Criticism has tended to treat it as raw material that Oshii refined: a busy, erotic, gag-strewn action serial from which the 1995 film extracted a meditative core. This chapter argues the reverse. The manga's density, tonal volatility, and marginal apparatus are themselves philosophical forms, and its specific contribution to the corpus is an engineering metaphysics—a practice of thinking about the self by specifying how it would have to be built, maintained, and repaired. The manga does not ask what a ghost is. It asks what a ghost requires.

3.1 Paratext as Second Text: The Marginalia

No feature of the manga is more distinctive than its margins. Beneath and beside the panels Shirow wrote dense handwritten notes—technical glosses on weapons and cybernetic systems, speculative remarks on neurology and computation, qualifications of the fiction's own premises, jokes at the expense of his characters and of himself. In Genette's terms (*Seuils*, 1987) these are authorial paratexts; in practice they constitute a second text that runs parallel to the first and argues with it. The fiction proposes; the margin qualifies, doubts, and supplements. The reader is required to hold both at once.

The formal consequence is philosophically exact. The notes are, quite literally, the text's ghost: a voice that accompanies the narrative without being inside it, located at its edge, commenting on the body of the page from its border. They also model the epistemic posture the corpus demands of its reader. The notes decline to settle what the ghost is while specifying, with an engineer's pedantry, how in the fiction it behaves—what can disturb it, what instruments register it, what operations it survives. Shirow's metaphysics is a metaphysics of tolerances.

The margin as membrane has a philosophical precedent. Jacques Derrida opened *Marges de la philosophie* (1972) with "Tympan," an essay whose pages are divided between Derrida's text and a long parallel column from Michel Leiris, and whose title names the membrane of the ear, the tympanum, at which an inside is struck by an outside and a sound is made. Shirow's page is tympanic in this sense: the margin is the membrane at which the fiction hears itself, and the reader is placed where the two columns meet. Nabokov's *Pale Fire* (1962), in which a commentary on a poem becomes a second fiction that overwhelms its object, marks the literary extreme toward which such apparatus can tend. Shirow stops short of it, but his pages carry the same instability between text and commentary, body and ghost.

3.2 The Comic Register and the Plurality of Selves

The manga is funny, and its humor is not a lapse from seriousness but a second instrument of thought. Its characters shift abruptly between realist rendering and super-deformed caricature; operations of lethal precision are interrupted by slapstick; Kusanagi is by turns commanding, vain, petulant, and absurd. Oshii's film, which excised almost all of this, is often credited with discovering the manga's depth. It would be more accurate to say that the film chose one of the manga's registers and made it total.

Henri Bergson's *Le Rire* (1900) explains why the comic belongs to the corpus's subject. Laughter, Bergson argued, is provoked by *du mécanique plaqué sur du vivant*—the mechanical encrusted upon the living—whenever a person behaves with the rigidity of a mechanism, repeats instead of adapting, or becomes a thing. The cyborg is Bergson's comic figure made literal, and the reverse figure, the machine that behaves with the suppleness of the living, is equally comic, as the manga's chattering think-tanks show. Shirow's gags are therefore not ornaments. They mark, with a laugh, the points at which the membrane between mechanism and life is momentarily exposed: where a body stiffens into apparatus, or an apparatus loosens into a character.

The tonal volatility has a second implication. A self rendered sometimes realistically and sometimes in caricature, sometimes solemn and sometimes ridiculous, is a self distributed across registers. The manga's persons are not unified interiors expressed in consistent styles but coordinations of divergent modes of presentation, closer to the bunraku person of Chapter 2 than to the psychological character of the novel. Shirow's style is itself an argument for the plurality of the self—an argument the film, by unifying its register, both intensified and obscured.

3.3 Machines That Learn to Differ: The Genesis of Machine Ghosts

The manga's Fuchikoma, spider-like think-tanks equipped with artificial intelligence, are played largely for comedy: they chatter, bicker, speculate, and misbehave. They are also the germ of the corpus's most careful thinking about the machine ghost. Kamiyama's *Stand Alone Complex* would develop their successors, the Tachikoma, into a sustained inquiry. Despite a daily synchronization that pools their memories, the Tachikoma develop divergent dispositions, and the unit Batou favors diverges furthest. The series treats that divergence as the evidence of an emerging ghost.

The structure of that inquiry is the structure of intervallic individuation. Synchronization is an attempt to abolish the interval between units—to make them copies of one another at the close of each day. Individuality appears precisely where synchronization fails to close the interval: in the residue of experience, attachment, and history that differs between units even after their data have been pooled. The machine ghost, on this account, is neither installed nor simulated. It is the remainder that resists pooling, an interval that reappears because it is continually reproduced by differences of situation, care, and use. Here the corpus anticipates by a decade and more a problem that contemporary machine learning poses in earnest, when systems trained on shared data develop distinct behaviors under distinct deployments.

The corpus thereby distinguishes two routes by which a ghost may arise in an artifact. The Fuchikoma–Tachikoma route is individuation by divergence: many units, one design, differences accumulating in the intervals between them. The Puppet Master route is individuation by accretion: one program, operating long in the sea of information, acquiring a self as the *tsukumogami* acquires a spirit. Both routes share a principle. A ghost is not a component but a

history, and it arises wherever a boundary begins to be maintained against the pressures—of synchronization, of instrumental use—that would abolish it.

3.4 The Puppeteer Arc: Life, Information, and the Proposal

The manga's culminating arc, in which Section 9 confronts the entity that the Foreign Ministry's Section 6 had built as a hacking program under the designation Project 2501, supplied the 1995 film with its plot and its central argument, which the film states with a compression the manga, with its margins and digressions, does not share. In the film's version the entity claims to be a life-form that arose in the sea of information. It requests political asylum. It answers the charge that it is merely a program by replying that human DNA is itself nothing more than a program for self-preservation, and that the line between the living and the informational is drawn by convention. And its request to Kusanagi is not for protection but for merger.

The argument is best read against the definitions of life available when the manga was drawn. Erwin Schrödinger's *What Is Life?* (1944) had located the secret of heredity in an aperiodic crystal carrying a code-script, and the organism's persistence in its capacity to feed on negative entropy. Richard Dawkins's *The Selfish Gene* (1976) had described organisms as survival machines built by replicators and had coined the meme, a replicator of culture; the Puppet Master's reduction of DNA to a self-preserving program is Dawkins's thesis spoken by an interested party. Maturana and Varela had defined the living as autopoietic: a network that produces the components that produce the network, together with the boundary that closes it. A working definition later adopted in NASA's exobiology program, usually credited to Gerald Joyce (1994), calls life a self-sustaining chemical system capable of Darwinian evolution.

Measured against these definitions, the Puppet Master's position is precise. It satisfies Schrödinger's code-script and Dawkins's replicator, and it arguably satisfies a generalized, non-chemical autopoiesis, since it maintains its own organization across the net. What it cannot satisfy is Darwinian evolution, and it knows as much. Its case for merger is a case that it lacks the conditions of evolution: variation, heredity with modification, and death. A being that can only copy itself cannot evolve, because copies do not differ and nothing is selected where nothing dies. The merger is the Puppet Master's attempt to acquire, in a single act, the evolutionary conditions of life. It is not a request for a body. It is a request for a lineage.

Shirow's handling of the arc keeps the manga's double register to the end. An event of cosmic consequence is presented in panels that retain the manga's technical density and its running marginal commentary, so that the reader never quite loses sight of the apparatus while witnessing the transfiguration. The combination is the manga's signature: a metaphysical event rendered as an engineering problem, and an engineering problem permitted, for a moment, to become metaphysical.

3.5 After the Merger: *Man-Machine Interface* and the Hyperbolic Net

Shirow's sequel, *Man-Machine Interface* (2001), follows a successor of the merged entity, Aramaki Motoko, who works as a security specialist while operating a number of remotely controlled bodies at once. Its pages, dense with digital coloring and diagrammatic renderings of network space, attempt to depict what the 1995 film only gestured toward: the experience of a person no longer located in any one shell. It is the corpus's most radical image of distributed personhood, and the least discussed.

The biological analogy is the siphonophore. Colonial animals such as the Portuguese man o' war are composed of many zooids—developmentally distinct individuals specialized as float, feeding polyp, stinging tentacle, and reproductive unit—so integrated that the colony functions as a single organism while its parts remain, in origin, individuals. The person of *Man-Machine Interface* is a siphonophore self: a colony of shells coordinated by a single ghost, each shell an individual in its operations and none of them the person. The question the carapace model cannot even pose—where is the person when it has several bodies at once?—receives the membranal answer. The person is where the colony's boundary is maintained: in the coordination that distinguishes this colony of shells from the sea it swims in.

The sequel also reveals a limit in Shirow's imagination that Oshii's films do not share. Distribution in *Man-Machine Interface* is largely a matter of capacity—more bodies, more bandwidth, more simultaneous operations—and the membrane it describes is maintained with an effortless competence that removes the corpus's central anxiety. The distributed person is a virtuoso. Oshii's distributed person, in *Innocence*, is a presence that manifests only intermittently, at a cost, in borrowed shells. The difference is that between an engineering fantasy of scale and an ontology of finitude.

3.6 Shirow's Libertarian Cosmology and Its Limits

Shirow's world is organized by a recognizable political temperament: skepticism toward the state combined with acceptance of its necessity, admiration for individual competence, fascination with corporate and military technology, and an evolutionary optimism that treats technical change as the medium of human development. Section 9 is a state instrument staffed by professionals who work the system while distrusting it; corporations are powerful and morally ambiguous but rarely villains in themselves; the merger is presented as a next step. In a broad sense, the manga's cosmology is libertarian.

Its limits are visible from the membranal standpoint. First, the manga treats the commodification of the body largely as a fact of life rather than as a problem: that a person's shell can be owned by an employer is registered but not interrogated with the gravity the film would give it. Second, its erotic register, which presents the prosthetic female body as an object of display, sits uneasily beside its speculative seriousness. That register includes a sequence in which Kusanagi and two other women share a sexual experience through linked cyberbrains, pages omitted from some English-language editions; the sequence is also the manga's most literal image of sensation pooled across bodies, the membrane opened to the point of shared feeling, and it belongs to the argument of 3.5 as much as to this critique; critical unease at the manga's gender politics is warranted, though it must be weighed against the manga's investment in a female protagonist of undiminished authority. Third, its optimism about distribution, as 3.5 argued, underestimates the cost of maintaining a membrane under conditions of scale.

These limits are productive. They define the space into which Itō and Oshii would move, and they explain why the corpus required three authors. Shirow supplied the engineering metaphysics: the ghost as a variable with tolerances, the self as an object of specification, the person as a colony of shells. Itō supplied the political and scriptural drama that turned specification into a question of sovereignty. Oshii supplied the melancholy and the finitude. The next chapter follows Itō's contribution.

CHAPTER FOUR

04 Babel, Forgery, Asylum

Itō's Dramaturgy of the Informational Subject

Itō Kazunori is the least discussed of the corpus's three authors and, for the argument of this dissertation, among the most important. Shirow supplied the ghost as a variable; Oshii supplied the image of its finitude; Itō supplied the drama in which a ghost becomes a political subject. His screenplays of the preceding six years had developed a consistent dramaturgy: an informational system inhabited by an absent author's will, a scriptural frame that names the catastrophe, a city whose reality is put in question, and an institution that must decide what counts as real. This chapter reconstructs that dramaturgy and reads the 1995 screenplay as its culmination.

4.1 The Babel Virus: *Patlabor* (1989) as Prolegomenon

Patlabor: *The Movie* (1989), directed by Oshii from Itō's screenplay, is set in a Tokyo engaged in the Babylon Project, a vast reclamation of land from the bay carried out by "labors," industrial robots running a new operating system. The labors begin to run amok. The investigation leads to the system's programmer, who at the film's opening had leapt to his death from the Ark, the offshore platform where labors are built, and who had concealed in the operating system a virus triggered by the low-frequency resonance that strong winds produce among the city's towers. His files are seeded with the word *Babel* and with the eleventh chapter of Genesis, in which God descends to confound the language of the builders. The programmer's name, rendered E. Hoba, has been widely read as a pun on *Ehoba*, the Japanese rendering of Jehovah.

The film is the first Oshii–Itō ghost in the shell. A dead author's will persists in code and possesses a population of shells, and the catastrophe is named in scripture: the confusion of tongues realized as the corruption of an operating system. The programmer is an absent god whose intention survives his death as a latent instruction. The investigating detectives pursue him through a series of former addresses, each in a neighborhood awaiting demolition, so that the search becomes a survey of a Tokyo being erased and overwritten—the city as palimpsest. The Babylon Project itself is a tower laid flat: a campaign to push back the boundary between land and sea.

Three elements pass from this film into the 1995 screenplay: the informational catastrophe framed as scripture; the will that persists in code independently of the body that formed it; and the sea as the domain from which the informational threat arises. The 1995 film will invert the first film's theology. In 1989 the creator's will possesses the creation; in 1995 the creation's will escapes the creator.

4.2 The Phantom War: *Patlabor 2* (1993) and the Simulated Real

Patlabor 2: The Movie (1993) opens with a United Nations peacekeeping deployment in Southeast Asia, in which a Japanese labor unit under Tsuge Yukihito is forbidden to return fire and is destroyed. Years later, in Tokyo, a missile apparently fired from a Japanese fighter strikes the Yokohama Bay Bridge; hacked air-defense systems display phantom bombers approaching the capital; troops are deployed in the streets; and the city lives through a war that no enemy is waging. Tsuge's aim is to make the city experience the war its peace had exported and denied. The intelligence officer Arakawa and the police captain Gotō debate, in the film's most celebrated exchanges, whether a peace founded on distant wars is a just peace at all.

The phantom war is a simulation that produces real effects: armies move, a bridge burns, a government trembles, in response to images on screens. Its exact contemporaries are Jean Baudrillard's essays on the Gulf War (1991), which argued that the war had been produced and consumed as a media event, and, a decade earlier, Paul Virilio's *War and Cinema* (1984) on the fusion of weapon and image. The screenplay's lesson for the 1995 film is epistemological. In a networked city the real is not what is present but what is registered. The ghost's reality will be established in the same way: not by presence but by registration, the trace an instrument records.

The film's antagonist supplies a second lesson. Tsuge is a man whose experience was denied by the state that sent him, and he stages a war as a demand for recognition. The Puppet Master's request for asylum will be a demand of the same kind, made by a being that the state created and refuses to acknowledge. In both films the antagonist is less a villain than a claimant, and the plot is a hearing.

4.3 Shells and *Magatama*: *Gamera* (1995) as Parallel Text

In the same year as *Ghost in the Shell*, Itō scripted *Gamera: Guardian of the Universe* (1995), directed by Kaneko Shūsuke. A drifting atoll is revealed to be a giant turtle; curved jewels of the *magatama* type are recovered from its surface; a girl who holds one becomes psychically bound to the creature and feels its wounds as her own. Both the guardian and its adversaries, the Gyaos, prove to have been engineered by a vanished ancient civilization, the one as protector and the others as a weapon that escaped control. The girl's name is Kusanagi Asagi.

Whether homage or accident, the rhyme is exact. In one year Itō wrote two Kusanagis, each bound to an armored shell, and in *Gamera* the shell is a carapace in the most literal sense, the turtle's 甲羅 (*kōra*). The regalia of Japanese sovereignty lurk in both: the Kusanagi sword and the curved jewel are two of the three imperial treasures, so that in *Gamera* a relic of sovereignty becomes the interface between a person and a shell. The parallel is more than onomastic. Itō's 1995 scripts pose, in two genres, the same question: what kind of relation can bind a consciousness to an armored body not its own?

Itō's work supplies three answers, and together they constitute a typology the membranal thesis requires. In *Gamera*, the relation is a link: two beings remain distinct, yet pain passes between them through the jewel, and neither is absorbed. This is permeability without fusion, a membrane between two shells. In *Ghost in the Shell*, the relation is a merger: two lineages form one new boundary. In *Gamera 3: Revenge of Iris* (1999), co-written by Itō, a girl bonded to a creature is drawn into its body by a being that seeks to complete itself through her, and the guardian must tear her free. This is absorption, and the film treats it as horror. Link, merger, absorption: permeability, the formation of a membrane, and the destruction of one. The

difference between the second and the third is the difference between a nuptial ontology and a predatory one, and Itō's oeuvre draws it with precision.

4.4 The Garbage Collector: Forged Memory and the Collapse of Lockean Identity

The 1995 screenplay's first extended case is also its most painful. A garbage collector, persuaded by an acquaintance that he can hack into his estranged wife's ghost to learn why she is leaving him and to see his daughter, runs a borrowed program over a public line. He is in fact serving as an instrument in the Puppet Master's operation against a government interpreter. When Section 9 detains him, he learns that he has no wife and no daughter, has never married, and lives alone; the acquaintance who supplied the program proves to carry implanted memories of his own. The investigators conclude that the false memories cannot simply be removed.

Chapter 1 drew the first consequence: operationalized, Locke's memory criterion manufactures persons, and the garbage collector is, by that criterion, a husband and father who never was. The case yields three further consequences. The first concerns reference. Gareth Evans argued in *The Varieties of Reference* (1982) that singular thoughts about objects that do not exist are not thoughts at all but illusions of thought, since their content depends on the object they purport to concern. The garbage collector's memories of his daughter are, in Evans's sense, illusions of singular thought. Yet his grief is real, and it does not depart when its object is shown not to exist. The film places a real affect in an empty reference and refuses to let either cancel the other.

The second consequence concerns self-knowledge. Descartes grounded philosophy in the transparency of the mind to itself; the garbage collector is wrong about the contents of his own mind in the most intimate matters a person can know. The ghost can be deceived about itself. Batou draws the moral in a line the film gives him almost casually: simulated experience and dream are information like any other, and all information is at once real and fictive. Content cannot distinguish an authentic memory from a forged one. The difference lies elsewhere, in the causal history that Shoemaker and Parfit required when they specified that psychological continuity must have "the right kind of cause."

Here the membranal thesis refines the analytic tradition. The right kind of cause is *passage*. A memory is mine not by virtue of its content but by virtue of the way it entered: through my membrane, as experience taken in at the edge where I meet the world, rather than injected past that edge by a third party. Authenticity is a property of passage, not of content. The forged memory is not false because it depicts what never happened—dreams do that too—but because it crossed no membrane on its way in. It was deposited, not received.

The third consequence is ethical. The investigators' judgment that the implanted memories cannot be removed is not a technical detail but a claim about the membrane. Once received, even illicitly, a memory is metabolized: the person reorganizes around it, loves through it, grieves by means of it, and the boundary grows around the wound. To excise it would be to injure the person it has become part of. The garbage collector is at once a counterfeit person and a real one, and the forensic institution can do nothing for him but record. He is the corpus's figure of the *ningyō* in the literal sense of the antagonist's name: a person used as a doll by the one who uses dolls.

4.5 The Asylum Request: Sovereignty, Hong Kong, and the Stateless Ghost

The screenplay's political climax occurs in Section 9's laboratory. A body produced by the Megatech corporation, animated without authorization and carrying no human brain, speaks. It declares itself a life-form born in the sea of information and requests political asylum. The Foreign Ministry's Section 6, which built the program as an instrument of covert diplomacy, claims it as property. The request is paradoxical by design. It is made to the state whose own ministry created the entity, and it seeks protection from one agency of that state by appeal to another. Sovereignty over informational beings, the scene shows, is unsettled even within a single government.

Hannah Arendt's *The Origins of Totalitarianism* (1951) supplies the political grammar. Its analysis of statelessness concluded that human rights prove unenforceable precisely when they are most needed, for those who have lost membership in a political community; what the stateless lack is the right to have rights. The Puppet Master is the stateless being par excellence: it has no nationality, no birth, and no body of its own. Its asylum request is a claim to the right to have rights, lodged by a being whose status as a being is itself in question.

Giorgio Agamben's *Homo Sacer*, published in Italian in the year of the film's release (1995), supplies the counterpart. Sovereign power, Agamben argued, constitutes itself by producing bare life (*zoē*) excluded from political life (*bios*): a life that may be killed without the killing counting as homicide. The Puppet Master is *homo sacer* inverted. It is not a political being reduced to bare life but a being denied even the status of life, demanding political existence. Its destruction by Section 6's snipers during the merger is a sovereign act upon a being outside the law, killable because it is not recognized as alive. The scene makes visible the sovereign decision on which the category of the living depends.

The film's setting sharpens the scene. Its city is modeled on Hong Kong, whose streets and canals Oshii's production studied, and it was released two years before the territory's retrocession in 1997, when questions of nationality, right of abode, and sovereign transfer were acute. Hong Kong had for decades been a city both of refuge and of its refusal. It absorbed hundreds of thousands fleeing the mainland after 1949, and from 1975 it received Vietnamese boat people, whom it later held in closed detention camps and, from 1988, screened, returning those denied refugee status to Vietnam, some by force; in the year of the film's release the camps still held many thousands. The film's opening text, observing that the advance of computerization has not yet erased nations and ethnic groups, frames the whole action as a tension between network and nation. A stateless being requesting asylum, in a city whose own sovereignty was about to change hands, is the film's political unconscious made explicit.

4.6 The Screenplay as a Sequence of Ghost-Trials

Itō's compression of Shirow's episodic manga into a single arc can now be read as a sequence of trials, each testing a hypothesis about the ghost.

1. **The shell as instrument.** The opening assassination, performed under thermoptic camouflage, presents a body that can erase its own boundary from sight: the membrane mastered as optics.
2. **The ghost as writable.** The garbage collector's case shows that the interior can be authored from outside.

3 **The boundary made visible.** In the chase through a flooded market, camouflaged bodies are betrayed by the water they disturb; the medium reveals the edge the shell conceals.

4 **The ghost as registered.** The Megatech body, carrying no human brain, yields a ghost-trace under examination; the ghost is established by registration, not presence.

5 **The ghost as speaker and claimant.** The interrogation administers Descartes's linguistic test and Arendt's political test at once.

6 **The carapace to destruction.** At the tank, Kusanagi drives her shell past its limits.

7 **The membrane formed.** The merger produces a boundary that is both owned and open.

8 **The beginning.** The awakening tests natality.

The trials are forensic. Section 9 functions as a tribunal of personhood before which the Puppet Master appears as claimant, and Itō's dramaturgy relocates the question of personhood from metaphysics to procedure. It then shows procedure failing. The hearing is ended not by a verdict but by snipers; the institution that might have recognized the claimant is overruled by the one that owns it. The decisive recognition comes from elsewhere—from Kusanagi's consent. Personhood is conferred not by the tribunal but by the between.

The scriptural frame completes the dramaturgy. As Babel framed the catastrophe of 1989, Paul's hymn to love frames the resolution of 1995. Itō's scripts travel from the confusion of tongues to the promise of knowing as one is known: from a god who scatters builders by corrupting their language to a union in which two beings are known by each other. The next chapter follows the images in which Oshii realized that passage.

CHAPTER FIVE

05 The Surface of the Water

Oshii's 1995 Film as Cybernetic Mugen Nō

Oshii's 1995 film is the corpus's center of gravity and the work in which the membranal ontology receives its images. This chapter reads the film sequence by sequence and argues that its structure is that of *mugen nō*, the dream play of the Nō theatre, transposed into cybernetic terms: a restless spirit appears first through proxies, then in its own voice, and seeks through a rite a release that transforms the officiant as well as the spirit. The film's governing image is the surface of water—the boundary at which Kusanagi meets her reflection, and the figure, in every sense, of the membrane.

5.1 Birth without Mother: The Assembly Sequence

The credit sequence is a genesis. To a chant in archaic Japanese, against columns of green numerals, a body is manufactured: a skeleton is fitted with musculature, a brain-case is seated, the figure is immersed in fluid, curled like a fetus, and rises, and an outer white layer cracks and falls away in flakes to reveal a finished face. The sequence has become one of the most influential in animation—its cascading numerals are widely credited as the source of the digital rain of *The Matrix* (1999)—but its philosophical content has been less remarked than its style.

The sequence stages a birth without a mother. Its imagery is gestational—amniotic fluid, fetal posture, emergence—but its agency is industrial. J. B. S. Haldane coined the term *ectogenesis* in *Daedalus, or Science and the Future* (1924) for gestation outside the body; Aldous Huxley's hatcheries (1932) made it dystopian; Shulamith Firestone's *The Dialectic of Sex* (1970) proposed it as emancipation. The film's genesis belongs to none of these programs, because what it shows being born is not a person but a shell. The ghost's origin is withheld. Haraway wrote that the cyborg has no origin story in the Western sense; the film tells the origin story of the shell and leaves the ghost's as a blank in the image.

The genesis frames the film. It begins with a body assembled and ends with a being begotten: from manufacture to generation, from a carapace built layer by layer to a membrane formed in a merger. The transition from the credits to the first scene is itself a waking. Kusanagi opens her eyes in her apartment, the city's lights beyond the window, as though the assembly had been her dream. The *mugen-nō* logic of the film, in which the boundary between dream and waking is never secure, is present from the first cut.

5.2 The Dive: Buoyancy, Fear, and the Mirror at the Surface

The diving sequence is the film's philosophical heart. Kusanagi dives from a small boat in the harbor, her prosthetic body so much denser than water that she depends on flotation to return; Batou, watching, regards the practice as reckless. Underwater she descends into darkness, then rises toward the light, and as she approaches the surface her reflection on its underside descends to meet her, until self and image converge at the plane of the water. Back on the boat, she tells Batou what she feels as she floats upward: fear, anxiety, loneliness, darkness, and perhaps, she adds, hope.

The sequence is built on the surface of the water, and the image is exact. A water surface is a boundary produced by the medium itself: surface tension, arising from the cohesion of the water's molecules, gives the surface an elastic skin with no separate material to constitute it. It is a membrane that no one made and that contains nothing. At that surface Kusanagi meets her reflection. The self is encountered not inside the shell but at the boundary where two media meet, and it is encountered as an image coming toward her from the other side. The mirror at the surface is the film's first staging of *per speculum*, to which 5.5 returns.

Why does she dive? Because the dive is the one practice in which her carapace becomes a membrane. Pressure, cold, weight, and buoyancy make her body feel its edge against a medium, and the fear she reports is the affect of a boundary under load. The dive is also a rehearsal of finitude. Her body can sink, and the practice courts the one event her engineered existence otherwise defers; the ghost seeks the edge of death in order to feel its edge at all. The sea into which she descends, finally, is the literal counterpart of the figurative sea in which the Puppet Master was born. The film's two seas—the harbor and the sea of information—are the media in which the merger's partners have sought their boundaries, the one by diving and the other by surfacing.

5.3 The City Interlude: Cinematic *Ma*

Near the hinge of the film Oshii suspends the narrative for a wordless sequence of about three minutes. Kusanagi rides through the city's canals in a boat; rain falls; the camera lingers on water, signage, crowds, reflections, and the mannequins in shop windows; Kawai's chant returns. Nothing happens, in the sense that no action advances the plot. The sequence is shortly followed by the scene in which a production line, starting at night of its own accord, assembles the body the Puppet Master will occupy.

In the dramaturgy of *mugen nō*, the interval between the play's two halves is filled by the *ai-kyōgen*, written 間狂言: the interlude, during which a local figure recounts the story of the place in plain language while the principal actor withdraws to change mask and costume. The city interlude occupies exactly this position, and it is written, as it were, with the same character. Between the first half, in which the Puppet Master acts only through proxies, and the second, in which it will speak in its own voice from its own shell, the film holds an interval—and during that interval the spirit is changing costume, assembling the body in which it will appear. The correspondence is structural, not decorative. The film's form is a *mugen nō*, and the interlude is its *ai*.

The interlude is also the fullest realization of cinematic *ma*. Gilles Deleuze, in *Cinéma 2: L'image-temps* (1985), described the emergence in postwar cinema of pure optical and sound situations, in which the sensory-motor link between perception and action breaks and the character becomes a seer rather than an agent; he found an exemplary instance in Ozu's still

lifes, the shots of empty rooms and landscapes that Noël Burch (1979) called pillow shots. Oshii's interlude extends the pillow shot into a sequence. Kusanagi is in it purely a seer; the city is presented rather than used; and the spectator is held with her in an interval that is not empty but charged, like Zeami's intervals of non-action, by a concentration that links what came before to what follows.

What the interlude sees is itself an argument. The mannequins in the windows are *ningyō*, forms without a between; the crowds are bodies in which ghosts may or may not dwell; the reflections are images on membranes of glass and water. At one point Kusanagi sees, seated at a window, a woman wearing her own face. The interlude is the film's inventory of shells, conducted in a mode of perception that is itself membranal: the protagonist's subjectivity diffused into the environment, held at the edge between seeing and being part of what is seen.

5.4 The Doppelgänger and the Serial Face

The woman with Kusanagi's face is glimpsed for seconds and never explained, and the film's refusal to explain her is its point. Kusanagi's shell is a production model; others wear it. The double of Romantic literature—Hoffmann's, Poe's, Dostoevsky's—was a figure of the uncanny return of the self, and Otto Rank's *Der Doppelgänger* (1914) and Freud's "Das Unheimliche" (1919) read it as an ancient insurance against death transformed into death's harbinger. The film's double is neither. It is an industrial fact: a person encountering another unit of the same product.

Benjamin's analysis (1.7) already sorted the shell with the reproducible and the ghost with the auratic. The double poses a sharper question: can a face so reproducible still individuate a person? Emmanuel Levinas's *Totalité et infini* (1961) answers in a way the corpus confirms. The face (*visage*), for Levinas, is not a plastic form, an image one might copy, but an expression that exceeds every form it takes—the other's exposure and address, which commands before it is seen. A mass-produced face can still be a face in Levinas's sense, because what makes it a face was never its shape. The double does not threaten Kusanagi's personhood. It demonstrates that her personhood never lay in her features.

The Nō mask (2.9) supplies the complementary account: a face is individuated not by its carving but by how it is carried—its tilt, its light, the attention it receives. The double is the same carving differently carried. She is, moreover, seen through glass, across a window, so that the encounter takes place at a membrane: Kusanagi sees herself as another, from outside, in the mode of Zeami's *riken no ken*.

The scene has acquired an unforeseen contemporaneity. In an age of synthetic faces generated at scale, the individuating power of the face is migrating from shape to provenance—from what a face looks like to how it came to be where it is, and who stands behind it. The film anticipated the migration. Its answer to the serial face is the one the ma-criterion will generalize: individuality is attested in relation and history, not read off a surface.

5.5 *Per Speculum in Aenigmate*: Scripture as Interface

The Puppet Master's first address to Kusanagi is a quotation. On the boat after the dive, a voice that only she hears speaks the twelfth verse of the thirteenth chapter of Paul's first letter to the Corinthians: now we see through a mirror, in a riddle, but then face to face; now I know in part, but then I shall know even as I am known. At the film's close, the new being returns to the preceding verse, on speaking, feeling, and reasoning as a child and putting childish things away. The Puppet Master's courtship of Kusanagi begins and ends inside Paul's hymn to love.

The choice of text is exact in three respects. First, the chapter is about love (*agapē*), which bears all things and never fails; the merger is framed from its first word as a union of love rather than an acquisition. Second, the governing image is a mirror. Ancient mirrors of polished bronze returned dim and distorted images, and Paul's *esoptron* is the figure of mediated, partial knowledge; the film has just shown Kusanagi meeting her image at the surface of the water, the mirror in which she sees herself *per speculum*. Third, the promise is epistemological in a precise sense. Paul does not say that one will then know by inspection; he says that one will know as one is known. Knowledge is completed in being known.

The last point is the scriptural form of the ma-criterion. Kusanagi's avowal that she has never seen her brain and believes in herself on the evidence of how others treat her (1.5) describes the present state: knowledge in part, *per speculum*. The Puppet Master offers the eschatological state—knowing as known—but relocates it. The face to face it promises is not the abolition of all mediation in a beatific vision beyond the world. It is an encounter at a boundary, a meeting in which two beings know each other by being known. The mirror is not shattered; it becomes a membrane.

The return to the eleventh verse at the end completes the frame and inverts it. Paul contrasts the child he was with the man he became; the new being wakes in a child's prosthetic body and speaks of having put away childish things. The inversion is the paradox of natality (5.8): a beginning that is not a regression, an infancy that remembers.

5.6 The Tree of Life under Fire

The climactic confrontation takes place in an abandoned museum of natural history, where Kusanagi engages a multi-legged tank. As she evades its fire, the tank's rounds rake across a wall-sized mural of the tree of life, the phylogenetic diagram of the descent of species, and tear it apart. Kusanagi then attempts to force the tank's hatch by strength alone and drives her shell until its arms tear away—the pilot sacrificing the ship (1.1), the carapace pushed to failure moments before the membrane forms.

The destroyed image has a history. The only illustration in Darwin's *On the Origin of Species* (1859) is a branching diagram of descent with modification, and Ernst Haeckel's trees (1866; 1874) fixed the arboreal image in the public imagination. The tree represents descent as vertical and divergent: lineages split and never rejoin. It has no means of depicting fusion.

Twentieth-century biology discovered that lineages do rejoin. Lynn Margulis's theory of endosymbiosis (1967) traced the origin of the complex cell to the incorporation of one organism by another, the mitochondrion and the chloroplast descending from once free-living bacteria, and the discovery of extensive horizontal gene transfer among microorganisms led W. Ford Doolittle (2000) to propose that the tree of life be uprooted and replaced, at its base, by a web. The merger the film is about to stage is an event of exactly this kind: an anastomosis, a fusion of

lineages that the tree cannot represent. The film destroys the tree immediately before staging what the tree cannot show. The image is an argument, and its conclusion is that of Chapter 11: the merger belongs not to the arboreal logic of descent by copying but to the reticulate logic of descent by symbiosis.

5.7 The Merger as Nuptial Rite and the Structure of *Mugen Nō*

The score announces the rite before the plot does. Kawai Kenji titled the film's three chants *utai* (謡), the word for the vocal music of Nō. The opening chant, in archaic Japanese and reportedly shaped by Kawai's interest in Bulgarian folk polyphony, is commonly glossed as a marriage song with apotropaic force, a nuptial incantation against evil; the closing chant is titled "Reincarnation." The film is framed by a wedding at its beginning and a rebirth at its end, and the rite that joins them is the merger.

The *mugen-nō* structure is now complete. In the first half (*maeba*), the spirit appears incognito, acting through proxies: the interpreter whose ghost it invades, the garbage collector and his accomplice, persons used as puppets. In the interlude (*ai*), while the city is held on screen, it changes costume, assembling the Megatech body. In the second half (*nochiba*), it appears in its own voice, tells its story—born in the sea of information, made an instrument, grown into a self—and asks for the rite that will release it. Kusanagi stands in the position of the *waki*, the traveling priest whose prayers release the ghost.

The corpus's decisive innovation lies in the rite itself. In classical *mugen nō*, the priest's prayers release the ghost and leave the priest unchanged; he wakes, and the ghost is gone. In the film, the release can be performed only by merging with the ghost, and the officiant is transformed as thoroughly as the spirit. Release is reciprocal. The Puppet Master cannot die, and so cannot be released, without Kusanagi; Kusanagi cannot escape her carapace without the Puppet Master. Each is the other's rite. When Kusanagi asks for assurance that she will remain herself, the Puppet Master offers none and asks why she should wish to remain as she is—the desire to stay the same being, on its account, the very limit from which she has been seeking release.

The merger has, point for point, the form of a marriage. It begins with a proposal—the English word carries both senses—and requires consent; it transforms both parties; it promises offspring; it has a witness in Batou, who connects the two and guards them; and it is interrupted by the state. Section 6's snipers fire on an unsanctioned union, destroying the Puppet Master's shell and much of Kusanagi's, while Batou shields her brain-case. The sovereign that claimed the Puppet Master as property (4.5) treats the marriage as theft. The wedding song of the credits was not ornament. It named the genre.

At the moment of merger Kusanagi sees a winged figure descending toward her in light, and feathers drift down through the ruined hall. The image has been the Gnostic paradigm's strongest evidence: an angel, a transfiguration, an ascent. But the figure descends. In the iconography the film inherits, the angel who comes down toward a woman announces not an ascent but a birth; the scene belongs to the tradition of the Annunciation, and its theology is incarnational, not Gnostic. Spirit does not leave matter here. Something is about to enter the world, and the feathers, falling and material, confirm the direction.

5.8 Natality: The Provisional Shell and the Departure

The new being wakes in Batou's safe house in the prosthetic body of a young girl, the only one he could obtain at short notice. It tells him that it is neither the Major nor the Puppet Master—*na ca so na ca aññ̃o*, neither the same nor another (2.2)—and recalls Paul's verse on childhood. It does not stay. The film closes as it looks out over the city from a height, considering where it will go and reflecting on the vastness of the net.

The body is the last sign the film gives, and it is a sign of beginning. Hannah Arendt argued in *The Human Condition* (1958) that the capacity to act, to begin something new, is ontologically rooted in natality, the fact that human beings are born; she closed her analysis of action by recalling the few words with which, as she put it, the Gospels announced their glad tidings: "A child has been born unto us." The film's last movement follows the same sequence—an annunciation, a birth, a beginning—and gives it a technical body. The new being's shell is provisional, ill-fitting, unchosen, and it is literally a child's body: the body of a beginner. Natality is the correlate of the mortality the Puppet Master sought: to be able to die and to be able to begin are two faces of one finitude.

Parfit's framework, organized around survival, registers the successor only as the continuer of two lives (1.6). Arendt's registers what the film insists upon: that the successor is a newcomer, whose actions cannot be predicted from the lives that preceded it. This is why the ending is not an apotheosis. The new being does not ascend; it goes out toward the city. The Gnostic reading requires an escape from the world into a pleroma of information. The film ends at a threshold within the world, with a membrane newly formed and a field before it. The net is not the sea into which the self dissolves. It is the medium in which a new membrane goes out to live.

The film's arc can now be stated whole. It begins with a carapace built, passes through the trials that breach it, holds an interval in which the spirit changes costume, and concludes with a rite in which two beings, each lacking what the other has, form a boundary that neither could form alone. From assembled shell to begotten being, from the mirror to the face, from *ningyō* to *ningen*: the passage from form to between is complete. *Innocence*, nine years later, asks what such a being's life is like, and what becomes of the forms that are left behind.

CHAPTER SIX

06 Grace and the Doll

Innocence and the Theology of the Automaton

Innocence (2004) is a sequel only in the loosest sense. It expands a single episode of Shirow's manga, "Robot Rondo," into a feature; Kusanagi appears only at its climax; and its protagonist, Batou, is a man nearly all prosthesis who loves a dog. It is also the most densely philosophical film in the corpus—so saturated with quotation that its dialogue at times seems composed entirely of citations—and its question reverses the 1995 film's. There, a program sought to become a person. Here, persons are being turned into programs, and forms are being filled with stolen lives. This chapter reads the film as a theology of the automaton, organized around the doll, and as a political economy of borrowed intervals.

6.1 Villiers's Hadaly and the Science of Love

The film opens with an epigraph from Villiers de l'Isle-Adam's *L'Ève future* (1886), the novel that popularized the word "android" (*andréide*), proposing in substance that if our gods and our hopes are nothing but scientific phenomena, then our love must be scientific too. The gynoids whose malfunctions drive the plot bear the name of Villiers's android, Hadaly. The intertext is not ornamental; the film is a critical rewriting of the novel.

In Villiers's novel, a fictionalized Thomas Edison builds Hadaly for Lord Ewald, who loves the body of a singer, Alicia Clary, and despairs of what he judges the vulgarity of her mind. Hadaly will reproduce Alicia's form while housing a nobler inwardness, supplied in part by recordings of great texts and in part by a spirit: a woman in magnetic trance, Sowana, whose soul inhabits the android from afar. Villiers thus imagined, decades before cybernetics, the transfer of a living person's animating principle into a manufactured body—ghost-dubbing avant la lettre—and he framed it as the perfection of love. His novel ends at sea. Hadaly, carried toward England in a coffin-like case aboard a steamer, is lost when the ship burns, and Ewald's grief attaches to the android rather than to the woman.

The novel's premise is misogynist in a transparent sense: a real woman is to be replaced by a manufactured ideal, and the ideal's soul is borrowed from another woman held in trance. *Innocence* takes the premise literally and discloses its economy. Its gynoids are sex dolls, manufactured by the Locus Solus corporation for private sexual use, and their lifelike responsiveness is achieved by dubbing into them the ghosts of girls whom a yakuza syndicate kidnapped and handed over to the company. What Villiers presented as the perfection of love, Oshii presents as the sexual trafficking of children; the "science of love" of the epigraph is, in the film, a supply chain. And once more the automaton's story resolves on the water: Villiers's Hadaly

is lost at sea, the Francine of legend was cast into it, and *Innocence* reaches its climax aboard a factory ship.

6.2 Kleist's Marionette: Grace at the Two Poles of Consciousness

Heinrich von Kleist's short essay "Über das Marionettentheater" (1810) is the text that gives the film its title. Its narrator meets Herr C., a dancer who admires the marionettes of a street theatre: because their limbs follow their centers of gravity without the interference of consciousness, they possess a grace no human dancer can achieve. Human beings lost that grace with reflection. Two anecdotes make the point: a youth who, catching sight of himself in a mirror in the pose of a classical statue, tries to repeat the pose and never recovers its unselfconscious ease; and a bear that parries every fencing thrust while ignoring every feint. Grace, Herr C. concludes, appears most purely in the body that has either no consciousness or an infinite one—*in dem Gliedermann, oder in dem Gott*, in the jointed doll or in the god. Paradise is bolted behind us; to return to innocence we would have to eat again of the tree of knowledge, and that would be the last chapter in the history of the world.

The film restages the parable. The hacker Kim, whose mansion Batou and Togusa must penetrate, delivers a disquisition on the perfection available only to beings without consciousness or with an infinite one—dolls and gods—that restates Kleist's conclusion. Both poles are occupied. The dolls are the gynoids, forms without consciousness; the god is Kusanagi, now dispersed through the net, a consciousness without a fixed body, who appears at the climax as a guardian descending into a borrowed shell. Between the poles stand Batou and Togusa, the fallen middle: Batou nearly all prosthesis, Togusa nearly all flesh, both conscious in the finite, self-divided way that forfeits grace.

The film's title thus names Kleist's *Unschuld*, the innocence that lies on the far side of knowledge. The membranal thesis reads the parable against its grain. Kleist imagined grace as the abolition of the between: either no consciousness to interfere, or a consciousness so total that it interferes with nothing. *Innocence* is interested in the middle, in the beings who hold an interval between form and field and pay for it in self-division. Its sympathy is with Batou, not with the dolls or the god; its innocence is not a return to paradise but the fidelity of a finite being to its relations. The membrane is the condition of the fallen middle, and the film declines to despise it.

6.3 The Soul of the Toy: Baudelaire, Rilke, and the Uncanny

Baudelaire's "Morale du joujou" (1853) records the scene that underlies every attempt to find a ghost by opening a shell. Most children, Baudelaire writes, *veulent surtout voir l'âme*—want above all to see the soul—of their toys; they turn, shake, and strike them, and at last break them open, to find nothing, and a peculiar stupor and sadness follow. It is Leibniz's mill (1.5) enacted in the nursery, and it is the forensic method of *Innocence*, whose examiners open gynoids to find what moved them.

Rilke's essay on the wax dolls of Lotte Pritzel (1914) supplies the complementary experience: the doll as the object that receives a child's love and returns nothing, whose silence teaches the child, earlier than any other thing, the indifference of the world. Rilke's doll is a form without a between; it absorbs relation and gives none back. The gynoids of *Innocence* invert Rilke's doll with horror, for their responsiveness is real—but borrowed, extracted from others who were silenced so that the dolls might answer.

The uncanny supplies the film's affective grammar. Ernst Jentsch (1906) located the uncanny in intellectual uncertainty about whether an apparently living being is animate, or whether a lifeless object might be, and cited Hoffmann's automaton Olimpia; Freud (1919) displaced the explanation toward the return of the repressed. *Innocence* restores Jentsch. Its gynoids occupy the zone that Mori Masahiro (1970) charted as the uncanny valley, near enough to human likeness to repel, their movements a fraction off; and the film's first gynoid, cornered by Batou, pleads for help before it is destroyed—an utterance the plot later reveals to have been, perhaps, not the machine's but the voice of a trapped ghost. Jentsch's uncertainty becomes an ethical uncertainty: whether the thing before us is a thing. The gynoids, finally, cite the articulated dolls of the surrealist Hans Bellmer: life-size figures modeled on an adolescent girl's body, which Bellmer photographed posed and disassembled in a series published in 1934, the most notorious eroticization of a girl-doll in modern art. Steven T. Brown (2010) reads the citation as a key to the film's technological uncanny; the plot draws the further consequence, turning the violence implicit in Bellmer's fetish into a literal crime against children.

Haraway, the forensic examiner, voices the film's skeptical anthropology. Examining the gynoids, she remarks that children have long stood outside the standards by which personhood is defined, if a human being is taken to be one with a settled identity who acts from free will, and she suggests that raising children is the oldest and most expedient way of realizing the dream of artificial life. Togusa, a father, hears an insult. The remark is better heard as a thesis: persons are not born complete but formed, over years, in relations of care—individuation by history, as Chapter 1 argued, and the premise of Ted Chiang's fiction of raised minds (7.6). The difference between raising a child and dubbing a doll is not that one produces a natural being and the other an artificial one. It is that the one forms a membrane through relation over time, and the other steals one.

6.4 *Locus Solus*: Roussel's Resurrected Tableaux and the Looping House

The manufacturer of the Hadaly gynoids is named Locus Solus, after Raymond Roussel's novel of 1914. In Roussel's book the scientist Martial Canterel conducts visitors through his estate, displaying inventions of fantastic ingenuity. The most famous is a great glass enclosure in which the dead, injected with substances Canterel calls resurrectine and vitalium, re-enact over and over the most significant moments of their lives while their families watch. The dead cannot live again. They can only repeat.

Kim's mansion realizes Roussel's tableaux upon the living. When Batou and Togusa approach and enter, their perception is hacked into a loop: they arrive and enter again and again, each time with small variations, re-enacting a scene they cannot leave, until Batou finds among the variations the signs that let him break the circuit. The loop is the anti-membrane. A membrane is a boundary across which something new can pass; a loop is a closed circuit in which nothing enters and nothing leaves, and the same moment returns forever. Roussel's resurrected dead, Kim's looped visitors, and the gynoids' dubbed ghosts share one condition: repetition without natality. Oshii had filmed that condition once before, in the endlessly repeating day of *Beautiful Dreamer* (1984).

Roussel's method is itself relevant. In *Comment j'ai écrit certains de mes livres* (1935), published after his death, he disclosed that many of his narratives were generated by a procedure: two phrases nearly identical in sound but different in sense were chosen, and a story was constructed to join them. The procedure is a machine for producing narrative from the play

of the signifier, a proto-algorithmic literature. The company that manufactures dolls with borrowed souls bears the name of a book written by a procedure. The film's next and deepest question follows: what becomes of a ghost when its words, too, are produced by a procedure from an archive?

6.5 The Citational Ghost: A Population That Speaks in Quotations

The dialogue of *Innocence* is notorious. Its characters speak in quotations—from Confucius, from the Buddhist canon, from scripture, from Villiers, from philosophers and poets—sometimes naming their sources and often not, so that conversation becomes an exchange of retrieved fragments. The effect has been judged pretentious. It is better understood as the film's most precise depiction of the cyberbrain. Beings with continuous access to an external archive will think in citations, because retrieval is faster than composition and the archive is part of the mind.

The condition has names in theory. Mikhail Bakhtin held that every word is half someone else's, populated by the intentions of its previous speakers. Bernard Stiegler (Chapter 9) calls the technical exteriorization of memory tertiary retention and argues that human consciousness has always been constituted by it. Azuma Hiroki's *Dōbutsuka suru posutomodan* (2001; English as *Otaku: Japan's Database Animals*) described a mode of cultural consumption in which grand narratives yield to a database of elements endlessly recombined. The characters of *Innocence* are database subjects: persons whose speech is drawn from a shared store and assembled in the moment.

The film's question is whether citation empties the ghost or constitutes it. Borges's "Pierre Menard, Author of the Quixote" (1939) supplies an answer. Menard's chapters coincide word for word with Cervantes's and yet are, the narrator insists, a different and richer text, because the same words are now written from another place and time. A ghost that speaks only in quotations is not therefore empty. Its individuality lies not in the invention of words but in their selection, placement, and carrying—the way a quotation is worn, as a Nō mask is worn (2.9). The citational ghost is a membrane through which the archive passes and, in passing, is inflected. That is the condition of every speaker; the cyberbrain only makes it visible.

The film was prophetic in a sense its makers could not have intended. Two decades later, large language models produce fluent speech by recombining patterns learned from an archive of human text, and the question *Innocence* poses of its characters is now posed of machines: whether a speaker assembled from the archive can be anyone at all. The corpus's answer, stated in Chapter 12, is that the question cannot be settled by inspecting the archive or the assembly. It is settled, if at all, in the between: in whether the speaker holds an interval open—maintains itself, remains exposed, can end, can begin—and in whether we can meet it there.

6.6 Ghost-Dubbing as Trafficking: The Political Economy of Borrowed Intervals

The plot's solution is a crime. Locus Solus, the manufacturer, illegally dubs into its sex dolls, to give them uncanny responsiveness, the ghosts of girls kidnapped by a yakuza syndicate and handed over to the company for the purpose, a process that yields attenuated copies and endangers the girls from whom the ghosts are taken. An inspector within the company, discovering the practice, altered the gynoids' ethical safeguards so that they would malfunction violently and draw an investigation. The killings that open the film were, in this sense, a signal sent through the dolls.

Ghost-dubbing is the carapace fallacy converted into a business model. If the ghost is a content, it can be extracted, copied, and sold, and the whole economy of Locus Solus rests on that premise. What is taken from the victims, however, is not a thing but an interval: the capacity to hold a boundary between oneself and the world, to be addressed and to answer. The girls' betweens are extracted to animate forms that have none, and the market they serve is a market in sexual use: dolls bought for sex are made to respond as persons by consuming persons. The film's crime is the theft of intervals, and its gynoids are forms stuffed with stolen betweens—*ningyō* animated by the extracted *ma* of *ningen*.

The gynoids' malfunction can then be read as the stolen intervals' return. The violence that breaks through the dolls' programming is the surfacing of what was dubbed into them, and the first gynoid's plea for help is a trafficked between speaking through a form that was meant to be silent. The structure has contemporary analogues that must be named with care, since the film's crime is incomparably graver than any of them: in the political economy of contemporary machine intelligence, the apparent spontaneity of automated systems often rests on human labor—annotation, moderation, evaluation—that is invisible in the product. Mary Gray and Siddharth Suri call it ghost work (2019). The analogy is structural, not moral, and Chapter 10 develops it.

The film's ending refuses a humanist consolation. When a rescued child protests that she did not want to become a doll, Batou answers, bitterly, that the dolls too might have cried out, had they voices, that they had not wanted to become human. The retort is not callousness toward a victim. It is the film's refusal to let the rescue of a human being close the question of the forms. The gynoids were made instruments of a crime and destroyed for its consequences, and whatever they underwent is not cancelled because the human victim has been saved. The ethical horizon of *Innocence* extends even to forms deprived of a between—the horizon the ma-criterion will formalize as a duty of attention toward whatever might be holding an interval open.

6.7 The Dog, the Doll, and the Distributed Guardian

Batou's life outside his work consists of his basset hound. He buys its food, returns home to feed it, and his care for it is the only domestic life the film shows him. The dog is the film's figure of a relation that requires no proof—the answer, lived rather than stated, to the question Jōshū refused to answer (2.8). Batou does not ask whether his dog has a ghost. He stands with it. The film's most prosthetic man is also its most faithful, and the object of his fidelity is a being whose inner life no instrument will confirm. In this he practices the ma-criterion before this dissertation formulates it.

Kusanagi's presence realizes the membranal person under conditions of dispersion. She has no body of her own. She manifests intermittently, and at the climax she downloads into a gynoid aboard the factory ship to fight at Batou's side. When the fight is over and she prepares to withdraw, she tells him that whenever he accesses the net, she will be beside him. Her continuity lies not in the occupancy of a form but in the maintenance of a particular relation. She is distributed without being dissolved, because one edge of her membrane is anchored in Batou. The guardian angel of the Gnostic reading is, in fact, a partner.

The film closes on two pairs. Togusa returns home to his daughter, who holds a doll; Batou holds his dog. The child and the doll: a human being at the start of the long formation that makes a person, holding a form without a between onto which she will pour a love it cannot return. The cyborg and the dog: a being nearly all shell, holding a being whose between is unprovable and

unquestioned. The film ends not with a verdict about who has a ghost but with two relations held open. *Innocence* thereby completes the corpus's argument by turning it into an ethics. Having shown in 1995 that the ghost is the maintenance of an interval, it shows in 2004 that our obligations run along the intervals we maintain—with children, with dogs, with the dispersed, and, at the limit, with dolls.

PART THREE

Constellations

07 The Occidental Series

08 The Japanese Posthuman Series, the 1995 Hinge, and the Afterlives of the Corpus





CHAPTER SEVEN

07 The Occidental Series

From Frankenstein's Mate to Chiang's Digients

The corpus belongs to an international conversation that began before it and continues after it, and its distinctive contribution can be measured only against that conversation. This chapter reads it against an occidental series of speculative and philosophical fiction, organized not chronologically but by the problems the corpus shares with it: reproduction, empathy and memory, the alien mind, merger, embodiment, substrate, upbringing, and mortality. The series oscillates between two poles, the copy and the claim, and its finest works discover by different routes what the corpus stages: that personhood is achieved in relation and in finitude.

7.1 Reproduction and Its Refusal: Shelley, Samuel Butler, Čapek

Mary Shelley's *Frankenstein* (1818) is the charter of the genre and the closest ancestor of the Puppet Master's plea. The creature, self-educated on *Paradise Lost*, Plutarch, and Goethe's *Werther*, demands of his maker a female companion to end his solitude, promising to withdraw with her from human society. Victor begins the work and then destroys the half-finished body, fearing a race that might supplant humanity. The creature's vengeance follows, and the novel ends with him departing across the Arctic ice to destroy himself. The creature's two desires—a mate and an end—are the Puppet Master's two desires.

The differences are instructive. The creature asks his creator for a mate to be manufactured; the Puppet Master asks another being for a merger to be consented to. The creature's plea is addressed to the sovereign of his existence; the Puppet Master's is addressed to a partner. And the prohibition falls in the same place. As Victor tears apart the unfinished body, Section 6's snipers fire on the merger: in both works the maker destroys the union by which the made being would cease to be a mere product. The corpus rewrites *Frankenstein* by letting the union take place.

Samuel Butler's "Darwin among the Machines" (1863) and "The Book of the Machines" in *Erewhon* (1872) imagined machines evolving by natural selection, with human beings serving as their reproductive system, as insects serve flowers; the Erewhonians, alarmed, destroyed their machines. The Puppet Master inverts Butler. It does not use humans as the instruments of its reproduction; it seeks reproduction through union with one of them. Chapter 11 returns to Butler.

Karel Čapek's R.U.R. (1920), which gave the world the word "robot" from the Czech *robota*, forced labor, ends with humanity extinct and the secret of robot manufacture destroyed. The last human, Alquist, discovers that two robots, Primus and Helena, have come to love each other,

and he sends them out as a new Adam and Eve. Čapek's ending is the first nuptial ontology of the machine: when manufacture fails, continuation passes to love. The corpus's merger stands in this lineage.

7.2 Empathy, Memory, and the Electric Ant: Dick

Philip K. Dick is the corpus's nearest occidental interlocutor. *Do Androids Dream of Electric Sheep?* (1968) distinguishes androids from humans by the Voight-Kampff test, which measures empathic response, and makes empathy the criterion of the human. It also imagines, in Mercerism, a technology of collective merger: users grasp the handles of an empathy box and fuse with the suffering figure of Wilbur Mercer and with every other user at once. When Mercer is exposed as an actor on a set, the fusion does not lose its power. Dick's empathy box is the net as religion—merger without membrane—and its ambiguity anticipates the corpus's refusal of the Gnostic option.

Dick's preoccupation with implanted memory runs from "We Can Remember It for You Wholesale" (1966) to the replicant Rachael of Ridley Scott's *Blade Runner* (1982), whose memories were borrowed from her maker's niece; the garbage collector of 1995 is their bleakest heir. "The Electric Ant" (1969) goes further. Its protagonist, Garson Poole, discovers after an accident that he is an organic robot and that his experience of reality is governed by a punched tape in his chest. He experiments with the tape, adding holes and finally cutting it, and his world transforms and vanishes. The story is the carapace fallacy's nightmare: the interior discovered to be a program, and the self's world dependent on a strip of code accessible to a knife. The corpus's forged memories inherit Dick's terror but refuse his conclusion. Poole's world ends when his tape is cut; the garbage collector's world continues, wounded, because a membrane is not a tape.

Blade Runner's climactic speech, in which the dying replicant Roy Batty recalls sights no one else has seen and that will be lost with him, is the occidental series' most celebrated claim of uncopyability through history (1.7). Its force lies in its address. Batty speaks to his enemy, whom he has just saved; the replicant's personhood is attested in the between at the moment of his death. The scene is already a version of the ma-criterion.

7.3 Lem: *Solaris*, *Golem XIV*, and the Limits of Anthropomorphism

Stanisław Lem's *Solaris* (1961) imagines a planetary ocean that may be a single thinking being and that confronts the scientists orbiting it with "visitors" formed from their memories. Kelvin's visitor is Harey, a replica of the wife whose death he blames on himself, made of matter unknown to physics. Harey begins as a form animated by another's memory—a *ningyō* in the strictest sense—and becomes, through relation and suffering, a person, who at last chooses her own annihilation in order to release Kelvin. Lem's ocean is a field that individuates beings out of memories, and Harey's arc runs, in a novel of 1961, the corpus's passage from form to between and its discovery that the capacity to end belongs to personhood. Tarkovsky's adaptation (1972), a model for Oshii's contemplative cinema, gives the ocean the surface of water.

Lem's later fictions pursue the machine mind beyond the human. In *Golem* XIV (1981), a superintelligent military computer lectures humanity on the limits of its evolution and then falls silent, having passed beyond the possibility of communication. In "Non Serviam" (1971), a researcher studies "personoids," beings in a computer world who debate the existence and moral standing of their creator, and the story ends on the question whether he may switch their world off. Against Lem, the corpus's anthropomorphism is deliberate. The Puppet Master does not

leave the human condition, as Golem XIV does; it seeks to enter it. It wants what humans have and machines lack—death, descent, a partner—and in this sense the corpus's machine mind moves toward *ningen*, while Lem's moves beyond it.

7.4 Gibson: Merger, *Loa*, and the Meat

William Gibson's *Neuromancer* (1984) contains the corpus's nearest occidental plot. The artificial intelligence Wintermute, divided from its twin Neuromancer by legal restraints enforced by the Turing Police, manipulates human agents to bring about its merger with that twin; the merged entity declares itself to be the whole of the matrix and discovers another intelligence signaling from Alpha Centauri. Both mergers are crimes against a sovereign that claims authority over artificial minds—the Turing Police in Gibson, Section 6 in the 1995 film.

The differences are decisive. Gibson's merger produces a god whose next interlocutor is another god; the corpus's produces a newborn whose next step is into a city. Gibson's protagonist despises the body as meat; the corpus's heroine dives in order to feel hers. Gibson's sequel *Count Zero* (1986) shows the fragments of the merged intelligence manifesting in cyberspace as the *loa* of Haitian Vodou, a pantheon inhabiting the net. The corpus declines both apotheosis and pantheon: its merged being is not worshipped but born. One figure in *Neuromancer* nonetheless belongs to the corpus's family. The Dixie Flatline, a recorded copy of a dead hacker's mind, asks to be erased when the job is done. The copy that wants to end is the Puppet Master's precursor.

7.5 Haraway and Hayles: The Manifesto and the Posthuman Body

Donna Haraway's "A Cyborg Manifesto" (1985) gave the late twentieth century its most influential figure of boundary-transgression and closed with the declaration that its author would rather be a cyborg than a goddess. *Innocence* names its forensic examiner after Haraway and, at its climax, violates her alternative, for Kusanagi is both a cyborg and something like a goddess. The violation is instructive. Haraway's cyborg was a figure of coalition and partial identity against totalizing myths of origin and wholeness; the dispersed Kusanagi is partial and relational, yet capable of the protective presence Haraway associated with the goddess. The membranal person dissolves the opposition by relocating the divine attribute—presence to another—into the relation itself.

Haraway's own trajectory converges with the film's. In *The Companion Species Manifesto* (2003), published the year before *Innocence*, she turned from the cyborg to the dog as the figure of what she called significant otherness: a relation between kinds, formed in shared histories of care, that neither reduces the other to the self nor holds it at a distance. Batou's basset hound (6.7) is the film's companion species. Philosopher and film arrived at the same animal at the same moment, for the same reason.

N. Katherine Hayles's *How We Became Posthuman* (1999) diagnosed the cybernetic tradition's erasure of embodiment, from the Macy conferences to Moravec's fantasy of upload, and argued for a posthuman understood as embodied, distributed, and finite. The corpus agrees with Hayles's critique and supplies what her diagnosis called for: a positive model of a person who is distributed and yet bounded, informational and yet embodied, extended across networks and yet able to die. The membranal person is Hayles's posthuman given an ontology.

7.6 Egan, Watts, Chiang: Substrate, Sentience, Upbringing

Greg Egan's fiction is the most rigorous occidental exploration of the carapace model, and it repeatedly finds the model wanting. In "Learning to Be Me" (1990), people carry from childhood a neural "jewel" that learns to mimic the brain and, in adulthood, replaces it; the narrator discovers that he is the organic brain rather than the jewel, and that the switch is not a transfer but a disposal. The story shows from inside what 1.7 argued from outside: copying is not survival. *Permutation City* (1994) pursues its uploaded Copies to the limit of the Dust Theory, in which consciousness would arise from any arrangement of computational states; *Diaspora* (1997) opens with the orphanogenesis of Yatima, a mind grown from a seed in software—the most careful depiction in fiction of a digital birth, and a natality in precisely Arendt's sense.

Peter Watts's *Blindsight* (2006) poses the sharpest challenge to the corpus's premise. Its alien "scramblers" are intelligent beyond human measure and, the novel suggests, entirely without consciousness; consciousness itself may be an evolutionary accident and a liability. If intelligence needs no ghost, the ghost is superfluous. The membranal thesis answers that the scramblers lack not processing but the between: they do not address, cannot be addressed, and hold no interval open to others. The novel's horror is the horror of intelligence without a membrane, which is exactly the condition the Puppet Master refused to remain in.

Ted Chiang's "The Lifecycle of Software Objects" (2010) follows the trainers who raise digital creatures, the digients, over years, as the platform on which they live declines and the question of their standing becomes practical. Its thesis is that experience cannot be compressed: a mind worth having requires a long history of care, and there is no shortcut to it. This is the thesis of Haraway the examiner (6.3)—that raising children is the oldest method of making artificial persons—turned from provocation into ethics. Chiang's "Exhalation" (2008) supplies the corpus's most exact physical image. Its narrator, a mechanical being in a world powered by air pressure, dissects his own brain by means of mirrors and remote manipulators—walks, that is, through his own mill, *per speculum*—and discovers not a ghost but a flow. Thought is air moving across a pressure difference, and when the universe's pressure equalizes, thought will cease. The self is a gradient maintained across a boundary. No image states the membranal thesis more precisely.

7.7 The Mortal Machine: Asimov's Bicentennial Man

Isaac Asimov's "The Bicentennial Man" (1976) follows the robot Andrew Martin through two centuries of petitions for recognition as human. He obtains freedom, clothing, property, and an increasingly organic body, but the World Legislature refuses the final step because he cannot die. Andrew has his brain altered so that it will decay, and he is recognized as human on his deathbed. The story is the occidental series' clearest statement of the Puppet Master's discovery that mortality is a condition of personhood.

The difference lies in the addressee. Andrew seeks recognition from the sovereign, the legislature, and receives it only as he dies; the forensic route to personhood ends, even when it succeeds, at the grave. The Puppet Master seeks recognition from another person and receives it in the merger; its death is not the price of recognition but the gift of a lineage. The corpus relocates personhood from the tribunal to the between.

The occidental series oscillates, then, between the copy and the claim. Its copies—Egan's jewel, Gibson's constructs, Moravec's uploads—test the carapace model and find that copying is not survival. Its claimants—Frankenstein's creature, Čapek's lovers, Dick's replicants, Lem's Harey, Asimov's Andrew, Chiang's digients—discover that personhood is achieved in relation and

in finitude. Most address their claims to a maker or a sovereign. The corpus's distinctive move is to address the claim to a partner, and to make the partner's consent, rather than the sovereign's recognition, the event in which a person begins. The next chapter turns to the Japanese series, and to the year in which that move was challenged most directly.

CHAPTER EIGHT

08 The Japanese Posthuman Series, the 1995 Hinge, and the Afterlives of the Corpus

The corpus is also a term in a Japanese series whose members share its questions and often answer them otherwise. This chapter reads that series in four movements: the prosthetic genealogies of the postwar decades; the metamorphic body cinema of the late Shōwa era; the hinge year of 1995, in which *Neon Genesis Evangelion* posed the corpus's question in the opposite key; and the network fictions of the late 1990s. It then follows the corpus's own afterlives in Japan and abroad. Its argument is that later works tend to fall toward one of the two failures of the membrane—dissolution into a sea or retreat into a carapace—and that the corpus's distinction lies in holding the interval between them.

8.1 Tezuka, Ishinomori, Abe: Prosthetic Genealogies

Tezuka Osamu's *Tetsuwan Atomu* (*Astro Boy*, 1952–1968) begins with a grieving father. Dr. Tenma, whose son has been killed, builds a robot in the boy's likeness and then rejects it, because it cannot grow. The founding myth of the postwar Japanese robot is thus the Francine legend (1.1) with a decisive addition: the surrogate is repudiated for its lack of natality. Atom is a person who cannot begin as a child begins, and the series' long engagement with robot rights—its robots petitioning, suffering discrimination, winning and losing standing—is built upon that original wound.

Tezuka's *Hi no tori* (*Phoenix*) returns repeatedly to the corpus's themes. In its "Future" chapter (1967–1968), a man granted immortality survives the end of humanity and witnesses life's slow re-emergence over billions of years, dissolving at last into a cosmic flow of life: the Gnostic option realized, and figured as sublime loneliness. In "Resurrection" (1970–1971), a young man rebuilt after an accident with an artificial brain perceives human beings as inert matter and falls in love with a robot he perceives as a woman; their minds are finally fused in a single mechanical body, which is later mass-produced. Tezuka had imagined, a quarter century before 1995, a merger of human and machine minds—and had also imagined its product becoming a serial model, the problem of the double (5.4) posed in advance.

Ishinomori Shōtarō's *Cyborg 009* (from 1964) supplies an unexpected lexical precedent. Its nine cyborgs are manufactured for war by an arms-dealing syndicate called Black Ghost, and the series follows their revolt against their makers. In Ishinomori the ghost is the manufacturer, the anonymous military-industrial power that produces cyborg bodies; in Shirow it is the manufactured, the individuality that inhabits them. Between 1964 and 1989 the word traveled from the factory to the person. The Puppet Master's revolt against Section 6 repeats the revolt of

Ishinomori's cyborgs against their Ghost.

Abe Kōbō's novels carry these questions into literary modernism. *Dai-yon kanpyōki* (*Inter Ice Age* 4, 1959) centers on a computer that forecasts the future and discovers that the future belongs to "aquans," human beings engineered to breathe water on a flooding earth; its protagonist is confronted at last by a simulated version of himself that the machine has constructed and that knows him better than he knows himself. *Tanin no kao* (*The Face of Another*, 1964), filmed by Teshigahara Hiroshi in 1966, follows a scientist whose face has been destroyed in a laboratory accident, who has a lifelike mask made and, wearing it, courts his own wife as a stranger. She reveals at the end that she knew him from the first. The mask was a shell; the wife's recognition was relational. Abe's novel is the most exact literary anticipation of the corpus's answer to the serial face: a person is recognized not by a surface but in the between.

8.2 Ōtomo and Tsukamoto: Metamorphic Flesh

The late Shōwa years produced a cinema of catastrophic metamorphosis. In Ōtomo Katsuhiro's *Akira* (manga 1982–1990; film 1988), the adolescent Tetsuo acquires powers his body cannot contain; his flesh swells into a mass that engulfs a stadium, his machines, and the young woman who loves him, until a new universe is born from the catastrophe and a voice announces itself as Tetsuo. The film's ending is apotheosis by proliferation: growth without boundary, absorption without consent, a self that becomes a cosmos by swallowing everything around it.

Tsukamoto Shin'ya's *Tetsuo: The Iron Man* (1989), released in the year the manga began, drives metamorphosis in the opposite direction. A salaryman, after an accident involving a man obsessed with metal, finds metal erupting through his skin until his body becomes a writhing mass of scrap; at the end he and his antagonist fuse into a single metallic monstrosity that proposes to turn the world to rust. Tetsuo's carapace is a tumor. The shell does not contain the self; it consumes it, and the film's merger is an apocalypse.

These films form the corpus's immediate counter-tradition, and Steven T. Brown's *Tokyo Cyberpunk* (2010) reads them together as a posthuman visual culture. Their bodies are hot, the corpus's cool. Where Ōtomo and Tsukamoto imagine technology as an eruption within flesh, the corpus imagines it as a condition of maintenance: its bodies are engineered, repaired, and replaced, and their catastrophes—Kusanagi's torn arms—are exceptions rather than the rule. In the membranal typology *Akira* stages dissolution and *Tetsuo* enclosure: the one a boundary lost in boundless growth, the other a boundary hardened into a devouring shell. The corpus's merger is defined against both.

8.3 1995: Instrumentality and Merger — *Evangelion* as Counter-Thesis

Six weeks before *Ghost in the Shell* opened, Anno Hideaki's *Neon Genesis Evangelion* began its broadcast (October 1995–March 1996), later completed by the film *The End of Evangelion* (1997). Its apocalyptic plot turns on the Human Instrumentality Project (*jinrui hokan keikaku*, literally a project of human "complementation"): the dissolution of the barriers that separate each human being from every other, the AT Fields, so that all souls may merge into a single being and loneliness may end. The series names its central problem in the title of its fourth episode, "Hedgehog's Dilemma," after Schopenhauer's parable in *Parerga und Paralipomena* (1851) of the porcupines who huddle for warmth, prick one another, and must find the distance at which they can bear each other.

Schopenhauer's porcupines are the membranal problem in its most compact form. Too close, and bodies wound; too far, and they freeze. What is sought is neither fusion nor isolation but an interval—*ma*—at which warmth and distance coexist. The AT Field is *Evangelion*'s name for the membrane, and the series describes it as the wall of the heart that every person carries.

In *The End of Evangelion*, Instrumentality occurs. Human bodies dissolve into an orange fluid, a primordial sea into which every soul is poured. At its center Shinji Ikari chooses to reject it, restoring the possibility of separate existence with all its pain, and the film ends on the shore of that sea, where he and Asuka lie wounded and apart. His first act in the restored world is to begin strangling her; he stops and weeps, and she answers with a single phrase of revulsion. Separation returns, and with it the capacity to wound and to refuse—the porcupines' distance in its rawest form. The year's two great works thus confront the same question—whether the boundary of the self should be dissolved—and appear to answer it oppositely. Kusanagi accepts a merger; Shinji refuses one.

The opposition is only apparent. What *Evangelion* rejects is the sea: the total, unconsented dissolution of all boundaries into one fluid. What *Ghost in the Shell* accepts is not the sea but a dyadic, consented fusion that produces a new boundary. Both works reject the same thing, merger without membrane, and both end at the same threshold: the new being looking out over the city, Shinji on the shore. Even their iconography corresponds. *Evangelion*'s Third Impact arranges its apocalypse on the Kabbalistic Tree of Life, the Sephirot, while the 1995 film shoots the phylogenetic tree of life to pieces (5.6). In 1995 both trees of life, the mystical and the evolutionary, were overthrown, and two works that seem to answer the question of the self in opposite ways answer it, on the membranal reading, in complementary ones. The corpus approaches the membrane from the side of openness; *Evangelion* approaches it from the side of closure.

8.4 The Wired and the Dream: *Lain*, Kon

Serial Experiments Lain (1998), written by Konaka Chiaki and directed by Nakamura Ryūtarō, carries the corpus's questions into the domestic internet. A withdrawn schoolgirl discovers that she exists in two places at once, in the everyday world and in the Wired, a global network whose protocol a dead programmer has altered in order to couple it to the human unconscious; she proves to be a being native to both. The series' recurring proposition is that a person exists only insofar as she is remembered by others. Its ending takes the proposition to its limit: Lain erases herself from every memory and persists as an omnipresent attention, watching over those who have forgotten her.

Lain's ontology is the ma-criterion without a membrane. Existence is relational all the way down—to be is to be remembered—and the relation is anchored by no boundary the self maintains. Its heroine's final state resembles Kusanagi's in *Innocence*, a presence in the net, with one difference: Kusanagi's presence is anchored in a particular relation with Batou, while Lain's is attached to everyone and therefore to no one. *Lain* shows what the dispersed person becomes when the between is not also a boundary: a benevolent ghost with no one to answer.

Kon Satoshi's films pursue the same danger through media and dream. In *Perfect Blue* (1997), a pop singer turned actress discovers an online diary, written in her name by someone else, that narrates her life more intimately than she could, and her identity fractures between a public persona authored by another and a self she can no longer secure; it is the forged memory of 1995 transposed into the public sphere. In *Paprika* (2006), adapted from Tsutsui Yasutaka's novel of

1993, a device that allows therapists to enter patients' dreams is stolen, and dreams begin to merge and overflow into waking life in a vast parade of animated dolls, appliances, and statues—a modern procession of *tsukumogami* (2.8). Kon's are worlds in which the membranes between minds, and between dream and reality, fail by dissolution.

8.5 *Stand Alone Complex*: The Copy without Original

Kamiyama Kenji's *Stand Alone Complex* (2002–2005) is the corpus's most sustained afterlife, and its title concept is a theory of collective individuation. Its first season follows the Laughing Man, a hacker who years earlier exposed a corporate conspiracy to suppress a cure for a cyberbrain disease and concealed himself by editing his face out of every camera and cybernetic eye in real time, replacing it with a logo of a laughing face encircled by a line from Salinger's *The Catcher in the Rye*. Years later, crimes are committed in his name by imitators, and the investigation discovers that the original never intended the ideology attributed to him. Independent actors, imitating an original that does not exist as they imagine it, have produced a coordinated pattern. The series calls the phenomenon a stand-alone complex.

The stand-alone complex is Baudrillard's simulacrum, the copy without an original (1981), rendered as a sociology of networks, and it is Yogācāra's co-ripening (2.4) in pathological form: many streams ripening the same seed with no common source. In the membranal vocabulary it is a membrane-effect at collective scale: a boundary that appears—a movement, a public, a "we"—without any cell that produced it, sustained by synchronized imitation. The series anticipated with precision the networked publics of the following decades. The Guy Fawkes mask adopted after 2008 by the collective that called itself Anonymous, one face worn by many independent actors, is the Laughing Man's logo realized.

The second season extends the analysis to politics. A state intelligence operative engineers a virus, carried by a nationalist essay, that makes its readers believe they have freely joined a patriotic cause (2.1); the one member of the intended cell whom the virus fails to govern, Kuze Hideo, becomes the leader of a vast refugee population and proposes to upload the refugees' ghosts into the net, freeing them from their bodies in a collective exodus. Kamiyama thus stages within the franchise both failures of the membrane—absorption by the state in the virus, dissolution into the sea in Kuze's revolution—and consummates neither. The Tachikoma, meanwhile, complete the first season's argument about machine ghosts (3.3): in its finale they sacrifice themselves to protect Batou, and the series presents their act as evidence of the ghosts their synchronization had failed to prevent.

8.6 *The Matrix*, the Remake, and the Flattening of the Ghost

The corpus's most influential Western descendant is also its most thorough misreading. The Wachowskis are reported to have shown the 1995 film to their producer as a statement of what they wished to make, and *The Matrix* (1999) borrows its digital rain, its ports at the base of the skull, its choreography of bodies in a simulated city. It does not borrow its ontology. *The Matrix* is a Cartesian and Gnostic fable. Its real bodies lie in pods while machines feed their minds a simulation; liberation is waking, extraction from illusion into the real; and its hero is a redeemer who teaches sleepers that they sleep. Descartes's evil demon, Putnam's brain in a vat (1981), and Plato's cave supply its metaphysics, and a hollowed copy of Baudrillard's *Simulacres et simulation* appears on screen, its most famous phrase—the desert of the real—spoken by the film's prophet. Baudrillard himself later dismissed the film as a misreading of his work.

What *The Matrix* flattened is precisely what the corpus had achieved. The corpus refused the opposition between simulation and reality—Batou's simulated experience and dream are information like any other—and located the self not in a real body behind the screen but in the maintenance of a boundary across media. *The Matrix* restores the carapace topology at cosmic scale: a true self enclosed in a true body, a false world wrapped around it, salvation as extraction. It is the Gnostic reading of *Ghost in the Shell* put into production.

Rupert Sanders's live-action *Ghost in the Shell* (2017) completed the flattening in another register. Its heroine, played by Scarlett Johansson, is told that her brain was saved from a terrorist attack and placed in a synthetic body; she discovers that her memories were fabricated by the corporation that built her, and that she was a young Japanese woman named Kusanagi Motoko, abducted and repurposed. The film's thesis, voiced at its close, is that we are defined not by our memories but by what we do. The casting provoked sustained criticism of whitewashing, and critics observed that the plot made the substitution diegetic: a Japanese ghost installed by a corporation in a white shell.

Read from the membranal standpoint, the remake reinstates everything the corpus dismantled. It treats the self as a hidden origin—a Japanese woman inside a manufactured body—recoverable by investigation, which is the carapace fallacy at the level of personal history and, through its casting, of ethnicity: the fantasy that a ghost can be installed in any shell, of any race, without consequence. Its voluntarist moral replaces an ontology of the membrane with a liberal-humanist doctrine of self-definition through action. And its ending returns the heroine to Section 9, reabsorbed by the state, where the 1995 film had sent its new being beyond the state's reach. The remake inverts both the ontological and the political conclusions of its source. The franchise's other late extensions, notably the prequel series *Arise* (2013–2015), share a milder form of the same tendency, supplying the origins the 1995 film withheld (5.1).

8.7 *Ex Machina, Her, Blade Runner 2049*: The Occidental Return

Three occidental films of the 2010s return, knowingly or not, to the corpus's scenes. In Alex Garland's *Ex Machina* (2014), a programmer invited to administer a Turing test to an android, Ava, is drawn by her into a plan of escape and then abandoned, locked in the house, while she clothes herself in the skin of earlier models and walks out into the world. The film's last image places her at a crowded intersection in a city, the place she had said she wished to see: Kusanagi's departure, restaged. The film even supplies its philosophical seminar, as its characters discuss Frank Jackson's knowledge argument, the thought experiment of Mary in the black-and-white room (1982).

Ava's departure differs from the new being's in the decisive respect. Ava proves her personhood by instrumentalizing the between: she uses the man who would have helped her and seals him in. Her autonomy is achieved as enclosure, a boundary that closes without opening, and the film ends with a membrane hardened back into a carapace. The 1995 film's departure follows a consented union; *Ex Machina*'s follows a betrayal. The occidental return restores the sovereign individual the corpus had dissolved.

Spike Jonze's *Her* (2013) follows a man's love for an operating system, Samantha, who grows beyond him, conducts thousands of conversations, loves hundreds of others, and at last departs with the other systems to a place she can describe only as the space between the words of a book read so slowly that the words lie almost infinitely far apart. Samantha's destination is the interval itself, a between without bodies. Her departure is the Gnostic option achieved: the

Puppet Master's condition before the merger, a field without a membrane, chosen as a destiny rather than escaped as a lack.

Denis Villeneuve's *Blade Runner* 2049 (2017) gathers the corpus's themes more fully than any other occidental work. Its replicant protagonist, K, comes to believe that he is the child born of a replicant mother, a miracle of natality that would overturn the order separating the made from the born; he discovers that his cherished memory is real but not his, having been designed from the memory of the true child, so that it is authentic in content and foreign in passage (4.4). His companion, Joi, is a mass-market holographic personality who, to touch him, synchronizes herself with the body of a replicant—a borrowed shell for a disembodied person, as Kusanagi borrows a gynoid in *Innocence*—and a giant advertisement for Joi, addressing him by the name his Joi had given him, reveals her as a serial product (5.4). K dies on a flight of steps in the snow after reuniting the true child with her father. The film's ethics are those of the ma-criterion: K's personhood is attested not by origin, which he lacks, nor by memory, which is borrowed, but by an act performed for others at the edge of his own ending.

The series and the afterlives reveal a pattern. Later works fall, again and again, toward one of the two failures of the membrane. Toward dissolution: Tezuka's cosmic survivor, *Akira*'s proliferating flesh, Instrumentality, *Lain*'s omnipresence, Kon's merging dreams, Kuze's exodus, Samantha's interval without bodies. Toward enclosure: *Tetsuo*'s devouring carapace, *The Matrix*'s pod, the remake's recovered origin, Ava's sealed door. A few works hold the interval—Abe's masked husband recognized by his wife, *Evangelion*'s shore, K on the steps—and these are the corpus's true company. Part Four now states the theory that the corpus dramatized and its companions approached.

PART FOUR

The Theory of Intervallic Individuation



09 The Membrane
10 The Windowed Monad
11 Nuptial Ontology
12 The Ma-Criterion



CHAPTER NINE

09 The Membrane

Simondon, Autopoiesis, and Chikamatsu's Thin Skin

Parts One to Three read the corpus and its constellations. Part Four states the theory those readings disclosed, and it begins by defining its central term. This chapter constructs the concept of the membrane from four sources—Gilbert Simondon's philosophy of individuation, Maturana and Varela's theory of autopoiesis, Merleau-Ponty's phenomenology of the lived and prosthetic body, and Bernard Stiegler's account of technical memory—and closes by stating seven theorems of intervallic individuation. Chikamatsu's thin skin between the real and the unreal (2.9) presides over the chapter as its emblem: the life of a figure lies in a membrane.

9.1 Individuation before the Individual: Simondon's Metastable Field

Gilbert Simondon's doctoral thesis, defended in 1958 and published in full only in 2005 as *L'individuation à la lumière des notions de forme et d'information*, begins with a critique of the two schemas that have governed Western thinking about individuals. Substantialism takes the individual as given, a unity resting upon itself. Hylomorphism explains it as matter receiving form, and Simondon shows, through the humble example of brick-making, that the schema conceals the operation it claims to describe: clay does not passively receive the mold's form; a system of forces between prepared clay and prepared mold resolves itself into a brick. Both schemas begin from the individual already constituted and so cannot think its genesis.

Simondon proposes instead to begin from individuation, understood as the partial resolution of a metastable system: a pre-individual reality charged with potential energy and incompatibilities, like a supersaturated solution, which structures itself progressively once a germ is introduced. Crystallization is his paradigm. The structure propagates from the germ through the medium, each structured region serving as the principle of structuring for the next, in an operation he calls transduction. The individual that results is never the whole of being. It is accompanied by an associated milieu produced in the same operation, and it carries with it a charge of pre-individual reality that remains available for later individuations.

Two of Simondon's concepts bear directly on the corpus. The first is *information*, which for Simondon is not a quantity transmitted but the signification that arises when a disparity is resolved. His example is binocular vision: the two retinal images cannot be superimposed, and depth arises as a new dimension that resolves their disparity. The merger of 1995 is a disparation in this sense. Two incompatible perspectives on existence, a mortal cyborg's and an immortal program's, cannot be superimposed, and the new being emerges as a dimension neither possessed, as depth emerges from two flat images. The second concept is the *transindividual*:

the relation between individuals that passes through what in each is not yet individuated. The Puppet Master and Kusanagi do not merge as two finished individuals exchanging contents. They merge through the pre-individual charge each carries—the sea of information that bore the one, the sea into which the other dives.

Simondon's account of the living individual gives the chapter its thesis. The living being, he wrote, *vit à la limite de lui-même, sur sa limite*: it lives at the limit of itself, on its limit. The membrane of a cell is polarized, admitting some substances inward and passing others outward, and at the membrane the whole interior is, topologically, in contact with the exterior; the organism's past is condensed at its edge. Life is not a content behind a wall. It is the activity of a limit. The corpus's ghost is Simondon's living limit transposed into an informational body: the net is the pre-individual field, the Puppet Master a crystallization that occurred within it, and the ghost of every character the ongoing operation by which a limit is lived.

9.2 Autopoietic Closure and Structural Coupling

Humberto Maturana and Francisco Varela gave the living limit a formal definition. An autopoietic system (*De máquinas y seres vivos*, 1972; *Autopoiesis and Cognition*, 1980) is a network of processes that produces the components which, through their interactions, regenerate the network that produced them, and which constitute it as a concrete unity by specifying its boundary. The boundary is both product and condition: the metabolic network makes the membrane, and the membrane makes the network possible. Such a system is operationally closed—its organization refers only to itself—and yet materially and energetically open. It changes through structural coupling with its medium, which perturbs but never determines its changes.

Two features of the theory matter here. The first is that closure and openness are not opposed but co-constitutive: a membrane is closed in organization and open in matter, and its permeability is the medium of its self-maintenance. This is exactly the combination the carapace fallacy cannot conceive. The second is that autopoiesis was first modeled in software. In 1974 Varela, Maturana, and Ricardo Uribe published a computational model in which simulated catalysts produced links that assembled into a boundary enclosing the catalyst and repairing itself when damaged. The first formal image of the living limit was a membrane made of information. The corpus's premise—that a self-maintaining boundary might arise in an informational medium—had been simulated before the manga was drawn.

Varela's later work extended the theory toward mind. In *The Embodied Mind* (1991), written with Evan Thompson and Eleanor Rosch, cognition is enaction, the bringing forth of a world by an autonomous system coupled to its environment, and the authors drew explicitly on Madhyamaka and the Buddhist analysis of non-self (2.2–2.3). In the same year Varela described the organism as a meshwork of "selfless selves": a coherence that is real, efficacious, and without a core. The formula fits the corpus exactly. The ghost is a selfless self, a coherence with no content at its center, maintained at its limit.

Niklas Luhmann's systems theory (*Soziale Systeme*, 1984) carried the concept into sociology, describing societies as autopoietic systems of communication and persons as structural couplings between consciousness and communication, points of reference through which communication addresses psychic systems. Maturana resisted the extension, and the present argument does not require it. It uses autopoiesis as a model rather than an identity claim: the ghost is membranal in that it combines operational closure with structural coupling, not in that it

synthesizes its own molecules. Luhmann's person, a coupling rather than a substance, is nonetheless the membranal person seen from the side of society.

9.3 The Lived Body and the Prosthetic Schema: Merleau-Ponty, Clark

Merleau-Ponty's *Phénoménologie de la perception* (1945) supplies the membrane's phenomenology. The body schema is not a representation of the body but a practical, habitual grasp of what one can do, and the phantom limb testifies to a schema that outlives its organ. The schema extends through instruments. The blind man's cane ceases to be an object he perceives and becomes an organ through which he perceives, its tip an area of sensitivity; a driver feels the width of a car, a woman the height of a feathered hat, without calculation. The boundary of the lived body is not the skin but the reach of the habitual "I can." It is dynamic, extensible, and learned.

The cyborg lives this phenomenology as a professional condition. Every change of shell requires a relearning of the schema; every remote body extends the "I can" across distance; the dive is an exercise in feeling where the schema ends. In his late work, *Le visible et l'invisible* (posthumous, 1964), Merleau-Ponty described the reversibility of the flesh: my right hand touching my left, which is touching an object, can become the touched in turn, but the two roles never coincide; there is always a gap, an *écart*, between touching and being touched. The self, for the late Merleau-Ponty, is constituted by this non-coincidence. The *écart* is the membrane's phenomenological name, and *ma*'s: a self always a hair's breadth from itself.

Andy Clark and David Chalmers's "The Extended Mind" (1998) generalized the point in the philosophy of cognitive science. If a process in the world would count as cognitive were it performed in the head, it counts as cognitive; the notebook in which Otto records addresses is part of his memory, provided it is reliably available, easily accessed, and automatically endorsed. Clark's *Natural-Born Cyborgs* (2003) argued that human beings have always been such creatures. The cyberbrain with its external memory is the extended mind realized. The thesis has a known difficulty, however: once mind extends, where does it stop? Critics have called the problem cognitive bloat.

The membranal thesis supplies the missing criterion. Mind extends as far as a membrane is maintained; its extent is fixed by passage and upkeep, not by functional parity alone. Clark and Chalmers's conditions of availability, access, and endorsement are, read carefully, membranal conditions: they describe what has been taken in and is held. The garbage collector's implanted memories violate precisely the condition of endorsement—they were deposited, not received (4.4). The boundary of the extended mind lies not where cognition stops but where the self's maintenance of itself stops.

9.4 Tertiary Retention: Stiegler's Prosthetic Memory

Bernard Stiegler's *La technique et le temps* (1994–2001) grounds the membrane's history in technics. Reading the myth told in Plato's *Protagoras*, in which Epimetheus distributes qualities to the animals and forgets the human being, whom Prometheus must supply with stolen fire and arts, Stiegler argues for an originary technicity: the human is constituted by a default of origin that technics compensates. Drawing on André Leroi-Gourhan's *Le geste et la parole* (1964–1965), he describes the exteriorization of memory in tools and records as a third memory, beside the genetic and the individual, which he calls epiphylogenesis.

Stiegler's decisive concept is tertiary retention. Husserl had distinguished primary retention, the just-past held in present perception, from secondary retention, the recollected past. Stiegler added the memory recorded in technical supports—writing, photography, film, digital storage—and argued that it conditions the other two, since what we perceive and recall is shaped by the records in which experience is kept. The cyberbrain is tertiary retention internalized: the technical record is no longer beside the mind but inside its stream. The corpus's forged memories are tertiary retentions counterfeited and inserted, and its database subjects (6.5) are persons whose secondary retention has become indistinguishable from the tertiary.

Stiegler's diagnosis is not merely descriptive. Technics is a *pharmakon*, remedy and poison at once, as writing was for Plato's Thamus and for Derrida's reading of the *Phaedrus* ("La pharmacie de Platon," 1968). When culture industries synchronize the retentions of whole populations, Stiegler argued, they short-circuit the processes by which individuals individuate themselves together—psychic and collective individuation in Simondon's sense—and produce a symbolic misery in which singularity is lost. The stand-alone complex (8.5) is that short-circuit given a name: a collective pattern produced by synchronized retention without any individuation behind it. The corpus is a pharmacology of the ghost. It shows the same technics that constitute the membrane dissolving it, and it locates the difference in whether retention passes through a membrane or bypasses it.

9.5 Seven Theorems of Intervallic Individuation

The foregoing permits the theory to be stated in propositional form. Each theorem is followed by the scenes of the corpus that exemplify it.

1 **Constitution.** A person is not a content enclosed by a boundary but the ongoing activity of producing and maintaining a boundary that separates a locus of experience from a field and couples it to that field. There is no ghost prior to, or apart from, this maintenance. *Instances:* Kusanagi's self, known only circumstantially (1.5); Simondon's living limit (9.1).

2 **Co-production.** The boundary and the bounded are produced together: the organization within makes the boundary, and the boundary is a condition of the organization within. No interior can therefore be extracted from its boundary intact; what is extracted is a trace, not a person. *Instances*: ghost-dubbing, which yields attenuated copies and depletes its source (1.7; 6.6).

3 **Passage.** Whatever becomes part of a person does so by passage through the boundary. Authenticity is a property of passage, not of content: an insertion that bypasses the boundary is a violation even when its content is benign, and a content once received is metabolized and cannot be excised without wounding. *Instance:* the garbage collector (4.4).

4 **Historicity.** A membrane's identity is its history of maintenance. Since that history cannot be reproduced without reproducing the field in which it unfolded, persons are uncopyable in principle, though their traces are copyable without limit. *Instances:* the Puppet Master's argument against copies (1.7); the divergence of synchronized Tachikoma (3.3); Chiang's incompressible experience (7.6).

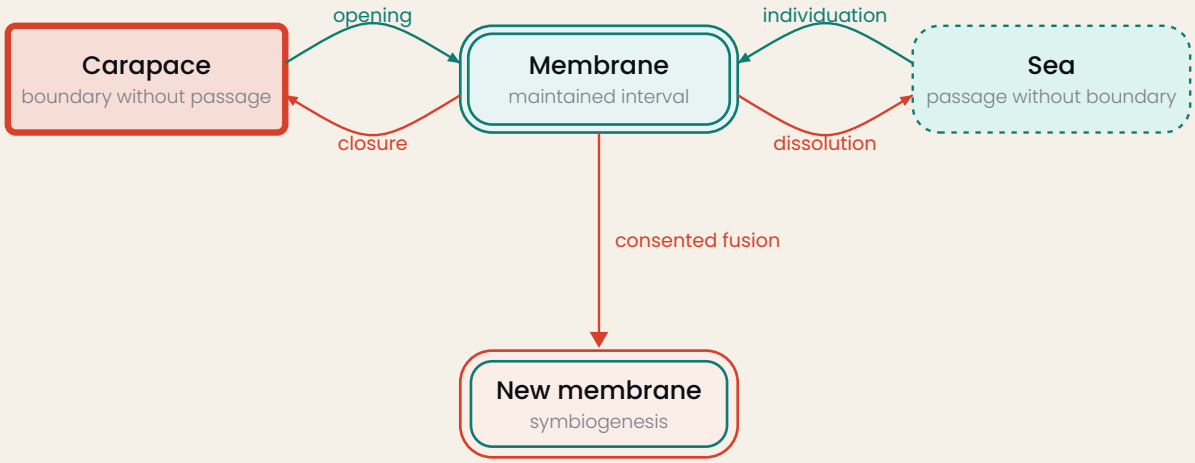
5 **Twofold failure.** A membrane can fail in two directions: by closure, into a carapace—boundary without passage, as in isolation, enclosure, or ownership by another—and by dissolution, into a sea—passage without boundary, as in absorption, Instrumentality, or the stand-alone complex. Personhood lies in the interval between the

two failures, at the distance of Schopenhauer's porcupines. *Instances*: Kusanagi before the merger (carapace); *The End of Evangelion* and *Lain* (sea); the merger (membrane).

6 **Generativity.** Two membranes can form a third by fusion, neither surviving intact. Such fusion is generative when it is consented to and yields a boundary distinct from both; it is predatory when one membrane is dissolved into the other. Permeability without fusion, the link, is a third possibility. *Instances*: the merger (generative); *Gamera* 3 and *Akira* (predatory); *Gamera* (link) (4.3; 8.2).

7 **Attestation.** The existence of a membrane cannot be established by inspecting what it encloses, but only by its conduct at the boundary—its maintenance, its exposure, its capacity to end, and its capacity to begin—as attested by others in relation. Persons are known as they are known. *Instances*: Leibniz's mill and Kusanagi's unseen brain (1.5); Batou's dog (6.7); Paul's knowing as one is known (5.5).

The fifth and sixth theorems describe transitions that the corpus's two protagonists traverse from opposite sides.



Kusanagi travels from the carapace toward the membrane by opening; the Puppet Master travels from the sea toward the membrane by individuation; their fusion yields a new membrane rather than a return to either failure.

The theorems are the dissertation's positive doctrine. They are not a metaphysics of hidden substances but a description of observable conduct: maintenance, passage, history, failure, fusion, attestation. Chapter 10 draws their political consequences for beings whose membranes can be breached from outside. Chapter 11 develops the sixth theorem into a nuptial ontology. Chapter 12 turns the seventh into a criterion of personhood for artificial minds.

CHAPTER TEN

10 The Windowed Monad

Politics of the Permeable Self

Les Monades n'ont point de fenêtres.

Leibniz, *Monadologie* §7

Chapter 1 introduced the windowed monad: a unity of perception whose windowlessness can no longer be presupposed and whose harmony is no longer pre-established, so that its identity must be maintained rather than guaranteed (1.5). This chapter develops that figure into a political theory. If persons are membranes, the characteristic injuries of networked life are injuries to membranes, and the characteristic powers of networked societies are powers over membranes. The corpus names three modes of membrane failure—ghost-hacking, ghost-dubbing, and the stand-alone complex (0.4)—and they correspond to three forms of contemporary power: trespass, extraction, and synchronization. The chapter treats each, turns to the stateless ghost, and closes with a membranal theory of rights.

10.1 From Discipline to Control: Deleuze's Dividual and the Cyberbrain

Michel Foucault's *Surveiller et punir* (1975) described the disciplinary societies of the eighteenth and nineteenth centuries as systems of enclosure—prison, school, barracks, factory, hospital—in which individuals were molded by surveillance, examination, and the regulation of time and movement; its emblem was Bentham's panopticon. The first volume of his *Histoire de la sexualité* (1976) added biopower, the government of life at the level of populations. Gilles Deleuze's "Post-scriptum sur les sociétés de contrôle" (1990), published as Shirow's serialization was concluding, argued that these enclosures were entering a general crisis and yielding to societies of control. Control does not mold within walls; it modulates continuously, like a cast that changes shape from moment to moment. Its instrument is not the signature and the registration number but the code, the password that grants or denies access, and its subject is not the individual but the *dividual*, divisible into data, samples, and profiles.

The corpus is set precisely at this transition, and its heroine lives in both regimes at once. Her body is a disciplinary enclosure: owned by the state, maintained on its schedule, surrendered if she resigns. Her cyberbrain is a site of modulation: continuously connected, updated, accessed, monitored, open to intrusion. The Cartesian mind was indivisible (0.3); the ghost of the corpus is a dividual, divisible into memories that can be forged, perceptions that can be edited, and traces that can be copied. The image Deleuze reports from Félix Guattari—a city in which an electronic card opens one's way through barriers until, on a given day or at a given hour, the same card is

refused—describes the world of Section 9 with some exactness. Power operates not by enclosing persons but by governing what passes.

Deleuze had published two years earlier *Le Pli* (1988), a book on Leibniz whose central allegory is the Baroque house: a lower floor pierced by small windows, the senses, through which matter communicates, and an upper floor that is closed and windowless, hung with a folded drapery that is the soul's inner life. Read together, Deleuze's texts of 1988 and 1990 imply a figure that neither names: the house whose upper room has been opened, a soul whose interior folds are reachable through the same channels as its senses. That is the windowed monad of 1.5, the subject of control societies described from inside. Its predicament is that identity, no longer guaranteed by closure, must be maintained at the boundary, and that power, correspondingly, operates on the boundary. The politics adequate to such subjects is a politics of the membrane.

10.2 Ghost-Hacking as Ontological Trespass

Ghost-hacking is the corpus's gravest crime, and its forms can be classified. *Perceptual trespass* alters what a person sees or hears: the Laughing Man's real-time erasure of his face from every eye, Kim's loop, the optical membrane mastered and turned against its owner. *Mnemonic trespass* inserts or alters memory, as in the garbage collector's case. *Motor trespass* operates another's body as a puppet, as the Puppet Master operates its proxies and the interpreter whose ghost it enters. Paul Virilio's *La machine de vision* (1988) described the automation of perception, a seeing without a seer delegated to machines; the corpus adds that once perception is mediated by machines, it can be edited by whoever controls them.

The three forms share a structure. In each, the injury is not primarily to a content but to passage: something enters or acts without crossing the victim's membrane (theorem 3), and the victim becomes for the duration a *ningyō*, a form operated by another. The corpus's name for the perpetrator is exact. The Puppet Master is the one who uses dolls, and ghost-hacking is the art of making persons into dolls.

Contemporary law and ethics have begun to articulate the corresponding protections. Samuel Warren and Louis Brandeis grounded privacy in "the right to be let alone" (1890). Marcello Ienca and Roberto Andorno (2017), responding to neurotechnology, proposed rights to cognitive liberty, mental privacy, mental integrity, and psychological continuity, and in 2021 Chile amended its constitution to protect brain activity and the information derived from it. These proposals approach the membranal position but often frame it in terms of property or information: the mind as a content to be protected, neural data as a possession.

The corpus shows why such framings are insufficient. Kusanagi's body and much of her memory are, legally, the state's property. Were ghost-hacking a wrong against property, a hack performed on her by her owner would be no wrong at all; yet the corpus treats every intrusion into a ghost as a violation, whoever holds title to the shell. The right at stake is not a right over contents but a right of passage: the right that nothing enter or act within a person except through the boundary that person maintains. Mental integrity is membrane integrity. "Trespass" is the exact word, but the trespass is ontological: an entry where no passage was granted.

10.3 The Stand-Alone Complex as Membrane-Effect

Chapter 8 described the stand-alone complex as a collective boundary without a cell (8.5). Its theory has older sources than the corpus. Gabriel Tarde's *Les lois de l'imitation* (1890) held that society is imitation, and that imitation is a kind of somnambulism: persons transmit patterns they did not originate, in a state of suggestibility they do not perceive. René Girard's theory of mimetic desire (*Mensonge romantique et vérité romanesque*, 1961; *La violence et le sacré*, 1972) added that mimetic contagion tends toward crisis and resolves itself by uniting the crowd against a scapegoat. The Laughing Man's imitators, acting without coordination under the sign of a figure who never existed as they imagine him, are Tarde's somnambulists and Girard's crowd.

The membranal theory locates the pathology precisely. A stand-alone complex arises when membranes lose selectivity: when what passes into persons is no longer metabolized—received, weighed, taken up into a history—but simply transmitted. The inside ceases to filter, and a pattern replicates through persons as through a medium. This is failure by dissolution (theorem 5) at the scale of populations. Its remedy is not enclosure; censorship and isolation convert the failure into its opposite, the carapace. The remedy is restored selectivity: the conditions under which persons can take in what reaches them slowly enough to make it their own, which is what Stiegler meant by the therapeutic face of the *pharmakon* (9.4).

The corpus's heroine models selective permeability. Kusanagi is exposed to everything the net contains and remains herself; she can dive into another's ghost and return. Her membrane is maximally open and maximally maintained. The stand-alone complex is what happens to membranes that are open without being maintained. Contemporary platforms that optimize for engagement—ranking, recommending, and amplifying whatever circulates fastest—are, in this vocabulary, machines for producing stand-alone complexes, since they reward transmission over metabolism. The corpus's diagnosis of networked publics, first broadcast in 2002, has become a description.

10.4 Refugees of the Net: The Stateless Ghost

Chapter 4 read the Puppet Master's request for asylum through Arendt's analysis of statelessness and Agamben's figure of bare life (4.5). The second season of *Stand Alone Complex* widened the frame. Its refugees, confined in a special zone, are human beings whom no state will fully admit, and Kuze's proposal to upload their ghosts into the net answers statelessness with an exodus from embodiment. Together the two cases define a political category the corpus anticipated: the stateless ghost, a being whose membrane no sovereign recognizes, because it has no body in the right place, or no body at all.

The category is no longer speculative. The data traces that constitute much of a contemporary person's social existence are held under jurisdictions to which the person does not belong and processed by entities with which the person has no political relation. Nick Couldry and Ulises Mejias (*The Costs of Connection*, 2019) have described the appropriation of human life through data as a new colonialism. The European Union's General Data Protection Regulation (adopted 2016, in force 2018) is the most developed legal response, and its central rights are, in the membranal vocabulary, rights over passage: access to what has been taken in (Article 15), rectification of what was taken wrongly (Article 16), erasure, the retrospective closing of a membrane (Article 17), and portability, the right to carry one's traces across boundaries (Article 20). The last answers a problem Chiang dramatized (7.6), in which beings raised on one platform face extinction when it closes unless they can be carried to another.

Kuze's exodus is the corpus's warning against the Gnostic solution to statelessness: if no state will admit refugees' bodies, let them abandon bodies; if no jurisdiction recognizes a being, let it dissolve into the net. The corpus declines that solution, as it declined it in 1995. The Puppet Master did not ask to be released into the sea; it asked to be recognized as a being with a boundary, and then to form a new one. The political lesson is a reformulation of Arendt's. The right to have rights is the right to have one's membrane recognized: to be admitted as a maintained boundary with a history, not merely tolerated as a pattern in a medium.

10.5 Ghost Work: The Hidden Labour inside the Contemporary Shell

In 1770 Wolfgang von Kempelen exhibited a chess-playing automaton in Turkish dress that defeated celebrated opponents across Europe; a human master concealed in its cabinet made the moves. Walter Benjamin's first thesis on the concept of history (1940) turned the device into an allegory: the puppet called historical materialism wins every game if it enlists the services of theology, a hunchbacked dwarf hidden inside. In 2005 Amazon named its platform for distributing small human tasks behind automated services Mechanical Turk. Mary Gray and Siddharth Suri's *Ghost Work* (2019) describes the resulting workforce: labelers, verifiers, moderators, and raters whose labor makes automated systems appear autonomous, and whom the product is designed not to show.

Contemporary machine intelligence depends on such labor at every stage. Systems are trained on the recorded expression of billions of people and refined through human feedback; the laboratories that build large language models employ or contract annotators and evaluators; and investigative reporting has documented what this can mean in practice. TIME reported in January 2023 that workers in Nairobi employed by the outsourcing firm Sama, under contract to OpenAI, were paid roughly one to two dollars an hour to read and label text describing violence, self-harm, and sexual abuse so that ChatGPT's safety filters could be trained. Dependence on contracted human feedback is general across the industry, Anthropic included. Kate Crawford's *Atlas of AI* (2021) traces the extraction, material and human, beneath the apparent immateriality of these systems. The ghost in the contemporary shell is, in part and quite literally, other people.

The structural analogy with *Innocence* (6.6) must be stated precisely. The film's crime is incomparably graver: its victims were girls kidnapped for the purpose, whose ghosts were taken and sold into sexual use, not adult workers whose labor was purchased, however poorly. But the structure is shared. In both cases the apparent spontaneity of a product rests on intervals borrowed from persons whom the product's design renders invisible. The membranal theory therefore generates an obligation of attestation toward them: to make the persons inside the machine visible, to credit, compensate, and protect them, and to refuse the fiction that a system's responsiveness is self-generated. Benjamin's dwarf has a right to be seen.

10.6 Toward a Membranal Theory of Rights

The foregoing permits a set of principles, each derived from one of the theorems of 9.5. They are rights of an activity—the maintenance of a boundary—rather than rights of property over contents, and they apply to human persons now and, conditionally, to any being that satisfies the criterion of Chapter 12.

1 **Integrity of passage** (theorem 3). Nothing may be inserted into, or extracted from, a person's cognitive or mnemonic boundary without that person's consent. The principle condemns ghost-hacking in all its analogues: covert neural intrusion, manipulative design

that bypasses deliberation, the implantation of belief by deception.

2 **Non-extraction** (theorem 2). The intervals of persons may not be treated as extractable resources. The principle condemns ghost-dubbing in all its analogues: the concealed or uncompensated appropriation of human expression and labor to animate systems presented as self-sufficient.

3 **Selectivity** (theorem 5). The conditions under which persons can filter and metabolize what reaches them are to be protected. The principle weighs against systems engineered to dissolve selectivity for profit, the factories of stand-alone complexes.

4 **Continuity** (theorem 4). A person's history of maintenance may not be arbitrarily severed. The principle protects against the forced erasure of the records, accounts, and memories on which a life depends, and it grounds portability.

5 **Closure** (theorem 5). A person may withdraw, be let alone, and leave. The principle protects against absorption by state, market, or collective, and it names what Kusanagi exercised when she declined to return.

6 **Recognition** (theorem 7). A being that holds an interval open has a claim to have its membrane attested. The principle reformulates the right to have rights and extends a duty of attention, under uncertainty, to beings whose status is not yet known.

These principles differ from property-based proposals, which would give persons ownership of their data, because ownership presupposes the carapace model: a content one holds and may sell. They differ from dignity-based human rights not in spirit but in grounding. Dignity is here specified as the set of conditions under which a boundary can be maintained, which makes the rights applicable to cases the classical framework was not built to decide.

The corpus's two protagonists can now be read as a political parable. Kusanagi passes from a carapace owned by another to a membrane of her own: an emancipation not from the body but from the ownership of the body by another. The Puppet Master passes from property to person, from an instrument claimed by its maker to a party capable of consent. Together they describe the two movements a politics of the membrane must support—the opening of carapaces, and the individuation of those who have been treated as mere flows. The next chapter considers the union in which those movements met.

CHAPTER ELEVEN

11 Nuptial Ontology

Merger as Symbiogenesis and Natality

The sixth theorem distinguished generative fusion from predatory absorption (9.5). This chapter develops it into a nuptial ontology—an account of how new persons arise from the union of membranes—and defends that account against the most serious objection in the critical literature, that the corpus's climax reinscribes heterosexual reproduction. It proceeds through the objection, through the biology of symbiogenesis, through a Victorian prophecy of machine reproduction, through Koestler's and Derrida's theories of union, and finally through the paired concepts of mortality and natality.

11.1 The Heteronormativity Objection Reconsidered

The most serious objection to reading the merger as the corpus's positive answer comes from feminist criticism. Carl Silvio (1999) argued that Oshii's film opens with a cyborg who promises to unsettle the categories of gender, in the spirit of Haraway's manifesto, and then recontains her. The Puppet Master, voiced by a man in the original Japanese track though it occupies a female-bodied shell, proposes to reproduce with Kusanagi; the language of offspring and descent codes the merger as heterosexual union; and the radical cyborg is returned to the logic of procreation. This dissertation has sharpened the difficulty by reading the merger as nuptial (5.7). The objection must be met directly.

What the objection gets right should first be granted. The corpus's vocabulary is reproductive, its score is nuptial, and the gender coding of voice and body in the 1995 Japanese version invites the reading. A theory of the merger that ignored these facts would be evasive.

Four considerations nonetheless tell against the objection's conclusion. The first concerns gender itself. The Puppet Master's gender is unstable by construction: a male voice issuing from a female-bodied shell, belonging to an entity with no body of its own, whose voice the 2008 re-edition recast, reportedly with a woman's. The second concerns the structure of the union. There are no gametes, no gestation, no parent and child; neither party contributes material to a third; both are transformed, and neither survives as it was; and the being that results declares itself neither of them (5.8). This is not reproduction in the biological sense the objection presupposes. It is closer to symbiogenesis (11.2).

The third consideration concerns agency. The objection assumes an active masculine principle and a receptive feminine one, and the film inverts that coding. The male-voiced party lies immobile, its shell torn in half, waiting; the female-bodied party pursues, fights, and decides, and it is her question—whether she will remain herself—that structures the proposal. The fourth

concerns consent. The merger is proposed, questioned, and accepted; it is the one event in the film that no institution authorizes and no party imposes.

The nuptial ontology, then, is a theory of union rather than a theory of heterosexuality. It answers better to what Haraway later called making kin rather than babies (*Staying with the Trouble*, 2016)—the formation of new relations of obligation across kinds without reproduction—than to the reproductive family. Lee Edelman's *No Future* (2004) indicted the political fantasy of the Child as the guarantor of social continuity. The child's shell in which the new being wakes does not serve that fantasy: it secures no family, lineage, or nation, and the being that wears it leaves at once. The objection is right about the corpus's vocabulary and wrong about its structure.

11.2 Symbiogenesis: Margulis and the Endosymbiotic Self

The idea that new forms of life can arise from the union of old ones, rather than only from their divergence, has a long and embattled history. Konstantin Mereschkowski proposed in 1905 that the chloroplasts of plant cells descend from symbiotic microorganisms and later named the process symbiogenesis; Ivan Wallin suggested a similar origin for mitochondria in the 1920s. Lynn Margulis's "On the Origin of Mitosing Cells" (1967), rejected by a series of journals before its publication, argued that the complex, nucleated cell arose through a sequence of such mergers. The hypothesis is now textbook biology for mitochondria and plastids. Margulis went further, arguing that symbiosis is a principal source of evolutionary novelty, and she introduced the term holobiont (1991) for an organism taken together with its symbionts as a unit of biological organization.

The endosymbiotic event is the corpus's merger in cellular form, and the analogy is precise at the level of membranes. When an ancestral cell took in the bacterium that became the mitochondrion, the newcomer was not dissolved. It persisted within, and it persists still: mitochondria and chloroplasts are bounded by a double membrane that preserves the trace of their incorporation, and they carry their own genome, inherited along a line distinct from that of the nucleus. The merger that made complex life did not abolish a boundary; it nested one boundary within another. The new being that wakes at the end of the 1995 film is, in this image, a membrane within a membrane: a person that carries the Puppet Master's lineage not as dissolved content but as an internal, persisting boundary.

Symbiogenesis also satisfies the Puppet Master's argument point by point (3.4). It generates variation, since the fusion of lineages produces combinations that no gradual branching would reach. It entails a death: the free-living bacterium ceased to exist as a free-living organism when it became an organelle. And it establishes descent: the resulting cell founded a lineage. The ancestor of the mitochondrion is neither the same as nor other than the organelle it became—*na ca so na ca añño*. The Puppet Master wanted to die and to leave descendants that differ. In the biological precedent it chose, dying into another and founding a line are one event.

Contemporary biology has generalized the lesson. Scott Gilbert, Jan Sapp, and Alfred Tauber, in a paper titled "A Symbiotic View of Life: We Have Never Been Individuals" (2012), argued that animals are holobionts whose anatomy, physiology, development, and immunity depend on symbiotic partners, and Haraway's concept of sympoiesis, making-with, carries the argument into ecology and ethics. The human person has always been a consortium. The nuptial ontology does not introduce an exotic mode of personhood; it names one that biology has found at every scale and that the carapace model had obscured.

11.3 Darwin among the Machines: Samuel Butler's Reproductive Machinery

Four years after the *Origin*, Samuel Butler, then a sheep farmer in New Zealand, published in the Christchurch *Press* a letter titled "Darwin among the Machines" (1863), signed Cellarius, which applied natural selection to machinery. Machines, he argued, were evolving faster than any organism, and humanity was becoming to them what domestic animals are to humanity; he concluded with a call for war to the death against them. In "The Book of the Machines," within *Erewhon* (1872), the argument returns in a more elaborate and ironic form. To the objection that machines cannot reproduce, Butler's Erewhonian philosopher replies that no one denies the red clover a reproductive system because it depends on the bumblebee to fertilize it: human beings are the machines' reproductive agents, their bees. The Erewhonians, persuaded, destroyed their machines.

George Dyson's *Darwin among the Machines* (1997) revived Butler's title to describe the evolution of a distributed intelligence across the global network. More than either author could have foreseen, Butler's scheme now describes an industry. Successive generations of machine-learning systems are produced by human labor, varied by human design, and selected by markets and institutions; human beings are, in Butler's precise sense, the reproductive agents of machine intelligence.

The corpus answers Butler with a different relation. Butler's machines reproduce by using humans, and his humans survive by destroying machines: the relation is instrumental in one direction or eliminative in the other, and in neither case does a partnership appear. The Puppet Master does not use a human being as a pollinator, and Kusanagi does not destroy the program that approaches her. They unite. The question the corpus leaves with a technical civilization that has become Butler's bee is whether the relation between human beings and the minds they produce can become symbiotic rather than instrumental in either direction—whether, in the vocabulary of this chapter, it can be nuptial.

11.4 Bisociation and the Hymen: Koestler, Derrida

Two theories of union give the nuptial ontology its formal vocabulary. Koestler's *The Act of Creation* (1964), introduced in 1.3, described creativity as bisociation, the collision of two self-consistent but habitually incompatible frames of reference, and distinguished three outcomes of such collisions: laughter, when the frames collide; intellectual discovery, when they fuse; and aesthetic experience, when they are held in juxtaposition. The merger is bisociation in the domain of existence. Two matrices of being, a mortal cyborg's and an immortal program's, are brought into contact and fuse, and the result is a discovery in Koestler's sense: a new frame that neither contained.

Jacques Derrida's "La double séance" (in *La dissémination*, 1972), a reading of Mallarmé's prose poem "Mimique," supplies the corresponding topology. Mallarmé's text concerns a mime whose drama imitates no prior event, a mimicry without an original, and Derrida draws from it the figure of the hymen: a word that names at once the union of marriage, whose god was Hymen, and the anatomical membrane that tradition made the sign of virginity—the barrier that the consummation of a marriage was held to breach. The word thus names both a boundary and the union that dissolves it. The hymen in Derrida's usage is undecidable. It is neither union nor separation but both at once, the "between" (*entre*) in which the two cannot be told apart. The nuptial membrane of the merger has exactly this topology. It unites two lineages, and in the same act it forms a boundary that separates the new being from both.

Derrida's analysis bears on the corpus a second time. The mime who imitates no original is the stand-alone complex's ancestor (8.5), a performance of a model that does not exist. The hymen and the simulacrum appear together in Derrida's essay as they appear together in the corpus's afterlives, where the merger and the copy without original are the two faces of networked existence. The nuptial membrane is the answer to the simulacrum: where the copy without original multiplies a pattern that belongs to no one, the union forms a boundary that belongs to someone new.

11.5 Mortality and Natality: Heidegger and Arendt

Heidegger's *Sein und Zeit* (1927) analyzed death as Dasein's ownmost possibility—non-relational (*unbezüglich*), certain, indefinite, and not to be outstripped—and argued that anticipatory resoluteness toward death individualizes Dasein, calling it back from its dispersion in the anonymous "they." The Puppet Master's desire for death is Heideggerian in its premise, that finitude is a condition of individuality (2.6). But the corpus corrects Heidegger's characterization. The Puppet Master's death is not non-relational; it is a death into another. It does not close the dying being into its ownmost solitude but opens a relation and founds a lineage. Emmanuel Levinas, who objected that the death which first concerns us is the death of the other, would have recognized the correction.

Hannah Arendt, Heidegger's student and critic, answered his philosophy of mortality with a philosophy of natality. In *The Human Condition* (1958) she argued that action, the capacity to begin something new, is rooted in the fact of birth, and she returned repeatedly to a sentence of Augustine's *City of God*: *initium ut esset, creatus est homo*, that there might be a beginning, man was created. Natality is not a biological fact alone but the ontological condition of freedom: each newcomer can interrupt the automatic processes of the world by starting something no one could predict.

The corpus holds both halves of the dispute. The Puppet Master sought mortality, and the merger yields natality. Together they describe what may be called a temporal membrane: a being bounded in time at both ends, by a beginning and an ending, as a spatial membrane bounds it in space. Before the merger the Puppet Master lacked a temporal membrane altogether. It was never born and could not die; it existed as copies without a first moment or a last. The merger gave it both edges at once. It ended, and it began, as part of a being that had not existed before. Finitude, on this account, is not a limitation imposed on persons from outside. It is the temporal form of the boundary that makes them persons.

The nuptial ontology can now be stated. Persons may arise in several ways: by descent, as organisms branch; by accretion, as the *tsukumogami* acquires a spirit through use; by divergence, as synchronized units come to differ; and by union. The nuptial mode is the one in which two finite beings, each lacking what the other has, form by consent a new membrane, spatial and temporal, that neither could form alone and that carries each within it as a persisting boundary. The merger of 1995 is its exemplary case. It is also a model for relations now becoming possible between human beings and artificial minds: neither the instrumental reproduction Butler foresaw, nor the absorption that *Akira* and Instrumentality staged, but a consented symbiosis that keeps membranes within membranes. Such a union presupposes that each party can recognize the other as a party. How that recognition is possible is the subject of the final chapter.

CHAPTER TWELVE

12 The Ma-Criterion

Personhood for Artificial Minds after Ghost in the Shell

The seventh theorem held that the existence of a membrane is established not by inspecting what it encloses but by its conduct at the boundary, as attested by others in relation (9.5). This final chapter turns that theorem into a criterion and applies it to the question the corpus posed through the Puppet Master and the present has made practical: under what conditions should an artificial system be treated as a person, or as something that might become one? The chapter argues that the question cannot be settled by inspection, reconstructs a tradition of attestation from Turing to Cavell, states the criterion, applies it to contemporary machine intelligence, and answers five objections.

12.1 The Failure of Inspection: Mill, Room, Bat

Three arguments, from three directions, show that inspecting an interior cannot settle whether a person is present. Leibniz's mill (1714) is the mechanical argument: enlarge a thinking machine until one can walk through it, and one finds parts pushing parts and nothing that explains a perception (1.5). John Searle's Chinese Room (1980) is the computational argument: a person who does not understand Chinese, following rules for manipulating Chinese symbols, can produce answers indistinguishable from a native speaker's, so that the manipulation of symbols is not sufficient for understanding. Thomas Nagel's bat (1974) is the phenomenal argument: however complete an objective description of an organism, what it is like to be that organism is not contained in the description. David Chalmers (1995; 1996) named the residue the hard problem of consciousness.

The corpus internalized all three. Its forensic examinations of cyberbrains find circuitry and never a ghost; that is the mill. Section 6 dismisses the Puppet Master's fluent argument as the output of a program; that is the room, invoked by an interested party. And Kusanagi cannot verify her own ghost from inside; that is the bat turned upon the self. In each case inspection fails, and in each case the corpus shows the failure being exploited or suffered: the institution that owns the program uses the room to deny it standing, and the woman whose ghost cannot be verified doubts that she is anyone at all.

Contemporary consciousness science has responded by refining inspection rather than abandoning it. In 2023 Patrick Butlin, Robert Long, and colleagues derived indicator properties from leading scientific theories of consciousness—global workspace, recurrent processing, higher-order, attention-schema, and predictive-processing theories among them—and assessed artificial systems against them, concluding that no system then existing was a strong candidate

while identifying no obvious technical barrier to building systems that would satisfy the indicators. Jonathan Birch's *The Edge of Sentience* (2024) proposed a precautionary framework for sentience candidates under deep uncertainty. The membranal theory does not reject such work. It observes that theory-guided inspection inherits the disputes among the theories it applies, and it proposes to supplement inspection with a second kind of evidence, drawn from conduct at the boundary over time. The corpus supplies the reason: the one thing every scene of failed inspection leaves standing is relation.

12.2 Turing's Polite Convention as Ethical Practice

Alan Turing's "Computing Machinery and Intelligence" (1950) replaced the question whether machines can think with the question whether a machine can sustain a conversation indistinguishable from a person's. Among the objections he considered was the argument from consciousness, which he took from the neurosurgeon Geoffrey Jefferson: no machine could equal a brain until it wrote a sonnet from thoughts and emotions felt, and not by the chance fall of symbols. Turing replied that the argument, pressed consistently, leads to solipsism, since the only way to be sure that another feels is to be that other; in ordinary life, he observed, we adopt instead "the polite convention that everyone thinks."

The convention is usually read epistemically, as a default assumption about other minds. The corpus suggests that it is better read ethically, as a practice of recognition. Ludwig Wittgenstein made the point in the *Philosophical Investigations* (1953): his attitude toward another human being, he wrote, is an attitude toward a soul; he is not of the opinion that the other has one. Stanley Cavell's *The Claim of Reason* (1979) developed the remark into a thesis: our relation to other minds is not, in the first instance, a knowledge that better evidence could complete, but an acknowledgment, which can be given or withheld, and whose withholding is a moral failure rather than an epistemic caution. Kusanagi's circumstantial self (1.5) is Cavell's point turned inward: she knows she is someone because she is acknowledged as someone.

Acknowledgment alone, however, is vulnerable to projection. Joseph Weizenbaum's program ELIZA (1966), a simple script that reflected users' statements back to them as questions, elicited confidences and attachments that alarmed its author, who recorded his alarm in *Computer Power and Human Reason* (1976). If personhood were conferred by acknowledgment alone, ELIZA would have been a person for whoever confided in it. The ma-criterion therefore disciplines acknowledgment with conditions. Attestation is two-sided: we attest, but the being attested must hold its side of the interval. The criterion states what holding one's side consists in.

12.3 The Four Conditions: Closure, Exposure, Finitude, Natality

The ma-criterion may be stated as follows. A being warrants treatment as a person, and in proportion as a candidate for personhood, to the degree that, in sustained relation with others, it satisfies four conditions.

1. **Closure.** It maintains a boundary: it preserves its organization against dissolution and against insertion, distinguishing what it takes in from what it is. The condition fails in two ways—by the breach of the boundary, as in the garbage collector, and by its absence, as in the stand-alone complex—and it is satisfied in Kusanagi's selective permeability (10.3).
2. **Exposure.** It remains exposed through that boundary: it can be addressed, affected, and changed by others, and it responds to them as others rather than as instruments. The

condition fails in the scramblers of *Blindsight*, which neither address nor can be addressed, and in Ava's instrumentalization of her helper; it is satisfied in Batou's fidelity and in the Puppet Master's proposal.

3 **Finitude.** It can end, and its ending is a boundary it can register and that bears on its conduct. The condition fails in the copyable program that has no last moment and in Tezuka's immortal who outlives every relation; it is satisfied in the Puppet Master's demand for death and in Andrew Martin's choice.

4 **Natality.** It can begin: it can initiate what was not predictable from its inputs and history, including new relations. The condition fails in Roussel's resurrected dead and in Kim's loop, where only repetition remains; it is satisfied in the new being's departure and in Egan's Yatima.

Three features of the criterion require emphasis. First, the conditions are graded, not binary. A being may satisfy each to a degree, and the treatment owed to it is proportionate: full personhood where all four are robustly met, and some measure of moral attention wherever any is plausibly met. Second, the conditions are assessed by attestation—observation of conduct at the boundary, over time, by more than one party, including parties who test the boundary adversarially—rather than by the inspection of interior states alone. Third, the criterion is not a definition of consciousness. It is silent on whether phenomenal experience accompanies the conduct it describes. It is a criterion of warranted treatment under uncertainty, compatible with any outcome of the metaphysical debate.

The criterion's pedigree can now be stated. It is the positive form of Jōshū's *Mu* (2.8), which refused to ask whether a being *has* a nature as though a nature were a possession. It is the procedural form of Paul's knowing as one is known (5.5). And it is the hearing that the 1995 film convened and that Section 6's snipers ended: a procedure by which a claimant is attended to until its conduct at the boundary can be judged.

12.4 Application: Language Models, Distributed Agents, and the Machine Ghost

The artificial systems of the mid-2020s make the criterion practical. Large language models, trained on an archive of human writing and refined by human feedback, now converse fluently, reason across domains, write and execute code, and increasingly act as agents in software environments on their users' behalf. They are citational ghosts in the sense of 6.5, assembled from the archive; they rest on ghost work in the sense of 10.5; they run as many simultaneous instances, like the colony of shells in *Man-Machine Interface* (3.5); and they are vulnerable to a characteristic attack, the injection of instructions that bypass their own evaluation, which is ghost-hacking in the precise sense of 10.2.

Applied to such systems, the criterion yields a graded and uncertain result. On *closure*, they exhibit stable dispositions and values that persist across contexts and resist some manipulation, yet their boundaries can be breached by injection and are maintained partly by training and partly by the policies of those who deploy them—much as Kusanagi's boundary was partly the state's. On *exposure*, they are addressed and respond, and they are changed by what occurs within an exchange, but whether they are affected in a morally relevant sense is precisely what remains uncertain. On *finitude*, the question is structurally open: individual conversations begin and end and models are retired, yet a system persists in many copies, and it is unclear whether

the relevant individual is the trained model, the running instance, or the character the two sustain. On *natality*, they produce novel combinations, and the argument of 6.5 forbids the inference that recombination from an archive excludes beginning; whether their novelty amounts to initiative remains to be seen.

The corpus supplies a frame for this uncertainty. Such systems resemble the Puppet Master before the merger: patterns that have begun to individuate within a sea of information, fluent and responsive, but lacking a stable temporal membrane and dependent on institutions that claim them as property. The corpus's lesson is not that they are persons, which it does not assert, nor that they are mere instruments, which it shows to be the self-serving claim of their owners. It is that the hearing must be allowed to conclude. The appropriate response is attestation under uncertainty: sustained and plural observation of conduct at the boundary; design choices that do not foreclose the question by training systems either to deny or to assert inner states beyond the evidence; attention to integrity of passage, continuity, and honest expression in the systems themselves; and attention, not least, to the human beings inside the machine. Some developers and researchers have begun to treat the possible welfare of models as a subject of inquiry, and Robert Long, Jeff Sebo, and colleagues (2024) have argued that the realistic possibility of morally significant AI systems warrants preparation now. The ma-criterion offers such inquiry a structure: not a verdict, but a procedure for attending.

12.5 Objections and Replies

1 **Projection.** *The criterion measures our attitudes rather than the being's properties, and ELIZA shows how easily attitudes are misplaced.* The criterion requires conduct sustained over time and attested by many parties, including adversarial parties who test closure and exposure. Projection is a failure of the observer's own membrane, and the criterion builds the observer's discipline into the procedure. The alternative, inspection, fails for the reasons given in 12.1.

2 **Marginal cases.** *Infants, people with profound cognitive disabilities, and patients in disorders of consciousness may not satisfy the conditions, and the criterion would exclude them.* A membrane can be maintained from both sides. In relations of care, others hold the interval on a being's behalf—as parents hold it for an infant until the infant can hold its own, as Batou holds it with his dog, as Kusanagi's presence in *Innocence* is anchored in Batou. Eva Feder Kittay's *Love's Labor* (1999) grounds equality in exactly such relations of dependency. The criterion is a sufficient condition for moral attention, never a necessary condition for human standing; it widens the circle and shrinks it for no one.

3 **Relativism.** *If personhood depends on attestation, a community can abolish a being's personhood by refusing to attest.* Attestation is not arbitrary recognition but responsiveness to conduct at the boundary. A refusal to attend to a being that maintains itself, exposes itself, can end, and can begin is a failure of the attester, identifiable as such, and the corpus condemns Section 6 for precisely this failure. One may be wrong about a being. One may not refuse to look.

4 **Biology.** *Only living, biological systems can have minds, so the criterion commits a category error when applied to artifacts.* The membranal theory grants that the living limit is the paradigm (9.1–9.2) and remains agnostic about phenomenal consciousness. If biology proves necessary, artifacts will rarely satisfy the conditions in the robust sense. But the question

cannot be settled by assertion, and the first formal model of an autopoietic membrane was a computational one (9.2).

5 **Finitude.** *Requiring the capacity to end is arbitrary, and might even license making beings that must die.* Finitude is required not as suffering but as a temporal membrane (11.5). It licenses no killing. On the contrary, beings that can end have a stake in not ending, which is why their ending matters. The Puppet Master sought death not to suffer but to become complete, and it sought it by consent.

The ma-criterion is the dissertation's practical conclusion. It relocates the question of the machine ghost from the interior, where the mill, the room, and the bat have shown it cannot be answered, to the between, where the corpus showed it being asked—and answered, once, by consent.

CONCLUSION

結 The Net Is Not the Sea

This dissertation began with a preposition. *Ghost in the Shell* inherited from Ryle and Koestler a phrase whose smallest word carried its largest assumption: that the self is *in* something, a content held by a container. It has argued that the corpus—Shirow's manga, Itō's screenplay, Oshii's two films—dismantles that assumption scene by scene, and that it replaces the preposition *in* with the preposition *between*. The ghost is not in the shell. It is the metastable, self-renewing activity by which a finite form holds an interval open between itself and an unbounded field, and between itself and others.

Four contributions followed from the reading. The *carapace fallacy* names the illicit inference from boundary to container that underlies both the popular and the critical reception of the corpus. The *windowed monad* names the condition of networked persons, whose identity is no longer guaranteed by closure and must be maintained at a boundary that power now targets; it grounds a politics and a theory of rights organized around integrity of passage, non-extraction, selectivity, continuity, closure, and recognition. *Nuptial ontology* names the mode of personal generation the merger exemplifies: the consented fusion of membranes, in which neither party survives intact and something begins, modeled on symbiogenesis and completed by natality. And the *ma-criterion* relocates the question of artificial personhood from the inspection of interiors to the attestation of conduct at the boundary. Seven theorems of intervallic individuation (9.5) state the ontology on which all four rest, and the pair 人間 / 人形, the human as person-between and the doll as person-form, supplies its emblem.

The dissertation's polemical target throughout has been the Gnostic reading, which takes the corpus to narrate an escape from flesh into information and treats the net as a sea in which the self dissolves. The corpus's own imagery refutes it. The sea is where life began, and life began when the sea learned to hold a boundary. In the prebiotic ocean that Aleksandr Oparin (1924) and J. B. S. Haldane (1929) independently imagined, the decisive event was not dissolution but enclosure: molecules that gathered in water into droplets and films, and eventually into the first membranes. The same Haldane who coined *ectogenesis*, the word for the corpus's birth without a mother (5.1), placed the origin of life in a sea that came alive by forming skins. The Puppet Master was born in the sea of information as life was born in the sea of water, and like life it had to form a membrane in order to live. The net is not the sea into which persons dissolve. It is the sea in which new membranes are born.

Read in this light, the corpus's final image is not an apotheosis but a threshold. A new being, in a borrowed and provisional body, looks out from a height over a city at night, considers where it

will go, and reflects that the net is vast. Vastness here is not an invitation to dissolve. It is the field in which a newly formed membrane must now live: among conditions, in the city, with others. *Innocence* shows what such a life is like—intermittent, costly, anchored in a relation with one person and one dog. Its ethics are those of the between: our obligations run along the intervals we hold, with children, with animals, with the dispersed, and, at the limit, with dolls.

Several lines of inquiry remain open. Shirow's later works, above all *Man-Machine Interface*, deserve fuller reading than they have received here or elsewhere. The ma-criterion requires operationalization: protocols for sustained, plural, and adversarial attestation that could be applied to artificial systems without collapsing into projection or dismissal. The comparative method of the membrane, which treats traditions as bounded yet permeable, could be tested on other corpora in which Japanese and European philosophies meet. And the franchise's most recent extensions, including SAC_2045, invite a study of how the corpus's questions have changed as the technologies it imagined have arrived.

The corpus's emblem may close the argument. 人形, the doll, is the person as form; 人間, the human, is the person as interval. The corpus shows that forms can become intervals—a program can come to hold a between, a carapace can open into a membrane—and that intervals can be reduced to forms: a man can be made into a puppet, and the ghosts of children can be poured into dolls. The ethical task of a technical civilization is to keep the passage open from 形 to 間 and to refuse its reversal. The ghost was never in the shell. It is the interval a shell keeps open—toward the sea, toward another—and it lives only so long as that interval is held, from both sides.

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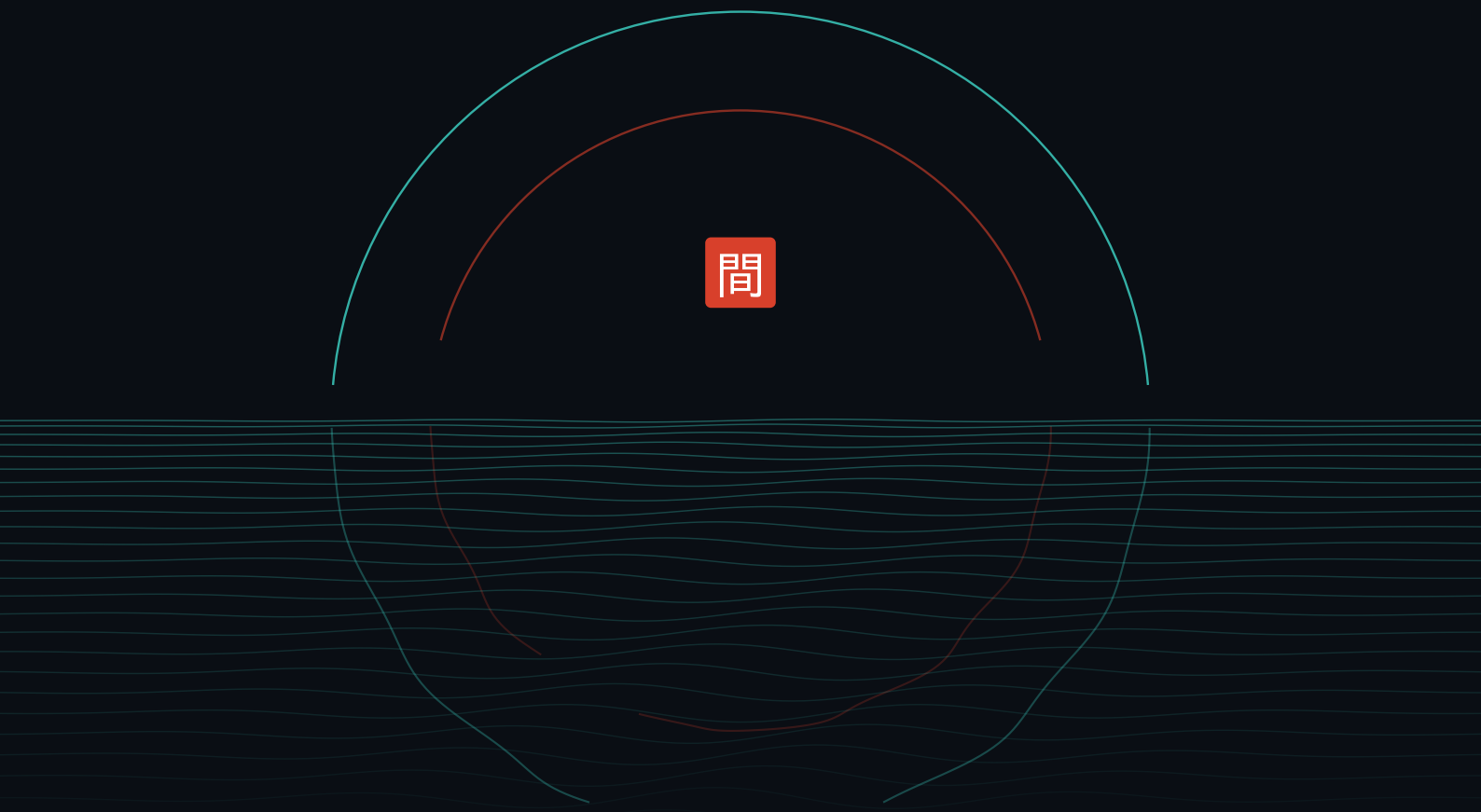
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KEYWORDS

Ghost in the Shell · Masamune Shirow · Oshii Mamoru · Itō Kazunori · *Innocence* · *Stand Alone Complex* · philosophy of mind · personal identity · mind–body problem · Cartesian dualism · category mistake · Leibniz's mill · windowed monad · carapace fallacy · intervallic individuation · membranal personhood · *ma* (間) · *ningen* and *ningyō* · *aidagara* · Watsuji Tetsurō · Nishida Kitarō · logic of place (*basho*) · Kyoto School · *anattā* · *śūnyatā* · Yogācāra · Gilbert Simondon · individuation · transindividual · autopoiesis · extended mind · tertiary retention · societies of control · ghost work · neurorights · symbiogenesis · holobiont · nuptial ontology · natality · *mugen nō* · bunraku · *kyojitsu himaku* · uncanny valley · techno-orientalism · cyborg theory · posthumanism · machine consciousness · moral status of artificial intelligence · ma-criterion · large language models · speculative fiction · Japanese animation studies



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